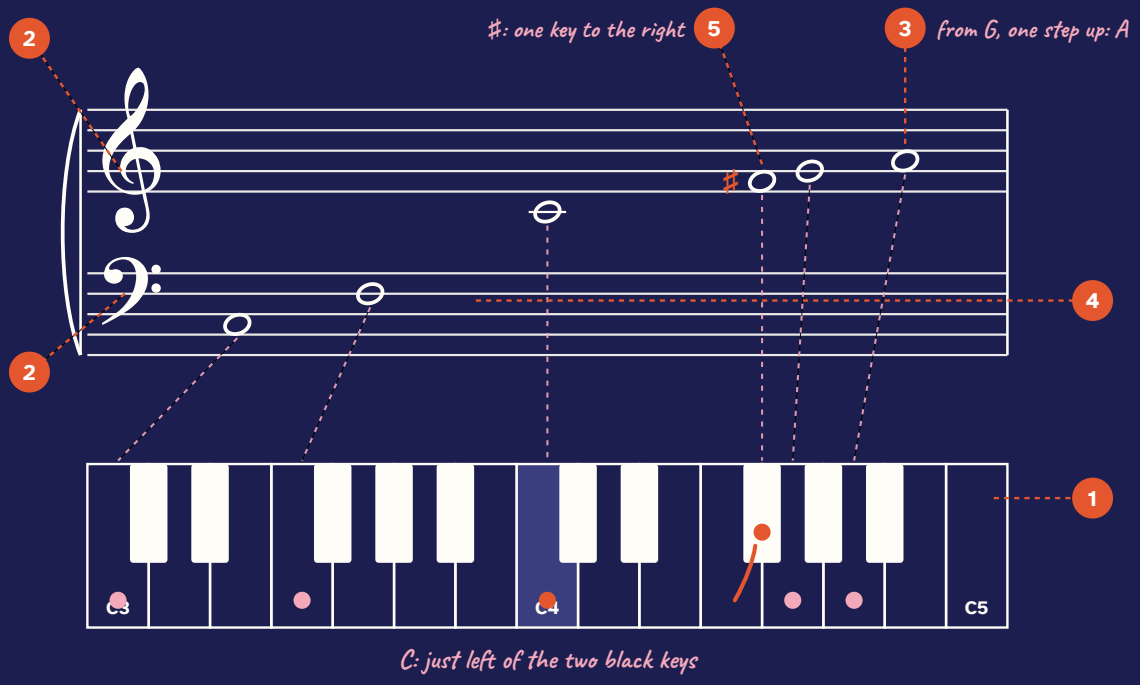
PART I · WHICH NOTE

Pitch

High or low? That is the first question every note answers. Start with the keys under your fingers, then find where they live on the staff and how sharps and flats move them. Everything in this part is in this one picture.



1 **The keyboard** p. 22
two and three black keys; C is left of the two

2 **The staff and the clefs** p. 25
the treble clef names G, the bass clef F

3 **Reading the notes** p. 28
find a landmark note, then step or skip from it

4 **The staff and the keyboard** p. 31
every line and space is one white key

5 **Sharps, flats and naturals** p. 34
a sharp moves one key right, a flat one key left

★ **Tips: when a note reads wrong** p. 37

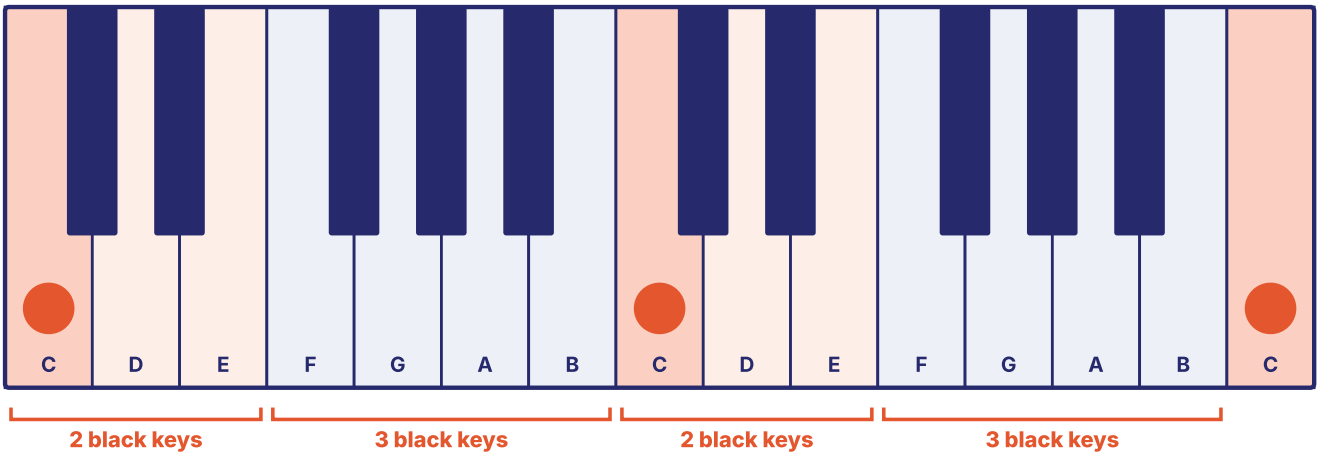
PART I · PITCH · 1

The keyboard

Keys to the **left** sound **lower**; keys to the **right** sound **higher**. The black keys come in groups of **two** and **three**, and that pattern repeats all the way along: it is how you find every note.

THE SECRET

Find the two black keys: C is always just to their left.



Every C sits just left of the two black keys

The white keys

Seven letters, **A B C D E F G**, then the letters start again, like the days of the week. **C D E** sit around each group of two black keys (C just left of it), **F G A B** around each group of three (F just left of it): the two tints on the picture.

Middle C and octaves

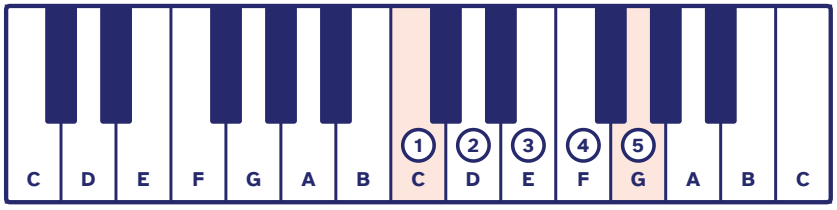
Middle C is the C nearest the middle of the keyboard. From one C to the next is an **octave**: eight white keys, counting both. Octaves are numbered from the bottom: middle C is **C4**, the C below it C3, the C above C5; every key up to the next B shares the number (D4, E4 … B4).

TERM	MEANING	FIRST MET
pitch	how high or low a sound is	Level 0
register	a part of the keyboard: low, middle or high	Level 0
semitone	from one key to the very next key, black or white	Level 0
tone	two semitones	Level 0
octave	from one note to the next with the same name (8 white keys, 12 semitones)	Level 0

PART I · PITCH · 1 · CONTINUED

EXAMPLE 1

Find G4 on the keyboard

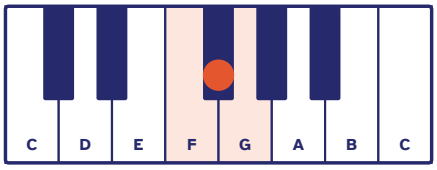


1. **Find middle C (C4):** the C just left of the two black keys nearest the middle.
2. **Count up the letters**, one white key each: C D E F G is five keys.
3. **Same octave**, so the number stays 4.

The key: **G4**. One octave higher, the next G, is G5.

EXAMPLE 2

Find F♯4

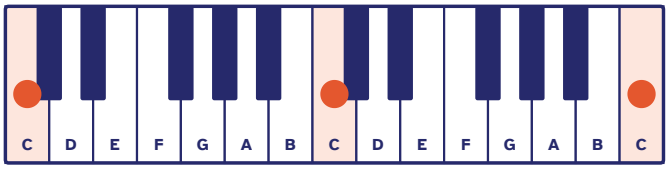


1. **Find F4:** the white key just left of the group of three black keys above middle C.
2. **Sharp means up:** move one key to the right.
3. **The very next key** is black: that is F♯.

F♯4 is the **black key** just right of F4, the first of the group of three.

EXAMPLE 3

Find C3, C4 and C5



1. **Find middle C (C4):** just left of the two black keys in the middle.
2. **Count seven white keys down:** C3.
3. **Count seven white keys up:** C5.

Every C sits just left of a group of **two** black keys, an octave apart.

TRY IT · 10 SECONDS



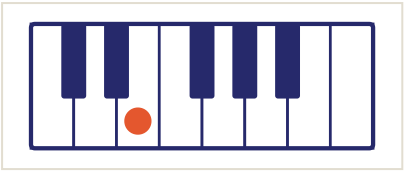
Play every C on the piano.

Start at the bottom and work up, saying “C” each time: left of the two black keys, every time. Then every F: left of the three.

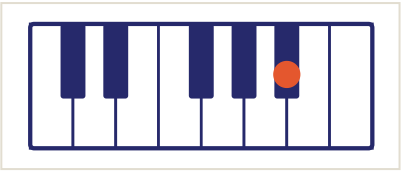
PART I · PITCH · 1 · CONTINUED

GAME

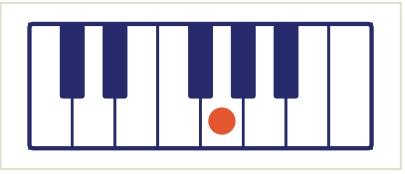
Name the key



1



2



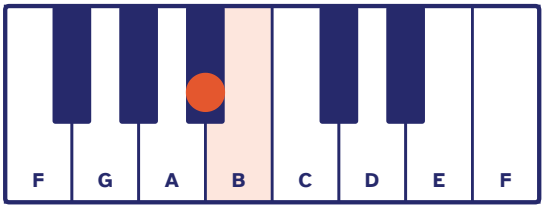
3

Each keyboard starts on C. Use the groups of black keys to name the red key.

Answer: 1 E (right of the two) · 2 the black key between A and B: A♯ or B♭ · 3 G (in the group of three).

EXAMPLE 4

Find B♭3

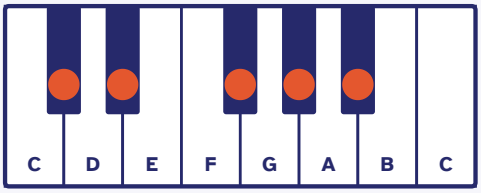


1. **Find middle C (C4)**, then the white key just below it: **B3**.
2. **Flat means down:** one key to the left.
3. **The very next key** is black: the top key of the group of three.

B♭3 is the **top black key** of the group of three just below middle C.

YOU KNOW THIS

Amazing Grace, on the black keys only



The whole tune fits on the **five black keys**: D♭ G♭ B♭ G♭ B♭ A♭ G♭ E♭ D♭. The groups of two and three are a little melody machine.

Try it. Start on the lower key of a pair of black keys and find the tune by ear.

THINK OF IT The black keys are like the little bumps on F and J of a computer keyboard: feel them and you know where you are without looking.

TRAP

× Calling every black key a sharp: each one has a flat name too (p. 34).

SEE ALSO

The staff and the clefs p. 25 · The staff and the keyboard p. 31

PRACTISE IT

Workbook Pitch 1 · Online: Library › Keyboard

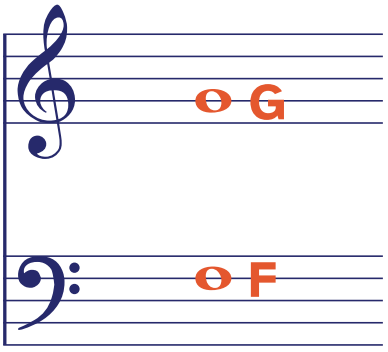
PART I · PITCH · 2

The staff and the clefs

Music lives on a **staff** (or stave): five lines and four spaces, counted from the **bottom**. Higher on the staff means higher in sound. The **clef** at the start is the code: it says which note each line and space stands for.

THE SECRET

The treble clef curls round G; the bass clef's two dots hug F.



The treble clef curls round G · the bass clef's dots hug F

CLEF	NAME	HOW IT SHOWS ITS NOTE	FIRST MET
𝄞	**treble clef** (G clef)	curls round the line for G (line 2)	Level 0
𝄢	**bass clef** (F clef)	its two dots sit either side of the line for F (line 4)	Level 0



Alto clef: middle C on line 3. Used by the viola.

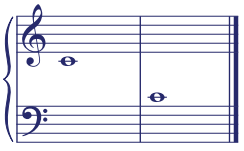


Tenor clef: middle C on line 4. Cello, bassoon and trombone, up high.

Other clefs. C clefs: the line through the centre of the clef is middle C. Piano music never uses them, but pianists meet them in theory exams and when reading the viola or cello part in chamber music or a score (Level 2).

The grand staff

Piano music uses two staves joined by a brace: the treble staff, usually for the right hand, above the bass staff, usually for the left. **Middle C** sits between them on a short line of its own, and can be written in either staff.

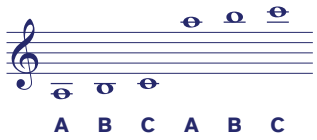


Middle C in the treble staff (left) and in the bass staff (right): the same key.

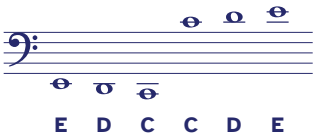
Ledger lines

Notes above or below the staff sit on short extra lines called **ledger lines**, spaced like the staff lines.

PART I · PITCH · 2 · CONTINUED



Treble: A, B, C below; A, B, C above.



Bass: E, D, C below; C, D, E above.

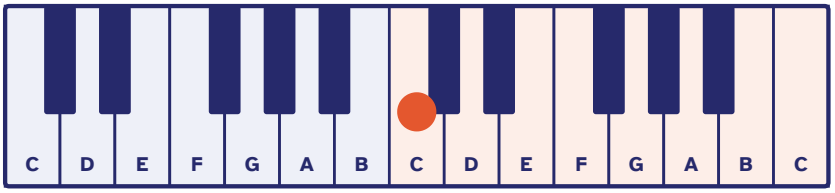
Octave signs

SIGN	MEANING
8va	an octave higher, as far as the dotted line
8vb	an octave lower
loco	back where it is written

First met: Level ½. Octave signs save writing many ledger lines: a high passage stays on the staff, with 8va above it.

EXAMPLE 1

Which staff does this note belong to?



1. **Below middle C** (blue): the bass staff, usually the left hand.
2. **Above middle C** (pink): the treble staff, usually the right hand.
3. **Middle C itself** (the red dot): either staff, whichever hand plays it.

Low keys: **bass clef**. High keys: **treble clef**. The two staves meet at middle C.

GAME

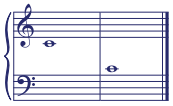
Treble or bass?

1. a note two keys above middle C
2. the lowest key on the piano
3. middle C
4. a high, high note

Which staff would each note be written on?

Answer: 1 treble · 2 bass · 3 either: it sits between the two · 4 treble, with ledger lines or 8va.

TRY IT · 10 SECONDS



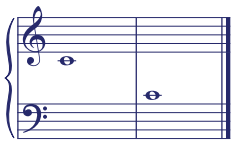
Point, then play.

Point to the line the treble clef curls round and play that G. Point to the line between the bass clef's dots and play that F. Middle C sits between them.

PART I · PITCH · 2 · CONTINUED

EXAMPLE 2

Find middle C on both staves

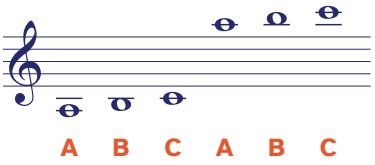


1. **Treble staff:** middle C hangs on a short ledger line just below it.
2. **Bass staff:** middle C sits on a ledger line just above it.
3. **Play both:** the same key, near the middle of the piano.

Two places on the page, **one key**: whichever hand plays it chooses the staff.

EXAMPLE 3

Read a high note on ledger lines
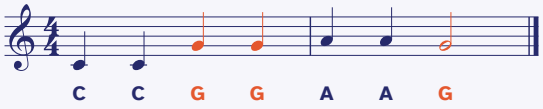


1. **The top line** of the treble staff is F.
2. **Step up:** G sits on top of the staff, A on the first ledger line.
3. **Keep stepping:** B above it, C on the second ledger line.

Two ledger lines above the treble staff: **C6**, two octaves above middle C.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The treble clef curls round **line 2: G**, the very note Twinkle leaps up to. Middle C hangs on its own ledger line below.

Try it. Point to the G line, then play C C G G A A G.

THINK OF IT A clef is a pair of glasses: the same five lines read as different notes, depending on which pair you put on.

TRAP

× Reading bass-clef notes as if they were treble: check the clef on every line.

SEE ALSO
Reading the notes p. 28 · The staff and the keyboard p. 31

PRACTISE IT
Workbook Pitch 2 · Online: Library › Staff and clefs

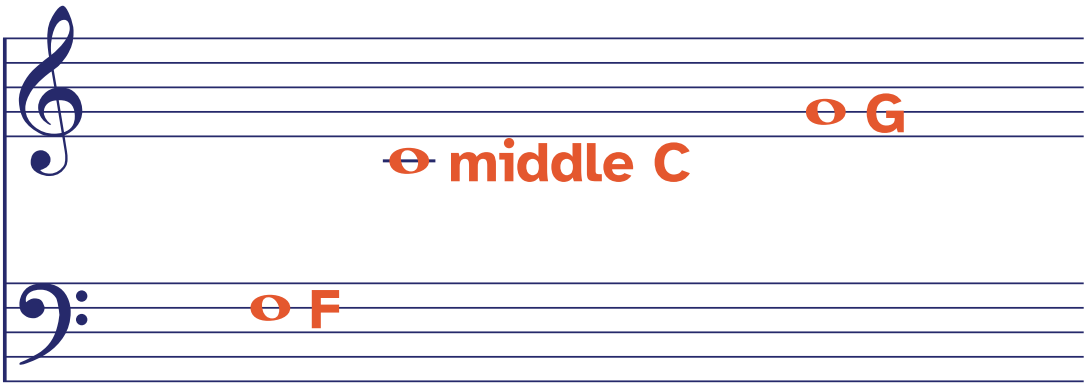
PART I · PITCH · 3

Reading the notes

Each line and space has one letter name. Learn them in two ways: by the patterns below, and by a few **landmark notes** you can find instantly and count from.

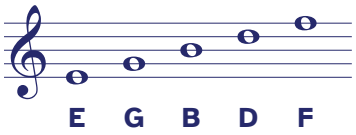
THE SECRET

Learn three landmark notes, then step or skip from them.



Find a landmark, then count

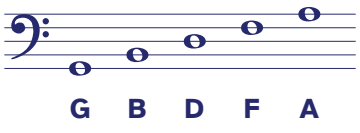
Find the three red **landmark notes** first: bass F, middle C, treble G. Every other note is a **step** (line to space) or a **skip** (line to line) away from one of them.



Treble clef: the lines



Treble clef: the spaces



Bass clef: the lines



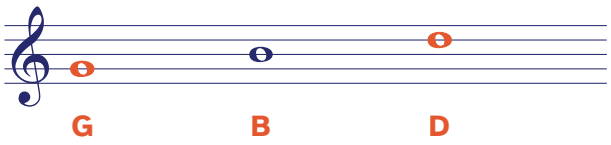
Bass clef: the spaces

STAFF	NOTES, BOTTOM TO TOP	TO REMEMBER
Treble lines	E G B D F	*Every Good Boy Deserves Fun*
Treble spaces	F A C E	spells *FACE*
Bass lines	G B D F A	*Great Big Dragon Flies Away*
Bass spaces	A C E G	*All Cows Eat Grass*

PART I · PITCH · 3 · CONTINUED

EXAMPLE 1

Read a note you are not sure of



1. **The clef:** treble.
2. **A landmark:** the G the clef curls round, on line 2.
3. **Skip up** line to line, one letter jumped each time: G, B, D.

The note on line 4 is **D**.

GAME

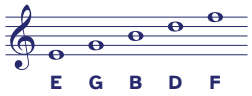
Fill the gap

1	2	3	4
E G _ D F	F A _ E	G B D _ A	A C E _

Treble lines, treble spaces, bass lines, bass spaces: one letter is missing from each.

Answer: 1 B · 2 C · 3 F · 4 G.

TRY IT · 10 SECONDS

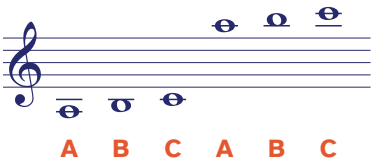


Say the treble lines up and down.

"E G B D F", then "F D B G E", out loud, as fast as you can without a slip. Then the spaces: "F A C E".

EXAMPLE 2

Read a ledger-line note



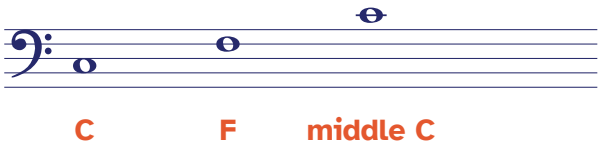
1. **Start at the staff edge:** the top line of the treble staff is F.
2. **Step up** one note at a time: G (space), A (first ledger line).
3. **Below the staff** count down the same way: D, C, B, A.

Ledger lines are the staff carried on: count by steps from the nearest line.

PART I · PITCH · 3 · CONTINUED

EXAMPLE 3

Read a bass-clef note from F

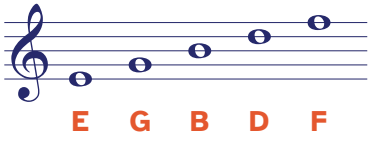


1. **The bass clef's dots** sit either side of line 4: that line is F.
2. **Step down** to the space below: E.
3. **One more step** to line 3: D.

Bass line 3 is **D**: two steps down from the clef's F.

EXAMPLE 4

Read the treble lines by skips
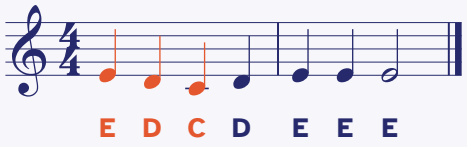


1. **The bottom line** is E.
2. **Line to line is a skip:** one letter is jumped, so the next line is G.
3. **Keep skipping:** B, D, F.

Skip, skip, skip: the lines read **E G B D F** without a rhyme.

YOU KNOW THIS

Mary Had a Little Lamb



E D C: **line, space, ledger line**, one step down each time. Read the shape and the tune walks down with it.

Try it. Read the notes aloud before you play them.

THINK OF IT Landmark notes are like lamp posts on a dark road: walk from one to the next and you are never lost.

TRAP

- × Naming bass-clef notes with treble names: in the bass, the bottom line is G, not E.
- × Counting ledger lines from the wrong end: start from the staff and count outwards.

SEE ALSO

The staff and the clefs p. 25 · The staff and the keyboard p. 31

PRACTISE IT

Workbook Pitch 2 · Online: Library › Staff and clefs

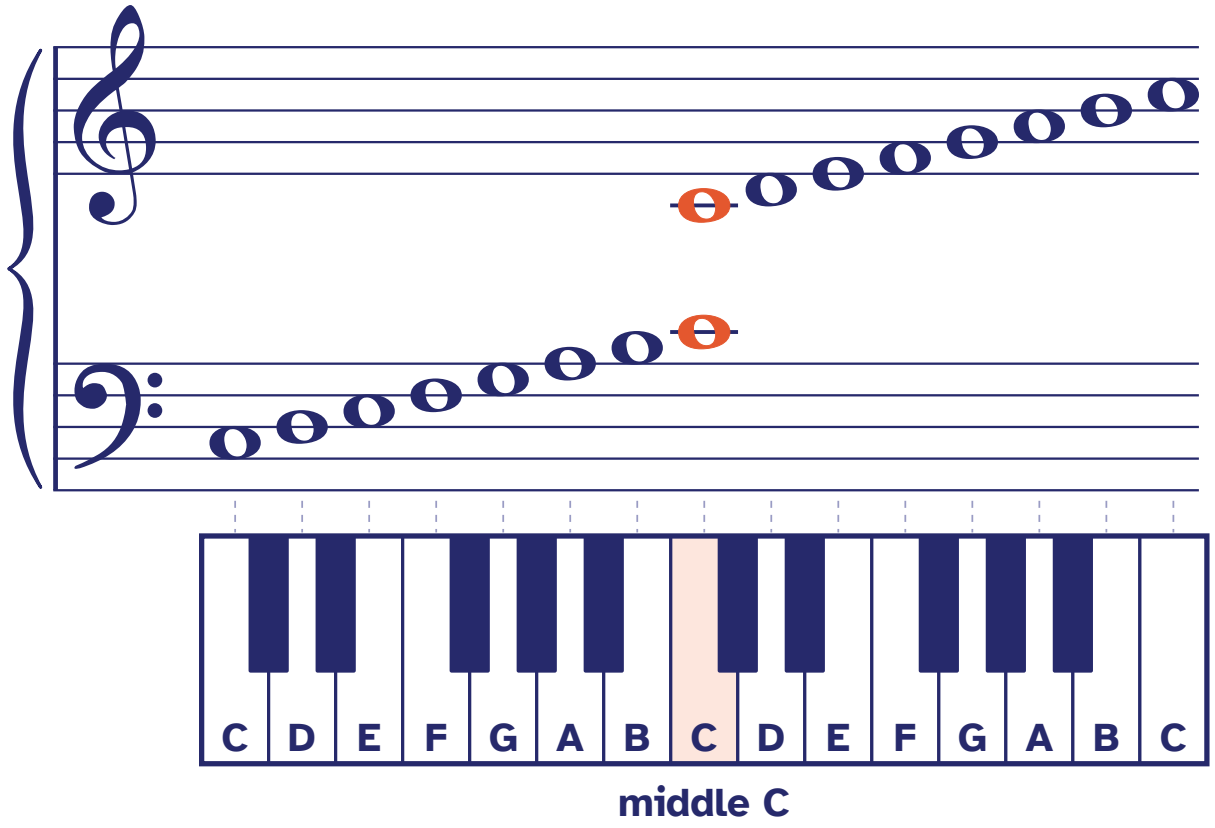
PART I · PITCH · 4

The staff and the keyboard

Every line and space of the grand staff is one white key. Read from left to right, the notes climb the staff exactly as the keys climb the keyboard: two octaves from C3 to C5, with middle C in red where the two staves meet.

THE SECRET

Every line and every space is one white key, in order.



What the map shows

Each step up the staff (line to space, space to line) is the next **white** key. The black keys have no line or space of their own: they are written with a sharp or a flat on a white key's line or space (p. 34).

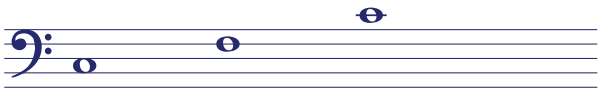
Two ways to write one key

Middle C can sit on a ledger line below the treble staff or above the bass staff. Both mean the same key; the music chooses the staff of the hand that plays it.

PART I · PITCH · 4 · CONTINUED

EXAMPLE 1

From the page to the key



1. **Name the note on the staff:** line 4 of the bass staff, between the clef's dots, is F.
2. **Count down from middle C:** C B A G F, the fifth white key down.
3. **Play it** with the left hand: that is F3.

The bass-clef F is **F3**, five white keys below middle C.

GAME

How many white keys?

1. middle C up to G
2. middle C down to F
3. middle C up to the next C

Count both ends. The staff climbs the same way: one line or space per white key.

Answer: 1 five (C D E F G) · 2 five (C B A G F) · 3 eight: an octave.

TRY IT · 10 SECONDS

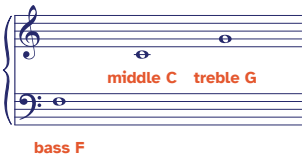


Play middle C, then walk up.

Say each white key as you play it, C D E F G A B C, and picture each one climbing the staff: line, space, line, space.

EXAMPLE 2

Landmarks either side of middle C



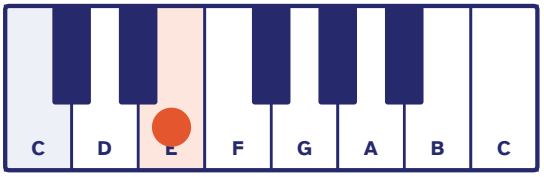
1. **Find middle C** on the map: between the two staves.
2. **Treble G** is five white keys above it.
3. **Bass F** is five white keys below it.

The landmarks sit at the same distance either side of middle C.

PART I · PITCH · 4 · CONTINUED

EXAMPLE 3

From the key to the page

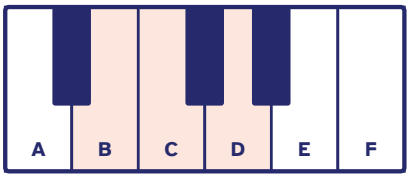


1. **The key:** E, two white keys above middle C.
2. **On the staff,** step up from middle C: D in the space below the staff, E on the bottom line.
3. **So the note sits** on the treble's bottom line.

E4 is the **bottom line** of the treble staff.

EXAMPLE 4

Where the two staves meet
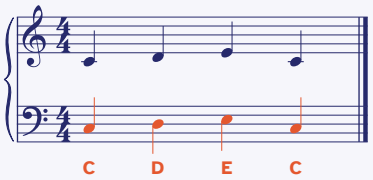


1. **The bass staff's top line** is A3.
2. **The treble staff's bottom line** is E4.
3. **The keys between,** B3 C4 D4, sit on ledger lines: in either staff.

B, C and D in the middle can be written in **either staff**: the hand that plays them chooses.

YOU KNOW THIS

Frère Jacques, in both hands



The left hand plays the **same letters**, C D E C, one octave lower, written in the bass clef.

Try it. Play it right hand, then left hand, then both together.

THINK OF IT The staff is a ladder laid on its side beside the keyboard: each rung, a line or a space, is the next white key.

TRAP

× Thinking the black keys have their own lines: they borrow a white key's line or space with a sharp or flat.

SEE ALSO
The keyboard p. 22 · Reading the notes p. 28

PRACTISE IT
Workbook Pitch 1–2 · Online: Library › Staff and clefs

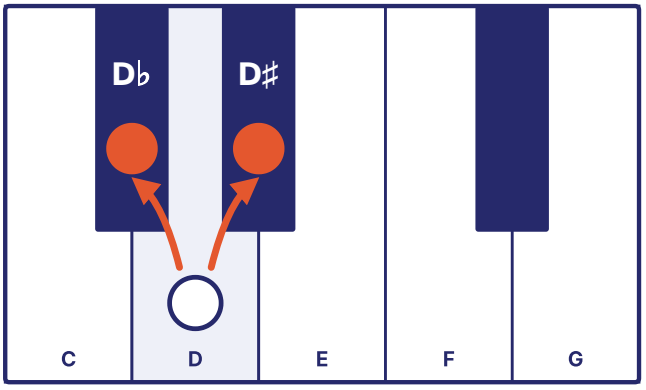
PART I · PITCH · 5

Sharps, flats and naturals

A sharp, flat or natural in front of a note (an **accidental**) nudges it, usually by a semitone. **Sharp**: one key right, usually black. **Flat**: one key left.

THE SECRET

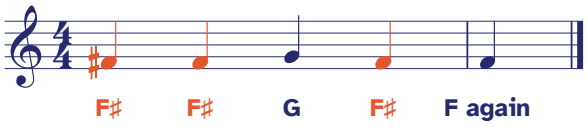
Sharp: the very next key to the right. Flat: the very next key to the left.



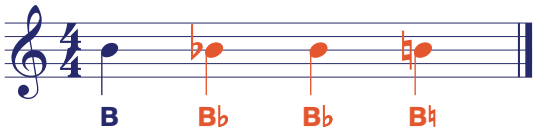
♯ one key right · ♭ one key left

SIGN	NAME	WHAT IT DOES	FIRST MET
♯	sharp	raises a note by a semitone	Level 0
♭	flat	lowers a note by a semitone	Level 0
♮	natural	cancels a sharp or flat	Level ½
𝄪	double sharp	raises by two semitones	Level 4
𝄫	double flat	lowers by two semitones	Level 4

Sharp till the bar line



The sharp still counts later in the bar; after the bar line, F is F again.



A natural cancels the flat for the rest of the bar.

Only that line or space. A sharp on the F in the first space does not change the F an octave higher.

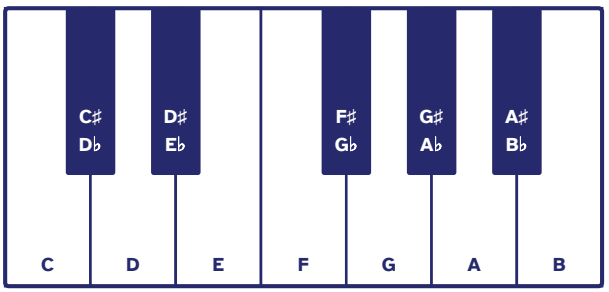
A tie carries it on. A note tied across the bar line keeps its accidental.

Reminders. A courtesy accidental, sometimes in brackets, just reminds you after a bar line.

PART I · PITCH · 5 · CONTINUED
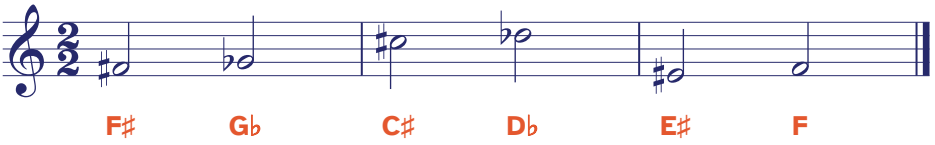
One key, two names

Every black key has two names: the key between F and G is both **F♯** and **G♭**. Notes that sound the same but are written differently are **enharmonic**. White keys can have sharp or flat names too: **E♯** is F, **C♭** is B.



EXAMPLE 1

One key, two names



1. **The first bar:** F♯, then G♭.
2. **Find them:** both are the black key between F and G.
3. **The last bar:** no black key sits above E, so E♯ is the white key F.

Same key, different names: these are **enharmonic** notes.

GAME

Which key is it?

1	2	3	4
C♯	E♭	B♯	F♭

Sharp: one key to the right. Flat: one key to the left. Which key do you press?

Answer: 1 the black key right of C · 2 the black key left of E · 3 the white key C · 4 the white key E.

TRY IT · 10 SECONDS
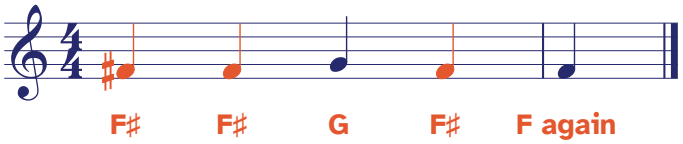


Play D, D♯, D♭.

D, then one key right (the black key), then one key left of D (another black key). Say the names as you play.

PART I · PITCH · 5 · CONTINUED

EXAMPLE 2

How long does the sharp last?

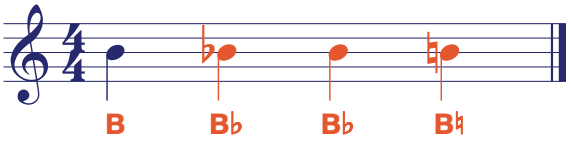


1. **The first note:** F♯.
2. **Later in the bar:** F is still F♯.
3. **After the bar line:** F is plain F again.

An accidental lasts **until the bar line**, then the key signature rules again.

EXAMPLE 3

A natural cancels the flat
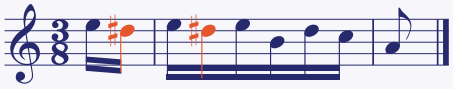


1. **B**: the white key.
2. **B♭**: one key to the left, and it stays flat for the rest of the bar.
3. **B♮**: the natural cancels the flat, back to the white key.

A natural **cancels** a sharp or flat until the bar line.

YOU KNOW THIS

Für Elise (Beethoven)



E, **D♯**, E, **D♯**: a semitone down and back. The sharp is written again in the new bar, because the bar line wiped it out.

Try it. Play E and D♯ back and forth: the black key just left of E.

THINK OF IT A key signature is the principal's notice on the school board: it speaks to every student, every day. An accidental is a teacher stopping at one student's desk: the message is for that note only, and it lasts until the next bar line.

TRAP

- × Forgetting that an accidental lasts until the bar line.
- × Carrying an accidental into the next bar: after the bar line, the key signature rules again.

SEE ALSO
Keys and key signatures p. 75

PRACTISE IT
Workbook Pitch 3 · Online: Library › Signs

PART I · PITCH · TIPS

When a note reads wrong

Most misread notes come from a handful of slips. Name the slip first, then fix only that note: playing the whole line again rarely finds it.

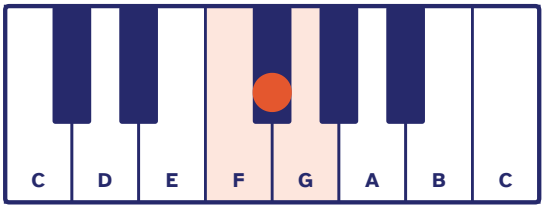
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
The left hand reads like the right	**the clef**: bass lines are G B D F A, not E G B D F	p. 25
You land an octave too high or too low	**a landmark**: middle C, treble G or bass F	p. 28
Notes above or below the staff slow you down	**the ledger lines**: count them from the nearest landmark	p. 28
A sharp or flat vanishes, or stays too long	**the bar line**: an accidental lasts only to the end of its bar	p. 34
You play the black key on the wrong side	**the direction**: a sharp goes right (up), a flat goes left (down)	p. 34

EXAMPLE

Same key, two names



1. **From F, go one key right:** the red black key is **F♯**.
2. **From G, go one key left:** you reach the same red key, **G♭**.
3. **One key, two names:** the music decides which name it uses.

If F♯ and G♭ both look odd, check the key: the name follows the notes around it, the key stays the same.

HOW TO

Three checks before you read a new line

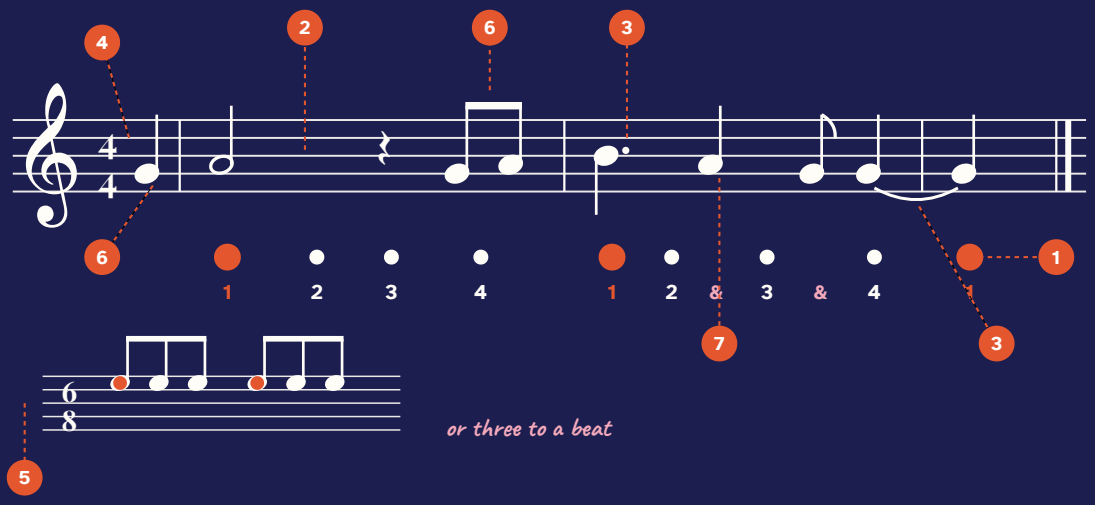
1. **The clefs:** which staff is treble, which is bass?
2. **The landmarks:** find middle C, treble G and bass F, and read the other notes from them.
3. **The accidentals:** mark each sharp, flat or natural, and see where its bar ends.

THINK OF IT Like reading a map: find the big landmarks first, then the streets around them. A note far from any landmark is the one most often misread.

PART II · WHEN, AND FOR HOW LONG

Time

Music never stands still. Underneath runs a steady **pulse**, packed into **bars**, and every note and rest gets its fair share. Everything in this part is in these two bars.



1 **Pulse, tempo and the bar** p. 39
the steady beat underneath; beat 1 is strongest

2 **Notes and rests** p. 42
the shape says how long; rests are counted too

3 **How long is a note?** p. 46
a dot adds half; a tie glues two notes into one

4 **Time signatures** p. 49
four seats in every bar, a crotchet in each

5 **Compound and irregular time** p. 52
beats in threes; bars of five or seven

6 **Rhythm: groups and upbeats** p. 55
beams show the beats; starting before bar 1

7 **Rhythm: off the beat** p. 58
a note between the beats that holds on

★ **Tips: when the rhythm slips** p. 61

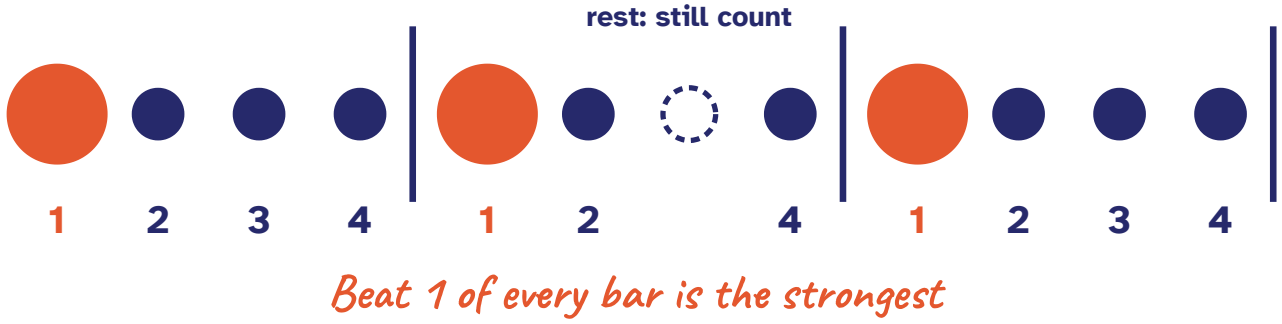
PART II · TIME · 1

Pulse, tempo and the bar

Put your hand on your heart: that steady beat is a **pulse**, and almost every piece has one. Its speed is the **tempo**. Beats travel in groups called **bars**, and beat 1 is the strongest.

THE SECRET

The beat never stops, not even in a rest. Feel it before you play.



Beat 1 of every bar is the strongest

Tempo

Tempo is measured in beats per minute. A **metronome mark** at the top of the music shows it: ♩ **= 60** means 60 crotchet beats a minute, one beat a second. ♩ **= 120** is twice as fast. Tempo words such as *Andante* and *Allegro* come with Expression, later in the book.

Strong and weak beats

BEATS	PATTERN	FEELS LIKE
2	**Strong** weak	a march
3	**Strong** weak weak	a waltz
4	**Strong** weak **medium** weak	most songs

Bars and bar lines

SIGN	NAME	MEANING
	bar line	divides the music into bars
	double bar line	marks the end of a section, or a change of key, time signature or tempo
	final bar line	marks the end of the piece
	repeat signs	play the passage between them again; if there is no start sign, go back to the beginning

PART II · TIME · 1 · CONTINUED

The beat in tunes you know

TUNE	BEATS IN A BAR	PATTERN
Twinkle, Twinkle, Little Star	4	**Strong** weak **medium** weak
Happy Birthday	3	**Strong** weak weak
The Blue Danube (a waltz)	3	**Strong** weak weak
Radetzky March	2	**Strong** weak

Counting

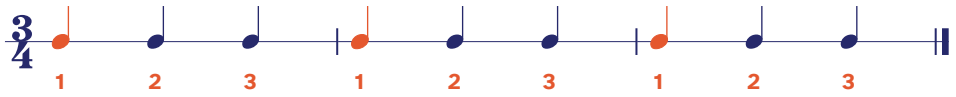


Three bars of 3/4. Every beat is counted, even inside a long note: the red number is where you play, the grey numbers are the beats you hold.

Count aloud at one speed, a little louder on beat 1. With a metronome, start slower than you think and speed up only when the passage is steady.

EXAMPLE 1

Is this a march or a waltz?



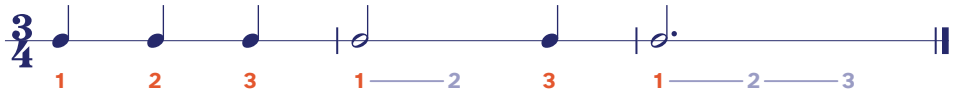
1. **Tap a steady beat** as you clap or listen.
2. **Find the strong one:** the red beats feel heavier.
3. **Count the gap** between strong beats: 1 2 3, 1 2 3.

Groups of three: a **waltz**, in 3/4. Groups of two would be a march.

EXAMPLE 2

Feel the beat before you play

♩ = 96



1. **Read the mark:** ♩ = 96, 96 crotchet beats a minute.
2. **Listen first:** set the metronome and let one full bar go by, counting “1 2 3” in your head.
3. **Step in** on the next beat 1, the way you step onto a moving escalator.

The first note lands **on the beat**, not a guess.

PART II · TIME · 1 · CONTINUED

GAME

How many beats in a bar?

1 2 3 4

The big dot is the strong beat. Count the beats in each bar: 2, 3 or 4?

Answer: 1 three (a waltz) · 2 four · 3 two (a march) · 4 three, twice.

TRY IT · 10 SECONDS



Hand on your heart. Tap the beat.

Say “1 2 3 4” with your taps, 1 a little louder. Keep going for ten seconds without slowing down.
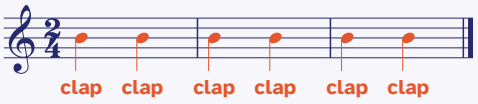
HOW TO

Find the beat of a new piece

1. **Read the time signature:** how many beats in each bar.
2. **Tap the beat** on your knee before you play, counting one bar aloud.
3. **Start slower** than the marked speed, and keep it steady to the end.

YOU KNOW THIS

Radetzky March (Johann Strauss I)



At the New Year concert the audience claps along on **every beat**. That steady clap is the pulse; the orchestra’s tune moves freely over it.

Try it. Put on any march and clap only the beat, not the tune.

THINK OF IT The beat is the stage and the notes are the actors: the stage is there before anyone walks on. So step in the way you step onto an escalator or take a plate off a sushi train: watch it move for a bar, then join in.

TRAP

- × Speeding up in easy bars and slowing in hard ones: the pulse never changes.
- × Counting only the notes you play: the beat goes on through long notes and rests.

SEE ALSO
Time signatures p. 49 · Tempo words: Expression (later)

PRACTISE IT
Workbook Level 0 · Pulse · Online: Metronome

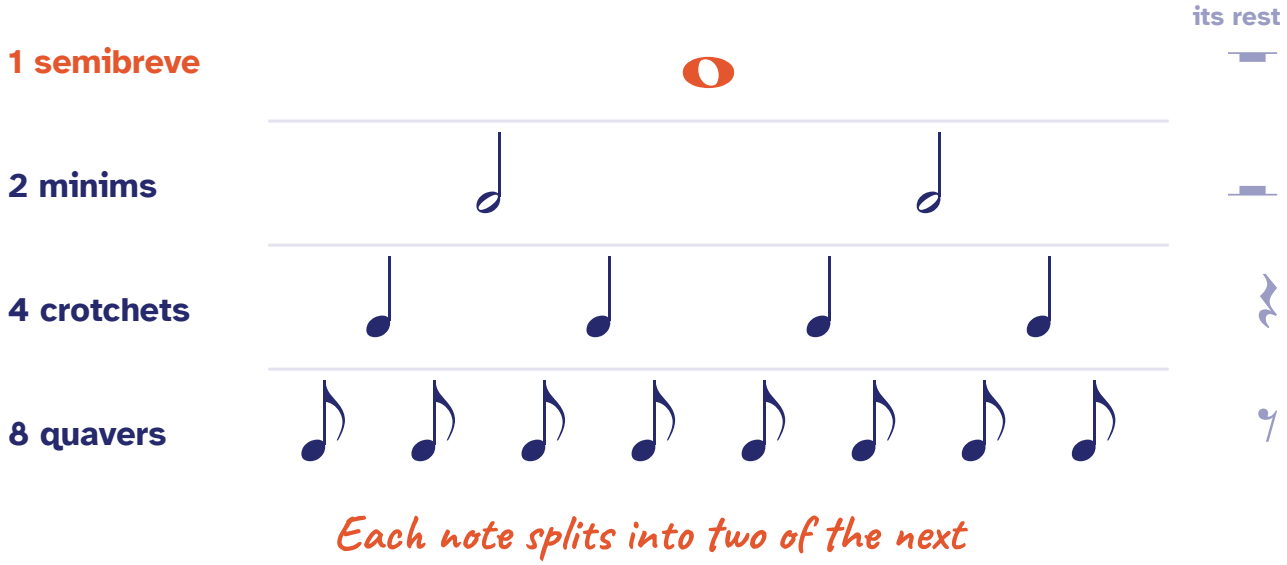
PART II · TIME · 2

Notes and rests

You can see how long a note lasts from its shape: each plain note is **half** the one above it. And every note has a twin **rest**, a silence you count just as carefully.

THE SECRET

Each note is half the one above it, and every note has a rest twin.

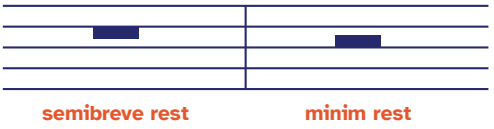


Each note splits into two of the next

Head, stem, flag

The **head** sits on a line or in a space. A **stem** goes up on the right of the head or down on the left. **Flags** on the stem halve the value each time; groups of quavers are joined by **beams** instead of flags.

Which rest is which?



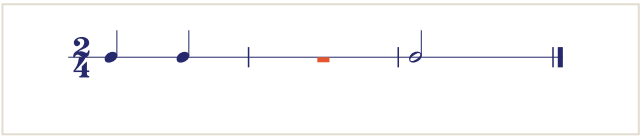
The semibreve rest **hangs** below the fourth line; the minim rest **sits** on the middle line.

THE SEMIBREVE REST'S SPECIAL POWER

Most rests last a fixed number of beats. The **semibreve rest** can also mean **a whole bar of silence** in any time signature: 4 beats in 4/4, but 3 in 3/4 and 2 in 2/4. It is then centred in the bar. No other rest can do this.

One more rule: keep the middle of a 4/4 bar visible. Silence on beats 2–3 is two crotchet rests, never one minim rest.





* With the crotchet as one beat, as in 2/4, 3/4 and 4/4. *Extra*: beyond Level 8.

PART II · TIME · 2 · CONTINUED

NOTE	REST	NAME	ALSO CALLED	BEATS*	FIRST MET
		**breve**	double whole note	**8**	Level 4
		**dotted semibreve**	dotted whole note	**6**	*Extra*
		**semibreve**	whole note	**4**	Level 0
		**dotted minim**	dotted half note	**3**	Level 0
		**minim**	half note	**2**	Level 0
		**dotted crotchet**	dotted quarter note	**1½**	Level ½
		**crotchet**	quarter note	**1**	Level 0
		**dotted quaver**	dotted eighth note	**¾**	Level 3
		**quaver**	eighth note	**½**	Level 0
		**semiquaver**	sixteenth note	**¼**	Level 2
		**demisemiquaver**	32nd note	**⅛**	*Extra*
		**hemidemisemiquaver**	64th note	**1/16**	*Extra*

PART II · TIME · 2 · CONTINUED

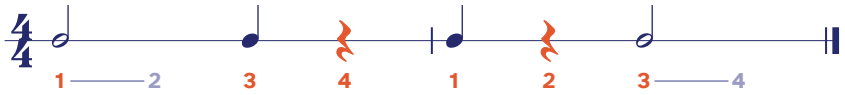
Why whole, half and quarter?

Think of each beat as a pizza. A bar of 4/4 is four pizzas. The note that fills the whole bar is the **whole note**; the others are a half, a quarter, an eighth and a sixteenth of it. Two quavers beamed together fill one pizza. A rest is the same share left empty: shaded with lines.

semibreve	whole note	4 pizzas · 4 beats	
semibreve rest	silence	4 empty pizzas · 4 beats	
minim	half note	2 pizzas · 2 beats	
minim rest	silence	2 empty pizzas · 2 beats	
crotchet	quarter note	1 pizza · 1 beat	
crotchet rest	silence	1 empty pizza · 1 beat	
quaver	eighth note	half a pizza · ½ beat	
quaver rest	silence	half an empty pizza · ½ beat	
semiquaver	sixteenth note	a quarter of a pizza · ¼ beat	
semiquaver rest	silence	a quarter of an empty pizza · ¼ beat	

EXAMPLE 1

How long is each sound, and each silence?

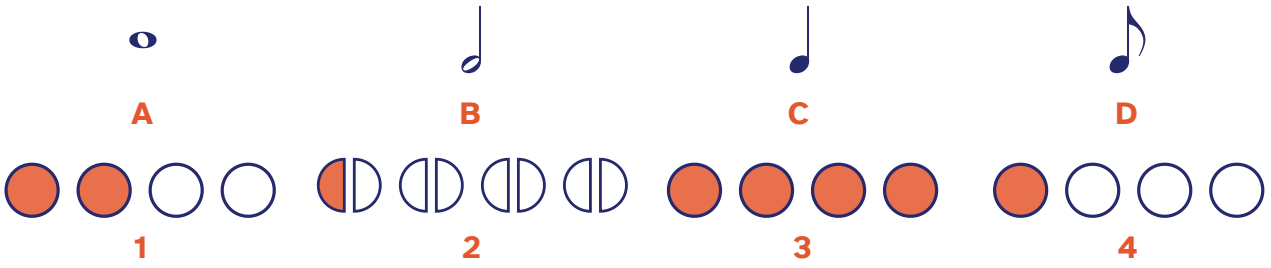


1. **Name each shape:** minim, crotchet, crotchet rest.
2. **Count its beats:** 2, 1 and 1.
3. **Count the rests too:** the red rests last as long as their notes.

Every bar adds up to **4 beats**, silences included.

GAME

Pizza match



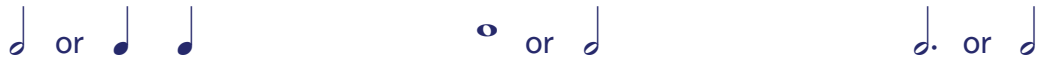
Match each note (A to D) with its pizzas (1 to 4). One pizza is one beat.

Answer: A 3, B 1, C 4, D 2.

PART II · TIME · 2 · CONTINUED

GAME

Which lasts longer?



1 2 3

Circle the longer side in each pair. One pair is a trick: both sides are the same length.

Answer: 1 the same (2 beats each), 2 the semibreve (4 beats), 3 the dotted minim (3 beats).

TRY IT · 10 SECONDS

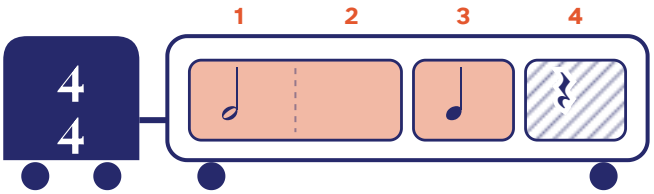


Clap once and hold for 2: a minim.

Now clap four crotchets (1 2 3 4), then one semibreve (clap, and hold all four beats). Say the counts out loud.

EXAMPLE 2

Fill a bar of 4/4



1. **The minim** takes two seats: beats 1 and 2.
2. **The crotchet** takes one: beat 3.
3. **One seat is left:** a crotchet rest, counted like a note.

Minim + crotchet + crotchet rest = **4 beats**: the bar is full.
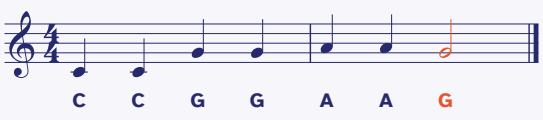
HOW TO

Learn the note shapes

1. **Hollow or filled:** hollow heads last longer (semibreve, minim).
2. **Stem or no stem:** the semibreve has none.
3. **Count the flags:** each one halves the note again.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



Six **crotchets**, then a **minim** at the end of the line: one beat, one beat … and two beats to breathe.

Try it. Clap the line and say “1 1 1 1 1 1 1-2”.

SEE ALSO
Pulse and the bar p. 39 · How long is a note? p. 46

PRACTISE IT
Workbook Level 0 · Note values · Online: Library › Notes

PART II · TIME · 3

How long is a note?

Need a note longer than its shape allows? Two signs stretch it: the **dot**, which adds half, and the **tie**, which glues two notes into one sound.

THE SECRET
A dot adds half. A tie glues two notes into one.
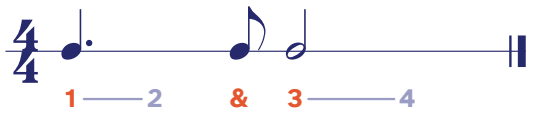


2 beats **2 + 1 = 3 beats**

The dot adds half

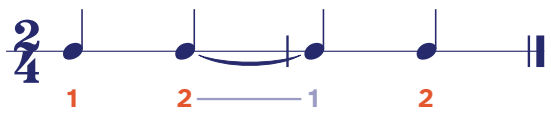
The dot

A dot adds **half the note's value**: a dotted minim lasts 2 + 1 = 3 beats; a dotted crotchet 1 + ½ = 1½. A **double dot** adds half, then a quarter: a double-dotted minim lasts 2 + 1 + ½ = 3½.



The tie

A curve joining two notes **of the same pitch** makes one longer sound: play the first, hold through the second. Ties can cross a bar line, where a dot cannot.
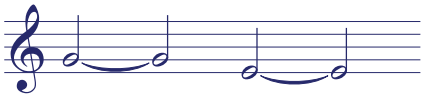


The dot, in pizzas

The dot is a helper that adds **half of whatever its note is worth**. Dark slices: the note. Lighter: what the dot adds. Lightest: a second dot, half of the first dot.

dotted minim		2 + **1** (half of 2) = **3 beats**
dotted crotchet		1 + **½** (half of 1) = **1½ beats**
dotted quaver		½ + **¼** (half of ½) = **¾ beat**
double-dotted minim		2 + **1** + ½ = **3½ beats**

PART II · TIME · 3 · CONTINUED

Tie or slur?



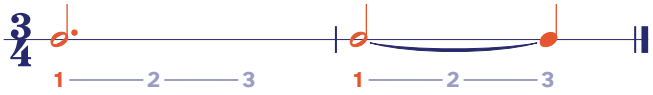
Tie: the same pitch. Play once and hold: G for 4 beats, then E for 4.



Slur: different pitches. Play every note, smoothly joined (legato).

EXAMPLE 1

Dot or tie: which is longer?
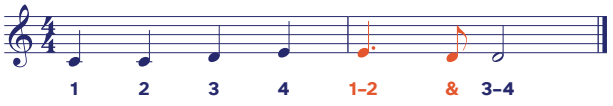


1. **The dotted minim:** 2 + half of 2 = 3 beats.
2. **The tied notes:** a minim tied to a crotchet, 2 + 1 = 3 beats.
3. **Play both:** one sound, held for three beats.

They are **the same length**. A tie can also cross a bar line, which a dot cannot.

EXAMPLE 2

A dotted crotchet you already know: Ode to Joy



1. **Bars 3–4 of Beethoven's Ode to Joy.** Sing it: "C C D E, E — D D".
2. **The red E is a dotted crotchet:** 1 + ½ = 1½ beats, held through "1 & 2".
3. **The quaver D** falls on the "&" of beat 2; the last D takes beats 3 and 4.

Long–short, then long: the dot is why the tune **leans** before it lands.

EXAMPLE 3

Which is longer: a dotted minim or a minim tied to a quaver?



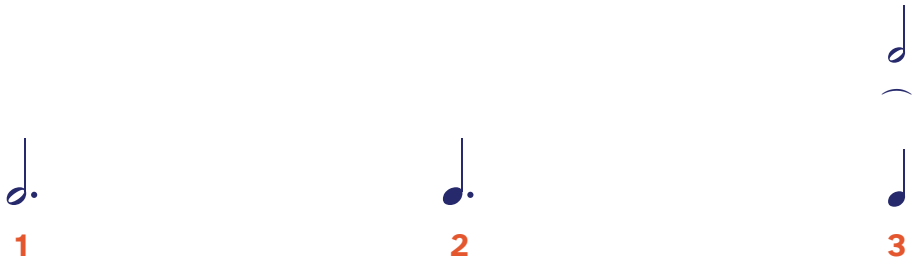
1. **Dotted minim:** 2 + 1 = 3 pizzas.
2. **Minim tied to a quaver:** 2 + ½ = 2½ pizzas.
3. **Compare:** 3 is half a beat more than 2½.

The **dotted minim** is longer, by half a beat.

PART II · TIME · 3 · CONTINUED

GAME

How many beats?



With a crotchet as one beat: how long does each one last?

Answer: 1 three (2 + 1) · 2 one and a half (1 + ½) · 3 three (2 tied to 1).

TRY IT · 10 SECONDS



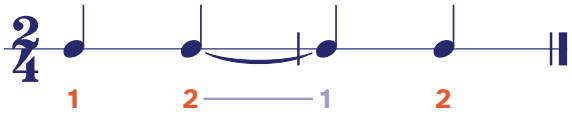
Clap once, hold for three.

Say “1 2 3” while you hold: that is a dotted minim.

Then clap a dotted crotchet: “1 &”, and let go on “2”.

EXAMPLE 4

Hold a tie across the bar line



1. **Play the first note** and count its beats.
2. **Keep holding** through the tie, counting the second note’s beats too.
3. **Let go** only when they are over. Never play the second note again.

A tie **adds the beats** of both notes into one sound.

THINK OF IT A tie couples two carriages: the passenger stays seated across the join, even over a bar line. A dot cannot cross the bar line.

TRAP

- × Putting the dot on the wrong note: a dot adds half of **its own** note.
- × Tying two different pitches: that curve is a slur, not a tie.

SEE ALSO

Time signatures p. 49 · Rhythm p. 55

PRACTISE IT

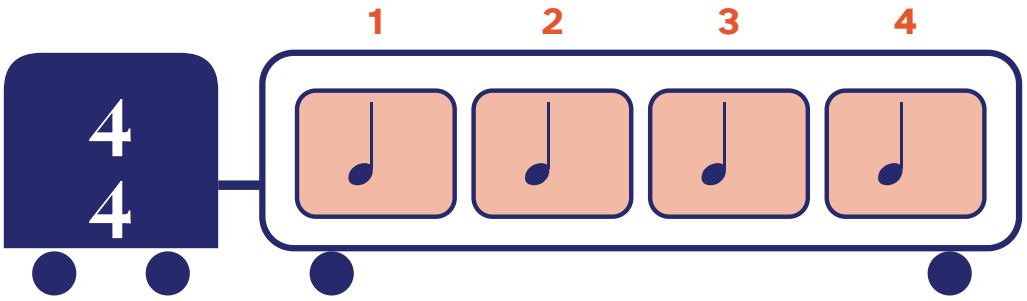
Workbook Level 0 · Note values · Online: Library › Notes and rests

PART II · TIME · 4

Time signatures

Two numbers, like a train ticket: the **top number** says how many seats (counts) each bar has; the **bottom number** says which note fills one seat.

THE SECRET
Top number: how many beats. Bottom number: which note gets one beat.



Top: how many seats · bottom: which note sits in one

The bottom number

BOTTOM	NOTE	ONE COUNT IS A	FIRST MET
2	𝅗𝅥	minim	Level 3
4	♩	crotchet	Level 0
8	♪	quaver	Level 1
16	𝅘𝅥𝅯	semiquaver	*Extra*

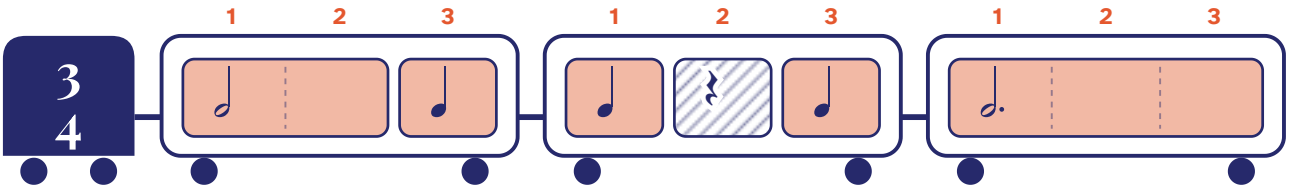
Simple time: each beat divides in two

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
2/4	2 crotchets	Level 0
3/4	3 crotchets	Level 0
4/4 𝄴	4 crotchets (the C sign: common time)	Level 0
2/2 𝄵	2 minims (the cut C: cut common time)	Level 3
3/2	3 minims	Level 3
4/2	4 minims	*Extra*
3/8	3 quavers	Level 3

PART II · TIME · 4 · CONTINUED

Think of a train

In 3/4 every carriage (bar) has exactly **3 seats**; each seat is one crotchet beat. Notes book seats: a minim takes 2, a crotchet 1. A rest books a seat and leaves it **empty** (shaded). Every seat in every carriage must be booked.



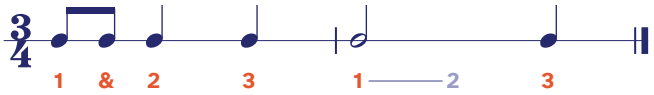
minim + crotchet · crotchet, crotchet rest, crotchet · dotted minim: 3 seats each

Changing time signature

A new time signature can appear in the middle of a piece, right after a bar line; it holds until the next one. A time signature is written only once, not on every line.

EXAMPLE 1

Read a time signature



1. **Top number 3:** three counts in each bar.
2. **Bottom number 4:** a crotchet is one count.
3. **Check a bar:** two quavers + crotchet + crotchet = 3.

3/4: **three crotchet beats** in every bar.

GAME

Which ticket?

1	2	3	4
3 crotchets	4 crotchets	6 quavers	2 minims

Each bar holds these notes. Write its time signature.

Answer: 1 3/4 · 2 4/4 · 3 6/8 · 4 2/2 (or 4/4).

TRY IT · 10 SECONDS

4
4

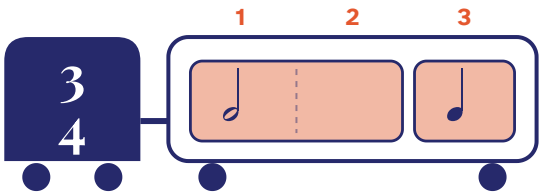
Open a piece you are playing.

Find the time signature, then count the seats in bar 1: do the notes add up to the top number? Try bar 2 as well.

PART II · TIME · 4 · CONTINUED

EXAMPLE 2

Count a bar of 3/4

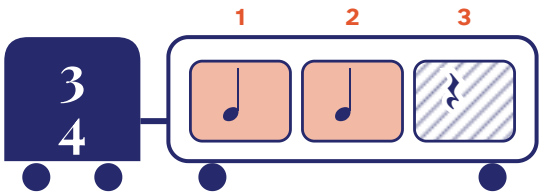


1. **The top number is 3:** three seats in every bar.
2. **The minim** books seats 1 and 2: say "1 2" while it sounds.
3. **The crotchet** takes seat 3.

Count **1 2 3**, and hold the minim through 2.

EXAMPLE 3

Where does the rest go?
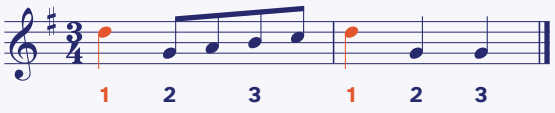


1. **Three seats** in 3/4.
2. **Two crotchets** fill seats 1 and 2.
3. **The rest** books seat 3: count it, do not skip it.

Rests book seats too: **1 2 (3)**.

YOU KNOW THIS

Minuet in G (from the Notebook for Anna Magdalena Bach)



Every bar holds **three crotchet beats**: 1 2 3, 1 2 3, the step of the dance. That is 3/4.

Try it. Count 1 2 3 aloud while someone plays it.

TRAP

- × Reading the bottom number as a count: it names the note that gets one beat.
- × Reading 6/8 as six beats: at most speeds it moves in two.
- × Mixing up 3/4 and 6/8: both hold six quavers, grouped differently (p. 52).

SEE ALSO

Pulse and the bar p. 39 · Compound time p. 52

PRACTISE IT

Workbook Level 0 · Time signatures · Online: Library › Time signatures

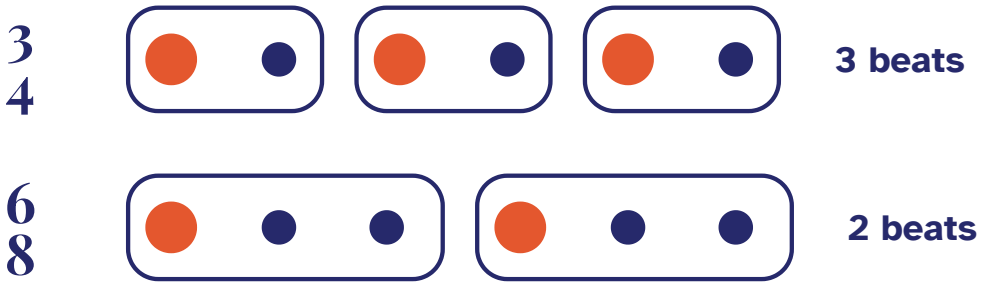
PART II · TIME · 5

Compound and irregular time

Compound time skips: each dotted beat splits into **three**. Irregular time keeps you guessing, with **five or seven** counts in uneven groups. (9/4, 12/4 and /16 signatures are rare.)

THE SECRET

Dotted beats split in three: 6/8 is two big beats.



Same six quavers, new groups

SIMPLE OR COMPOUND?

In **simple time** each beat divides into **two**. In **compound time** each beat is a dotted note and divides into **three**.

Quick test: a top number of 6, 9 or 12 means compound. Divide it by 3 to find the beats: 6/8 has 2, 9/8 has 3, 12/8 has 4.

Compound time

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
6/8	2 dotted crotchets	Level 1
9/8	3 dotted crotchets	Level 3
12/8	4 dotted crotchets	Level 3
6/4	2 dotted minims	Level 4

Counting 6/8



Slowly, count all six quavers. At speed, feel two beats, on 1 and 4: “1 & a, 2 & a”.

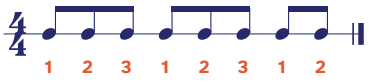
PART II · TIME · 5 · CONTINUED

Irregular time

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
5/4	5 crotchets: 3 + 2 or 2 + 3	Level 5
7/4	7 crotchets: 4 + 3 or 3 + 4	*Extra*
5/8	5 quavers: 3 + 2 or 2 + 3	*Extra*
7/8	7 quavers: 2 + 2 + 3 or 3 + 2 + 2	*Extra*

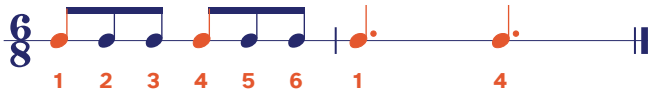
UNEVEN GROUPS IN 4/4

Many dances keep 4/4 but group the eight quavers 3 + 3 + 2: "1 2 3, 1 2 3, 1 2".



EXAMPLE 1

Count 6/8 in two



1. **Slowly, count all six:** 1 2 3 4 5 6.
2. **Lean on 1 and 4**, the red notes.
3. **Faster, count only those:** two beats, each a dotted crotchet.

6/8 has **two beats** in a bar, each split into three.

GAME

Two beats or three?

1	2	3	4
6/8	3/4	9/8	2/4

How many beats in a bar, and does each beat split in two or in three?

Answer: 1 two, in threes · 2 three, in twos · 3 three, in threes · 4 two, in twos.

TRY IT · 10 SECONDS

6/8

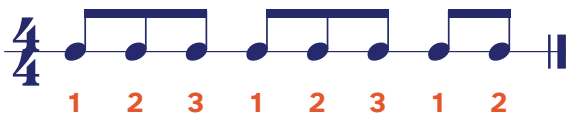
Sway left, sway right.

Say "1 2 3, 4 5 6" and sway once on 1 and once on 4. That is 6/8: six quavers, two big beats.

PART II · TIME · 5 · CONTINUED

EXAMPLE 2

Hear 3 + 3 + 2 inside 4/4



1. **Count eight quavers** in a bar of 4/4.
2. **Group them** 3 + 3 + 2, accenting 1, 4 and 7.
3. **Keep the beat** in your foot: it stays in four.

The bar is still **4/4**; only the accents are uneven.

EXAMPLE 3

Count 9/8



1. **The top number, 9,** divides by 3: compound time.
2. **9 ÷ 3 = 3** big beats in a bar.
3. **Each beat** is a dotted crotchet: three quavers.

9/8 is **three beats**, each split into three: “1 2 3, 4 5 6, 7 8 9”.
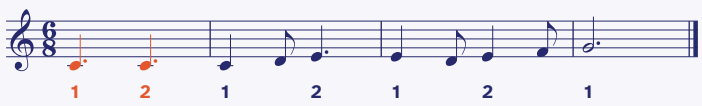
HOW TO

Feel compound time

1. **Say “pine-ap-ple”** on every beat and step only on “pine”.
2. **Clap the three syllables** while you step.

YOU KNOW THIS

Row, Row, Row Your Boat



Two big beats in every bar, each a **dotted crotchet**: “Row, row”. Inside each beat, “gent-ly” splits it into three. That is 6/8.

Try it. Sway twice per bar while you sing it.

THINK OF IT Like a skipping rope: two big swings, each with three little steps inside.

TRAP

- × Counting 6/8 as six slow beats at speed: feel two big ones.
- × Mixing up 3/4 and 6/8: the same six quavers, grouped differently.

SEE ALSO

Time signatures p. 49 · Rhythm: groups and upbeats p. 55

PRACTISE IT

Workbook Levels 1–4 · Compound time · Online: Rhythm clock

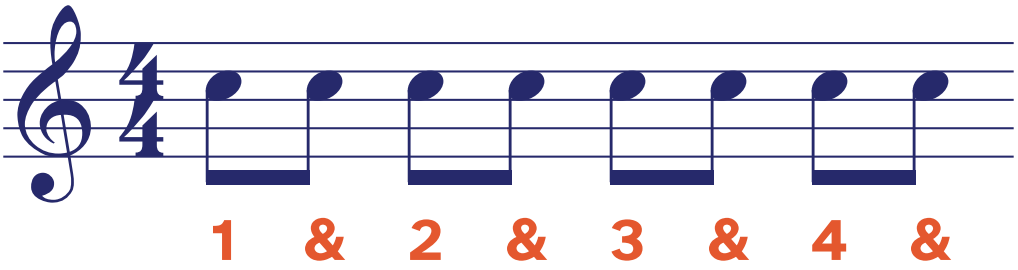
PART II · TIME · 6

Rhythm: groups and upbeats

Say your own name: long and short sounds, that's **rhythm**. The pulse marches steadily on; the rhythm dances around it. How the notes are grouped on the page shows where the beats fall.

THE SECRET

Beams group notes into beats; an upbeat borrows from the last bar.



One beam, one beat

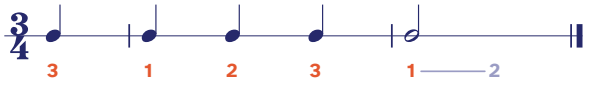
Grouping with beams

Beams join the notes of **one beat**, so the beats stay visible. In 4/4 a beam may cover beats 1–2 or 3–4, but never cross the middle of the bar (except in a deliberate grouping such as 3 + 3 + 2). In compound time, quavers are beamed in threes.

× A beam across beats 2 and 3 would hide the middle of the bar.

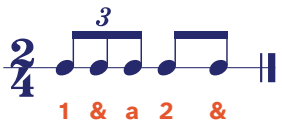
The upbeat (anacrusis)

A piece can begin **before** the first full bar, with one or more notes called an upbeat or anacrusis. The last bar is then short by the same amount, so the two together make one full bar. Count the missing beats in silence before you start.
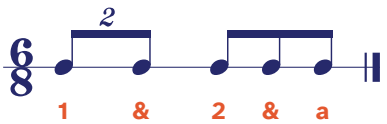


Count "1 2" silently, then play on 3.

Triplets and duplets



Triplet: three exactly equal notes in the time of two. First met: Level 2.



Duplet: two notes in the time of three, in compound time. First met: Level 2.

PART II · TIME · 6 · CONTINUED



Two quavers: one beat cut in 2



A **triplet**: one beat cut in 3



Four semiquavers: one beat cut in 4

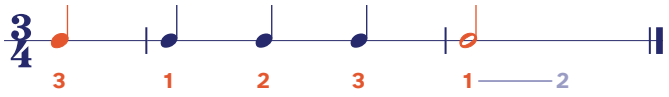
Counting smaller values



Crotchet, two quavers, four semiquavers and a triplet (“1 & a”): one beat each.

EXAMPLE 1

Balance the upbeat



1. **The upbeat:** one beat (count 3) before the first full bar.
2. **The last bar:** only two beats.
3. **Add them:** 1 + 2 = 3, one whole bar of 3/4.

The upbeat and the last bar **make one full bar** together.

GAME

May the beam join them?

1 **beats 1 and 2**

2 **beats 2 and 3**

3 **beats 3 and 4**

In 4/4, quavers are beamed so the middle of the bar stays visible. Which pairs may share one beam?

Answer: 1 yes · 2 no, that hides the middle of the bar · 3 yes.

TRY IT · 10 SECONDS
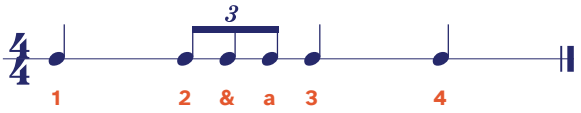


Clap “1 & 2 & 3 & 4 &”.

Now clap only the numbers, then only the &s, while your foot keeps the beat. The beat never moves; your claps do.

PART II · TIME · 6 · CONTINUED

EXAMPLE 2

Fit a triplet into a bar

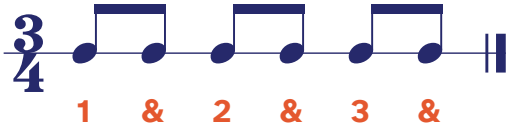


1. **Count the bar:** four crotchet beats, “1 2 3 4”.
2. **Find the 3:** on beat 2, three quavers share one beat.
3. **Say it:** “1, 2 & a, 3, 4”, keeping beats 3 and 4 steady.

The triplet squeezes three notes into **beat 2**; the beat itself never moves.

EXAMPLE 3

Beam six quavers in 3/4



1. **3/4 has three beats,** each a crotchet.
2. **Each beat** holds two quavers.
3. **So the beams go 2 + 2 + 2.** Grouped 3 + 3 they would look like 6/8.

In 3/4, six quavers are beamed **in twos**.

HOW TO

Practise triplets

1. **Say “trip-o-let”** on every beat, evenly.
2. **Clap the three syllables** while your foot keeps the beat.
3. **Switch** between two quavers (“1 &”) and a triplet (“trip-o-let”) every bar.

YOU KNOW THIS

Happy Birthday



“Hap-py” comes **before** the first strong beat: an upbeat. The strong beat lands on “birth”.

Try it. Sing it and clap only on “birth”, “day” and “you”: you feel where bar 1 starts.

THINK OF IT An upbeat is the breath before you speak: “and-ONE”. You lift on the upbeat so the strong beat can land.

SEE ALSO
Notes and rests p. 42 · Rhythm: off the beat p. 58

PRACTISE IT
Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

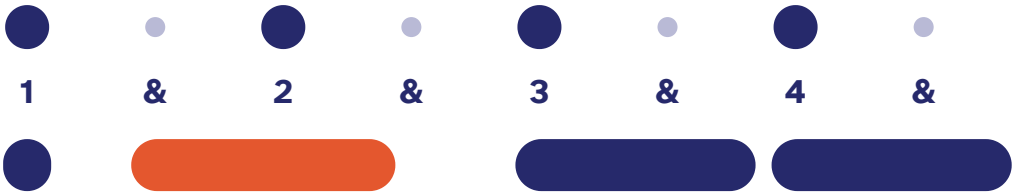
PART II · TIME · 7

Rhythm: off the beat

Now the fun part: rhythm that teases the beat. It only works if the beat underneath stays rock-steady.

THE SECRET

Keep the beat rock-steady and let the rhythm tease it.



starts on “&”, held over beat 2

A note between the beats

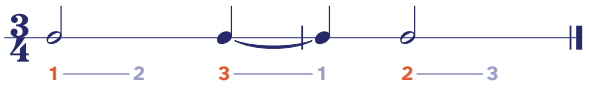
Syncopation



The crotchet starts on “&” and is held through beat 2.

Level 4. Feel beat 2 even though nothing is played on it.

Hemiola



Two bars of 3/4 felt as three minims.

Level 4. Common at cadences in Baroque dances.

Swing Level ½

Common from the first pieces: jazz, blues and many easy modern pieces are marked *swing*. Pairs of quavers are written evenly but played **long–short**, like the first and last notes of a triplet.



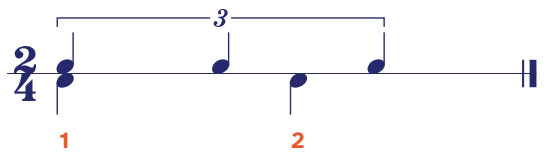
Written



Played

PART II · TIME · 7 · CONTINUED

Polyrhythm Level 7



Three against two: a crotchet triplet above, two crotchets below, starting together.

Rubato Extra

Time is stretched and then given back, for expression: the music leans into a phrase and catches up after it. The pulse always returns. Marked *rubato*, or simply part of the style in much Romantic music.

EXAMPLE 1

Count a syncopation



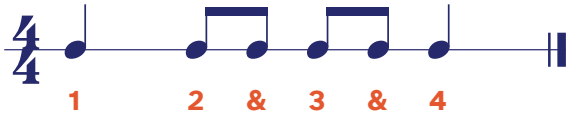
1. **Tap the beats:** 1, 2.
2. **Find the red crotchet:** it starts on the "&", between the beats.
3. **Hold it through beat 2,** then play bar 2 on the beats.

The accent falls **off the beat**: that is syncopation.

EXAMPLE 2

Swing this bar

Swing



1. **Find the quaver pairs:** beats 2 and 3.
2. **Play each pair long-short:** "doo-ba, doo-ba".
3. **Leave the crotchets alone:** beats 1 and 4 stay as written.

Only the **pairs** swing; the beat underneath stays even.

GAME

On the beat or off it?

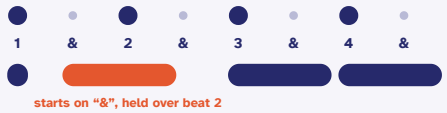
1	2	3	4
starts on 1	starts on &	starts on 2	starts on the & of 3

A note that starts on a number is on the beat; on an "&" it is off the beat. Which are off?

Answer: 2 and 4 are off the beat; 1 and 3 are on it.

PART II · TIME · 7 · CONTINUED

TRY IT · 10 SECONDS

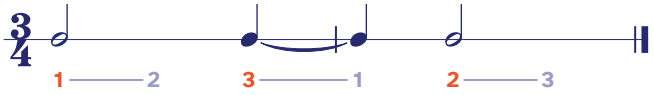


Foot on the beat, hands on the &.

Tap a steady beat with your foot. Clap only between the taps. If the foot starts to follow the hands, slow down.

EXAMPLE 3

Hear a hemiola



1. **Two bars of 3/4** make six beats.
2. **Group them in twos:** three minims.
3. **Count "1 2, 3 4, 5 6"** with a slight push on each minim.

The same six beats, felt as **three big ones**: a hemiola.

YOU KNOW THIS

The Entertainer (Scott Joplin)



The high **C** keeps landing just after the beat, not on it: syncopation, the bounce of ragtime.

Try it. Tap the beat with your foot and hum the tune: hear the C slip between the taps.

THINK OF IT Syncopation is a skip in your walk: a step that lands between the beats. Swing is walking with a spring in your step, long-short, long-short. Rubato is borrowing time like pocket money: whatever you spend, you pay back.

TRAP

- × Rushing the short notes after a long one: keep counting inside the long note.
- × Playing a triplet unevenly: three equal notes, not two quick and one long.
- × Letting syncopation drag the pulse: tap the beat with your foot.

SEE ALSO

Rhythm: groups and upbeats p. 55 · Words for how to play: Expression (later)

PRACTISE IT

Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

PART II · TIME · TIPS

When the rhythm slips

A rhythm that slips is nearly always a beat that was not counted. Find which beat, count it out loud, and the rest of the bar usually falls into place.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
It speeds up or slows down	**the pulse**: count one bar aloud before you start	p. 39
A rest gets skipped	**the rest**: it is counted just like a note	p. 42
Long notes come out short	**the hold**: count every beat of a minim, dot or tie	p. 46
The bar feels too long or too short	**the time signature**: how many counts, which note is one	p. 49
An off-beat note lands on the beat	**the "&"**: keep the beat in your foot and count it	p. 58

Four ways to steady a rhythm

1. **Count out loud**
   Say the counts while you play, not in your head: a slip you can hear is a slip you can fix.

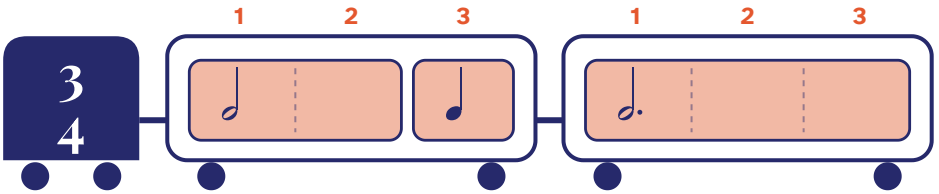
2. **Tap, then play**
   Tap the rhythm on the lid and say the counts; play only when the tapping is steady.

3. **Walk the beat**
   Step the pulse while you clap the rhythm, so the beat lives in your feet.

4. **Split the beat**
   For dots and quavers, count "1 & 2 &": the extra "&" shows exactly where each note goes.

EXAMPLE

Count the long notes



1. **Say 1 2 3** once through before you play, to feel the beat.
2. **The minim books seats 1 and 2:** keep saying "2" while it sounds.
3. **The dotted minim books all three:** let go on the next "1", not before.

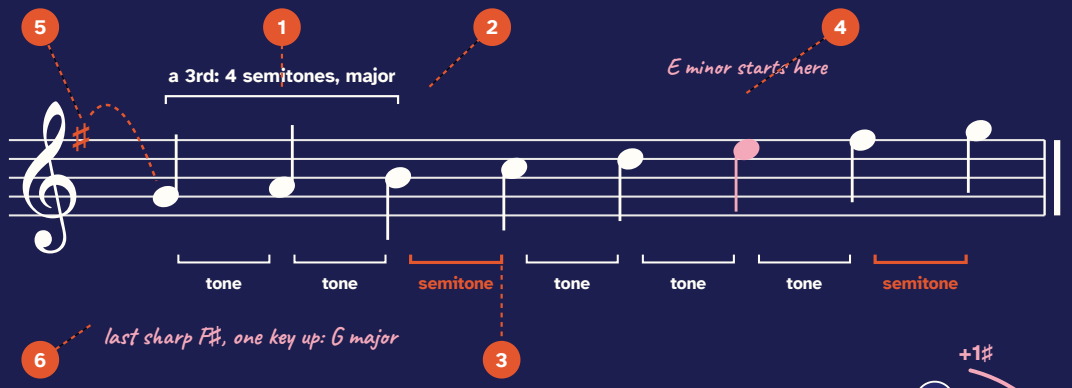
Long notes are where counting slips: **say every seat** they book.

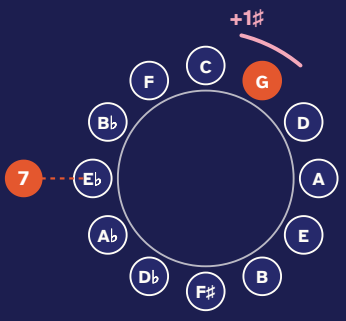
THINK OF IT Like a dance partner: the beat leads, the rhythm follows.

PART III · THE GROUND UNDER THE CHORDS

Scales and keys

Notes travel in families. Measure the distance between two notes, build the scales that whole pieces are made from, then learn to read the key a piece lives in. All of it is in one scale.





1 **Counting intervals** p. 63
count the letters, both ends: G A B is a 3rd

2 **Major, minor and perfect** p. 66
the same number, bigger or smaller

3 **Major scales** p. 69
T T S T T T S: the two short steps

4 **Minor scales** p. 72
the same notes from the 6th note up

5 **Keys and key signatures** p. 75
the sharps at the start apply all the way

6 **Finding the key** p. 78
last sharp, one key up; second-last flat

7 **The circle of fifths** p. 81
clockwise, one more sharp each step

★ **Tips: when a note sounds wrong** p. 84

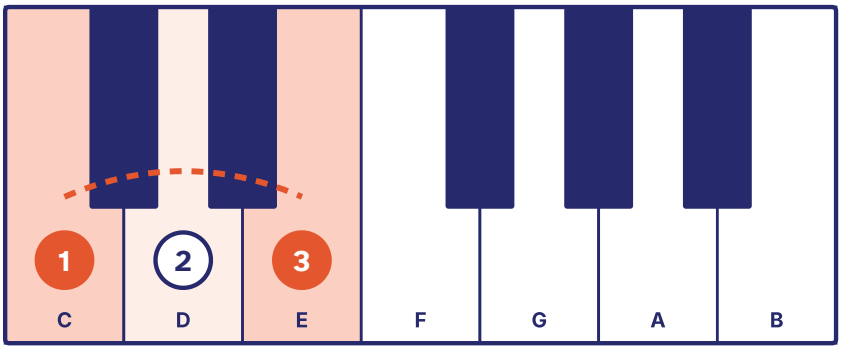
PART III · SCALES AND KEYS · 1

Counting intervals

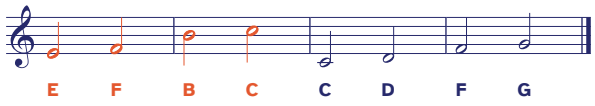
How far apart are two notes? That gap is an **interval**. The very next key is a **semitone**; two make a **tone**. To number it, count the letter names, both ends included.

THE SECRET

Count the letters, both ends included: C to E is a 3rd.



C to E: count the letters, 1 2 3, a 3rd

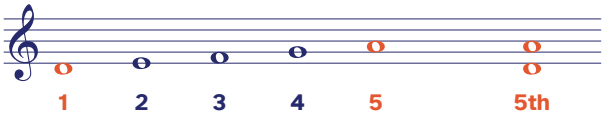


E–F and B–C are semitones; C–D and F–G are tones.



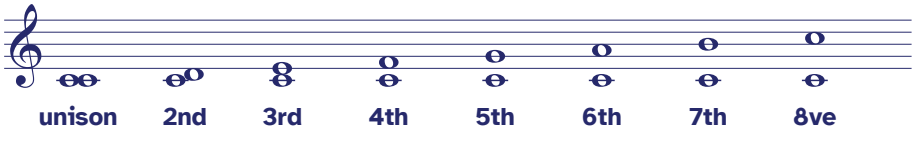
Melodic: one note after the other. Harmonic: together.

Counting the number



From **D** up to **A** (the red notes): D E F G A is five letters, so the interval is a 5th, whether the notes come one after the other or together.

Interval numbers from C

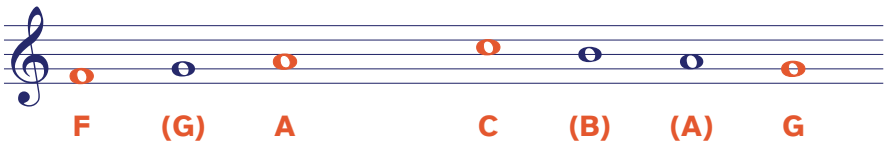


Your hand is a ruler: thumb on C, little finger on G: five fingers on five white keys, a 5th.

PART III · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 1

How far apart are these notes?

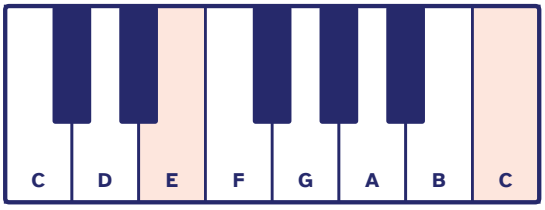


1. **Call the first note 1:** F is 1.
2. **Count every letter to the second note:** F G A is 3 letters: a 3rd.
3. **Going down works the same:** C B A G is 4 letters: a 4th.

F up to A is a **3rd**; C down to G is a **4th**. Sharps and flats never change the number.

EXAMPLE 2

Name an interval on the keyboard

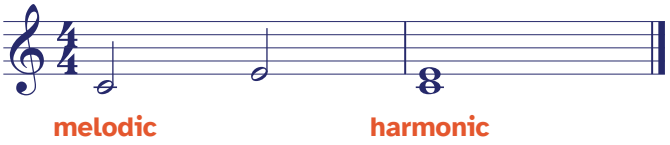


1. **Find both notes:** E and the C above it.
2. **Count the letters:** E F G A B C is six.
3. **Both ends count:** so the interval is a 6th.

E up to C is a **6th**.

EXAMPLE 3

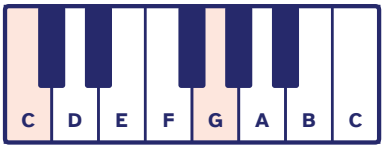
Melodic or harmonic?



1. **The first bar:** C then E, one after the other.
2. **The second bar:** C and E together.
3. **Count both:** C D E, a 3rd each time.

The same 3rd: **melodic** when played one after the other, **harmonic** when played together.

TRY IT · 10 SECONDS



Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

PART III · SCALES AND KEYS · 1 · CONTINUED

GAME

Count the letters

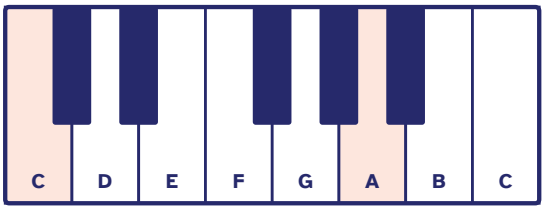
1	2	3	4
C up to G	D up to F	E up to the next E	G up to A

Count every letter, both ends included. What is each interval called?

Answer: 1 a 5th (C D E F G) · 2 a 3rd · 3 an octave (8) · 4 a 2nd.

EXAMPLE 4

Count an interval going down
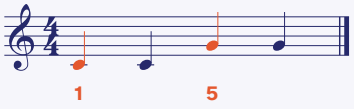


1. **Start on the top note:** A is 1.
2. **Count every letter down** to C: A G F E D C.
3. **Six letters:** a 6th.

A down to C is a **6th**, the same as C up to A.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The leap from “twin-” to “kle” is C up to G: C D E F G, **five letters, a 5th**.

Try it. Sing the first two notes of any song, then count the letters between them.

THINK OF IT Monday to Wednesday is three days, not two: you count the day you start on. Intervals count the same way.

TRAP

- × Counting only the gaps, not the notes: always count both ends.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

SEE ALSO

Major, minor and perfect p. 66 · Major scales p. 69

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Intervals

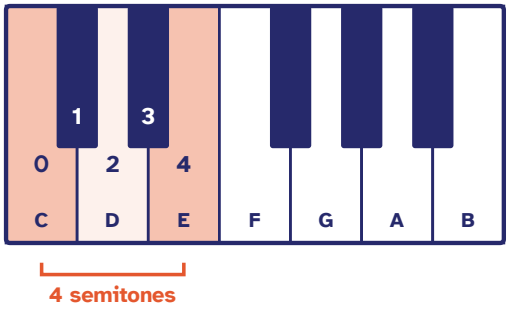
PART III · SCALES AND KEYS · 2

Major, minor and perfect

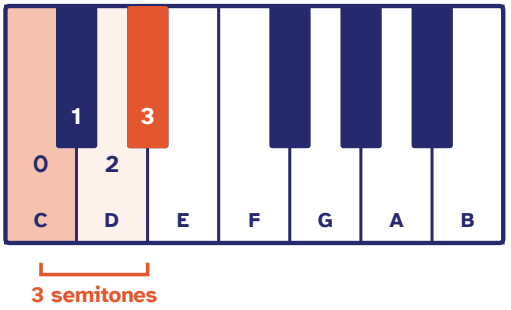
Two intervals with the same number can differ in size. C–E and C–E♭ are both 3rds, but one is a semitone wider. The **quality** (major, minor, perfect…) comes from counting **semitones**.

THE SECRET

Count the semitones: 4 make a major 3rd, 3 a minor 3rd.



From **C** to **E**: count every key, black and white. 4 semitones: a major 3rd.



From **C** to **E♭**: 3 semitones, one fewer: a minor 3rd.

One semitone less: a major 3rd becomes minor

Size in semitones

NUMBER	SMALLER	MAIN	WIDER	FIRST MET
unison		0 (perfect)		Level 1
2nd	1 (minor)	2 (major)		Level 1
3rd	3 (minor)	4 (major)		Level ½
4th		5 (perfect)	6 (augmented)	Level 1
5th	6 (diminished)	7 (perfect)		Level 1
6th	8 (minor)	9 (major)		Level 2
7th	10 (minor)	11 (major)		Level 2
octave		12 (perfect)		Level 1

2nds, 3rds, 6ths, 7ths
diminished ‹ **minor** ‹ **major** ‹ augmented

unisons, 4ths, 5ths, octaves
diminished ‹ **perfect** ‹ augmented

Each step on the ladder is one semitone. Six semitones (an augmented 4th or diminished 5th) is the tritone. Turned upside down, an interval's numbers add to 9 and major and minor swap: a major 3rd becomes a minor 6th.

PART III · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 1

Major or minor 3rd?



1. **Count the letters:** C D E is three, so it is a 3rd.
2. **Count the semitones:** C to C♯, D, E♭ is three.
3. **Three semitones** make a minor 3rd; four would be major.

C to E♭: a **minor 3rd**.

GAME

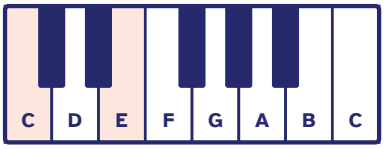
Perfect, major or minor?

1	2	3	4
C–G	D–F	C–A	E–A

Count the letters for the number, then the semitones for the size.

Answer: 1 perfect 5th (7) · 2 minor 3rd (3) · 3 major 6th (9) · 4 perfect 4th (5).

TRY IT · 10 SECONDS

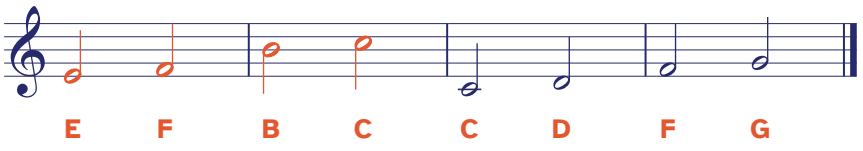


Play C–E, then C–E♭.

Bright, then dark. One key makes the difference: the major 3rd has four semitones, the minor 3rd three.

EXAMPLE 2

Tone or semitone?



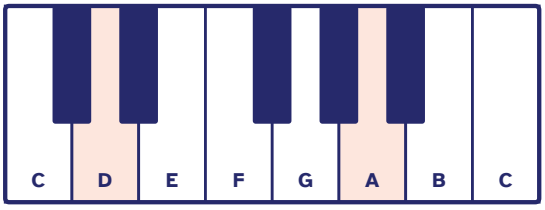
1. **E to F:** the very next key, no black key between: a semitone.
2. **C to D:** one key between (C♯): a tone.
3. **B to C** and **F to G**: check the same way.

E–F and B–C are semitones; C–D and F–G are tones.

PART III · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 3

Name D up to A

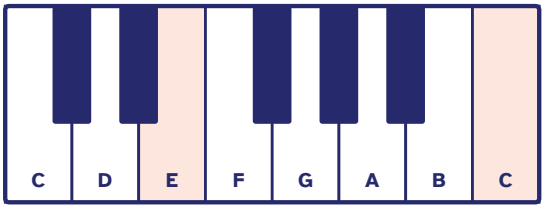


1. **The number:** D E F G A, five letters, a 5th.
2. **The size:** count every key from D to A, black and white: 7 semitones.
3. **7 semitones** is the main size of a 5th: perfect.

D up to A is a **perfect 5th**.

EXAMPLE 4

Name E up to C
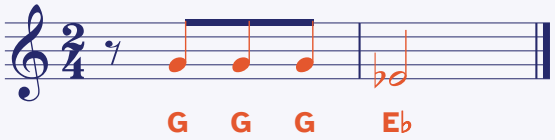


1. **The number:** E F G A B C, six letters, a 6th.
2. **The size:** 8 semitones from E to C.
3. **A major 6th** has 9, so 8 is one smaller: minor.

E up to C is a **minor 6th**.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to E♭ is four semitones, a **major 3rd**. Three semitones would be minor.

Try it. Count the keys from G down to E♭: F♯, F, E, E♭.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every key, black and white.
- × Calling a 4th or 5th “major”: they are perfect.

SEE ALSO

Counting intervals p. 63 · Chords in a key p. 92

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Intervals

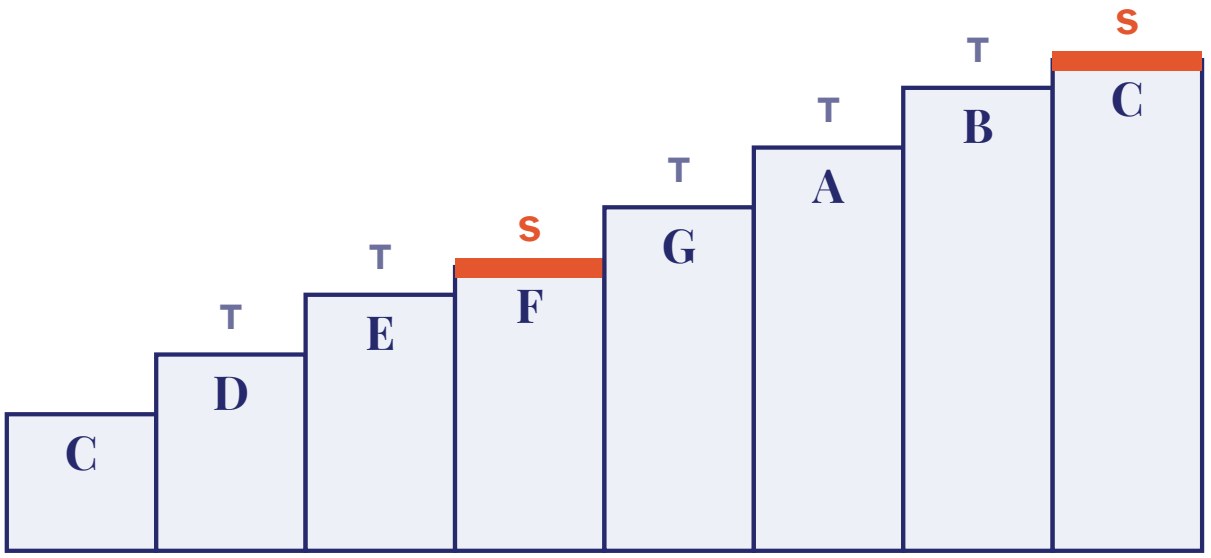
PART III · SCALES AND KEYS · 3

Major scales

Climb from a key note to the next one with the same name, following a fixed recipe of tones (T) and semitones (S): that is a **scale**. The recipe, not the starting note, makes it major or minor.

THE SECRET

Every major scale: tone, tone, semitone, tone, tone, tone, semitone.



Tone, tone, semitone, tone, tone, tone, semitone: the two short steps are 3–4 and 7–8

SCALE	STEPS	FIRST MET
major	T T S T T T S	Level ½
natural minor	T S T T S T T	Level ½
harmonic minor	T S T T S T+S S	Level ½
melodic minor (up)	T S T T T T S; down as the natural minor	Level 2
chromatic	every semitone	Level 2
pentatonic (black keys)	T T T+S T T+S	Level 0

Every step has a name

1 tonic · **2** supertonic · **3** mediant · **4** subdominant · **5** dominant · **6** submediant · **7** leading note

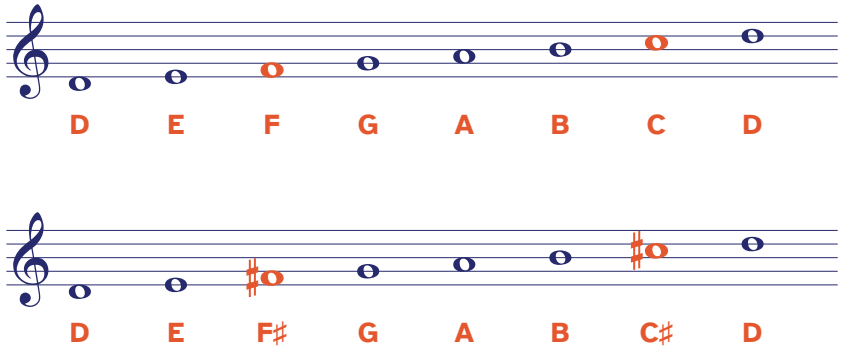
Major or minor?

The same key note can start a major or a minor scale: C major and C minor. The pattern decides. Major sounds bright and settled; minor, darker.

PART III · SCALES AND KEYS · 3 · CONTINUED

EXAMPLE 1

Build D major



1. **Write the eight letters** from D to D, every letter once.
2. **Check the steps** against T T S T T T S: E–F is too small, B–C is too small.
3. **Raise F and C** a semitone to fix them.

D major: **D E F♯ G A B C♯ D**, two sharps.

EXAMPLE 2

Build F major

1. **Write the letters** from F to F: F G A B C D E F.
2. **Check the steps** against T T S T T T S: A to B is a tone, but it must be a semitone.
3. **Lower B** to B♭: now A to B♭ is a semitone and B♭ to C a tone.

F major: **F G A B♭ C D E F**, one flat.

HOW TO

Practise a scale

1. **Hands separately,** slowly, saying the finger numbers aloud.
2. **Fingering for C, G, D, A and E major** (right hand): 1 2 3, thumb under, 1 2 3 4 5.
3. **Listen for evenness:** every note the same length and loudness.
4. **Then hands together,** one octave, then two.

GAME

Spot the wrong note

1

G A B C D E F G

2

D E F♯ G A B C D

3

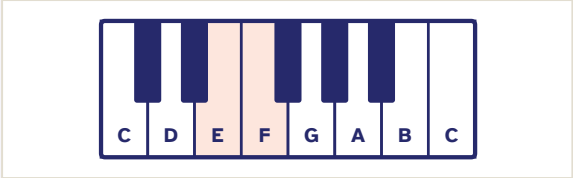
F G A B C D E F

Each major scale has one note that breaks the T T S T T T S pattern. Which, and what should it be?

Answer: 1 F should be F♯ · 2 C should be C♯ · 3 B should be B♭.

PART III · SCALES AND KEYS · 3 · CONTINUED

TRY IT · 10 SECONDS
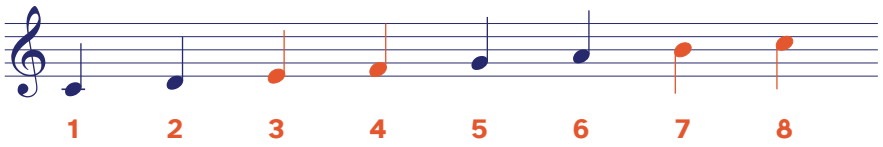


One finger, C to C.

Play the white keys from C up to C and say “tone, tone, semitone, tone, tone, tone, semitone” as you go. Feel the two short steps.

EXAMPLE 3

Where are the semitones?
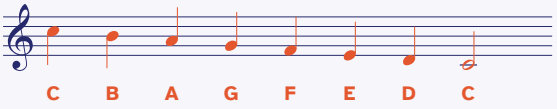


1. **Number the degrees** 1 to 8 from C.
2. **Look for no black key between:** E–F and B–C.
3. **Those are degrees 3–4 and 7–8.**

Every major scale has its semitones between **3–4** and **7–8**.

YOU KNOW THIS

Joy to the World



The first line is the **C major scale**, top to bottom: C B A G F E D C.

Try it. Play C major down from the top C and sing “Joy to the world” over it.

THINK OF IT A scale is a staircase with two short steps. Wherever you build it, the short steps sit in the same places: between 3 and 4, and between 7 and 8.

TRAP

- × Using the same letter twice: every letter appears once (F♯, not G♭, in G major).
- × Forgetting the semitone between 7 and 8: the leading note pulls up to the key note.

SEE ALSO
Minor scales p. 72 · Keys and key signatures p. 75

PRACTISE IT
Workbook Scales and keys 2 · Online: Library › Scales

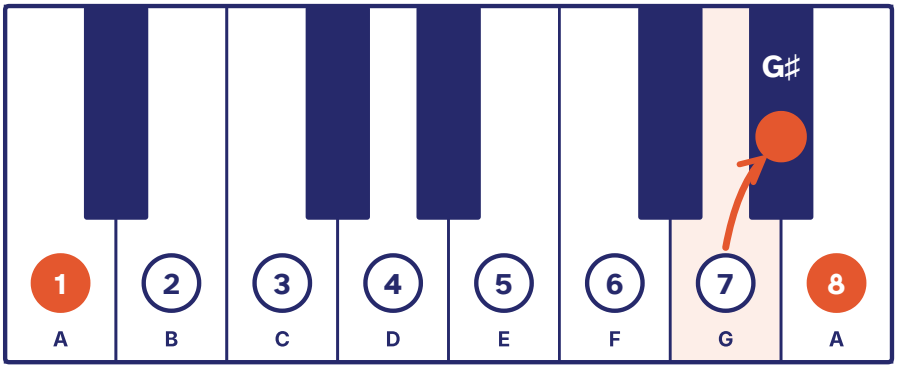
PART III · SCALES AND KEYS · 4

Minor scales

Every minor scale starts from the natural minor, the notes of its relative major from the 6th degree up. Two small changes give the harmonic and melodic forms, which sound stronger at the top.
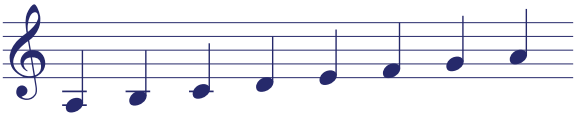
THE SECRET

Three minor scales, one home note: only the 6th and 7th change.
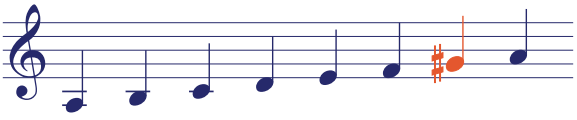


From A on the white keys · then raise the 7th: G♯

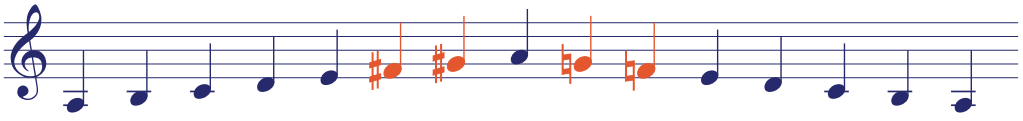
A minor in three forms



Natural: the notes of C major, from A to A. T S T T S T T.



Harmonic: the 7th raised a semitone (**G♯**). The gap F–G♯ is a tone and a half.



Melodic: the 6th and 7th raised going up (**F♯ G♯**), lowered coming down (**G♮ F♮**).

Relative minor

Same key signature, three semitones below the major: C major and A minor share every note.

Parallel minor

Same key note, different signature: C major (no sharps or flats) and C minor (three flats).

PART III · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 1

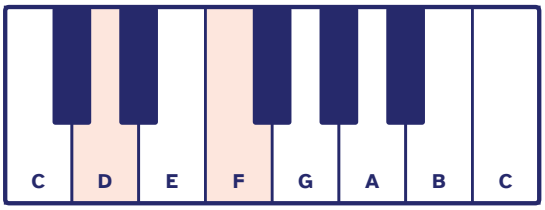
Build E harmonic minor

1. **Borrow G major's notes** starting on E: E F♯ G A B C D E.
2. **Raise the 7th:** D becomes D♯.
3. **Check the gap** C to D♯: a tone and a half.

E harmonic minor: **E F♯ G A B C D♯ E**.

EXAMPLE 2

Find the relative minor of F major



1. **Start on F,** the key note.
2. **Count three semitones down:** E, E♭, D.
3. **Check the signature:** D minor has one flat, B♭, like F major.

The relative minor of F major is **D minor**.

GAME

Which minor?

1

nothing raised

2

the 7th raised

3

6th and 7th raised going up, lowered coming down

Three forms of the minor scale. Name each one.

Answer: 1 natural · 2 harmonic · 3 melodic.

TRY IT · 10 SECONDS
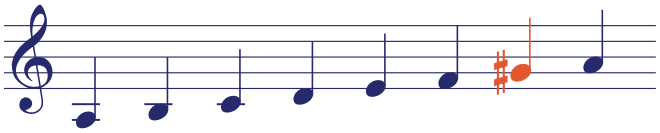


A to A on the white keys.

That is A natural minor: dark from the first note. Play it again with G♯ instead of G: harmonic minor, and the top pulls home.

PART III · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 3

Natural or harmonic minor?

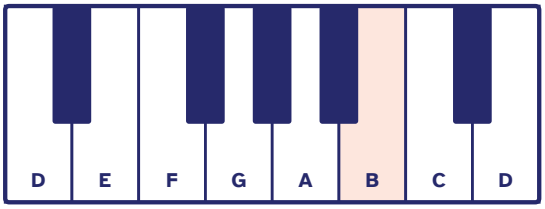


1. **Start on A** and read up to A.
2. **Look at the 7th:** G♯, raised.
3. **Nothing else changes** from the natural minor.

A raised 7th only: the **harmonic minor**.

EXAMPLE 4

Build D natural minor
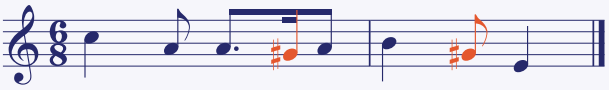


1. **Find its relative major:** three semitones up from D is F.
2. **F major has one flat:** B♭.
3. **Play F major's notes from D to D:** D E F G A B♭ C D.

D natural minor: **D E F G A B♭ C D**, with the same key signature as F major.

YOU KNOW THIS

Greensleeves



A minor, with a **G♯**: the raised 7th of the harmonic minor, leaning hard up to A.

Try it. Play it once with G♯, once with G: hear the pull disappear.

THINK OF IT The relative minor is the same clothes worn by a different person: the same notes, a new key note. The parallel minor is the same person in different clothes: the same key note, some notes changed.

TRAP

- ✕ Raising the 7th in the key signature: it is written as an accidental each time.
- ✕ Playing the melodic minor's raised notes on the way down.

SEE ALSO
Major scales p. 69 · Keys and key signatures p. 75

PRACTISE IT
Workbook Scales and keys 2 · Online: Library › Scales

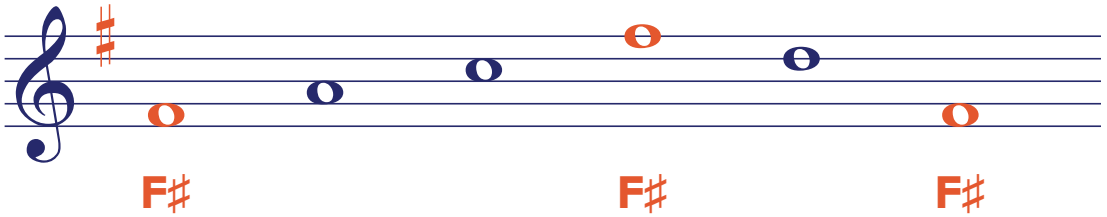
PART III · SCALES AND KEYS · 5

Keys and key signatures

Why write F♯ fifty times? The **key signature** at the start of each line says it once, for the whole piece. Its sharps or flats always line up in the same order.

THE SECRET

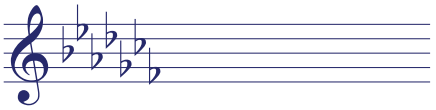
The key signature says once, at the start, which notes are always sharp or flat.



One sharp at the start: every F plays F♯



Sharps: F C G D A E B. The key note is a semitone above the last sharp.



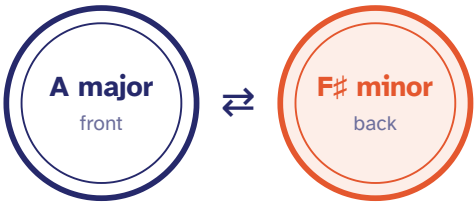
Flats: B E A D G C F. From two flats, the second-last flat names the key.

To remember the order: **F**ather **C**harles **G**oes **D**own **A**nd **E**nds **B**attle; backwards for the flats: **B**attle **E**nds **A**nd **D**own **G**oes **C**harles' **F**ather.

MAJOR	RELATIVE MINOR	SIGNATURE	FIRST MET
C major	A minor	—	Level ½
G major	E minor	1♯	Level ½
D major	B minor	2♯	Level 1
A major	F♯ minor	3♯	Level 3
E major	C♯ minor	4♯	Level 4
B major	G♯ minor	5♯	Level 4
F major	D minor	1♭	Level ½
B♭ major	G minor	2♭	Level 1
E♭ major	C minor	3♭	Level 3
A♭ major	F minor	4♭	Level 4
D♭ major	B♭ minor	5♭	Level 4

Six and seven sharps or flats (F♯, C♯, G♭, C♭ major and their minors) come at Levels 4–6.

PART III · SCALES AND KEYS · 5 · CONTINUED

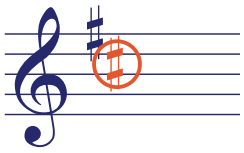


One signature, two keys

Every key signature belongs to **one major key and one minor key**, like the two sides of a coin: three sharps are A major or F♯ minor. The minor is the **relative minor**, three semitones below the major.

EXAMPLE 1

Name the keys



1. **Count:** two sharps, F♯ and C♯.
2. **The last sharp** is C♯; one semitone up is D.
3. **The relative minor,** three semitones below D, is B.

Two sharps: **D major** or **B minor**.

GAME

How many sharps or flats?

1

G major

2

D major

3

A major

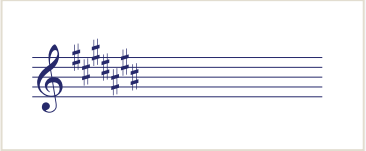
4

F major

Count the steps from C round the circle: clockwise adds a sharp, anticlockwise adds a flat.

Answer: 1 one sharp · 2 two · 3 three · 4 one flat.

TRY IT · 10 SECONDS
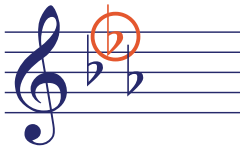


Say the sharps in order.

"Father Charles Goes Down And Ends Battle": F C G D A E B. Then backwards for the flats: B E A D G C F.

EXAMPLE 2

Read a flat key signature



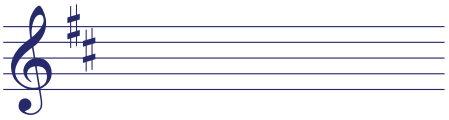
1. **Count the flats:** three, B♭ E♭ A♭.
2. **The second-last flat** is E♭: the key note.
3. **Its relative minor,** three semitones down, is C minor.

Three flats: **E♭ major** or **C minor**.

PART III · SCALES AND KEYS · 5 · CONTINUED

EXAMPLE 3

Read a sharp key signature



1. **Name the sharps:** F♯ and C♯.
2. **The last sharp is C♯;** one key up is D.
3. **Two keys share it:** D major, and its relative minor, B minor.

Two sharps: **D major** or **B minor**.

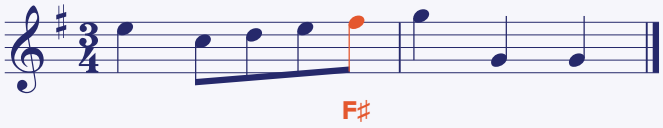
HOW TO

Write a key signature

1. **Sharps** always come in the same order, F C G D A E B, each on its own line or space.
2. **Flats** come in the reverse order, B E A D G C F.
3. **Write it again** at the start of every line, in both staves.

YOU KNOW THIS

Minuet in G



One sharp in the key signature: every F is F♯, like the one just before the top G. That sharp is why it is in G major.

Try it. Find every F in the piece and play it as F♯.

THINK OF IT A dress code on a party invitation: written once at the top, it applies to every guest all night. Only an accidental, a word to one guest, overrules it, and only until the bar line.

TRAP

- × Writing a key signature in the wrong order: sharps always F C G D A E B.
- × Forgetting that a signature names two keys: a major and its relative minor.
- × Reading it once and forgetting it: glance at the key signature at the start of every line.

SEE ALSO

Major scales p. 69 · Finding the key p. 78 · The circle of fifths p. 81

PRACTISE IT

Workbook Scales and keys 2 · Online: Library › Keys

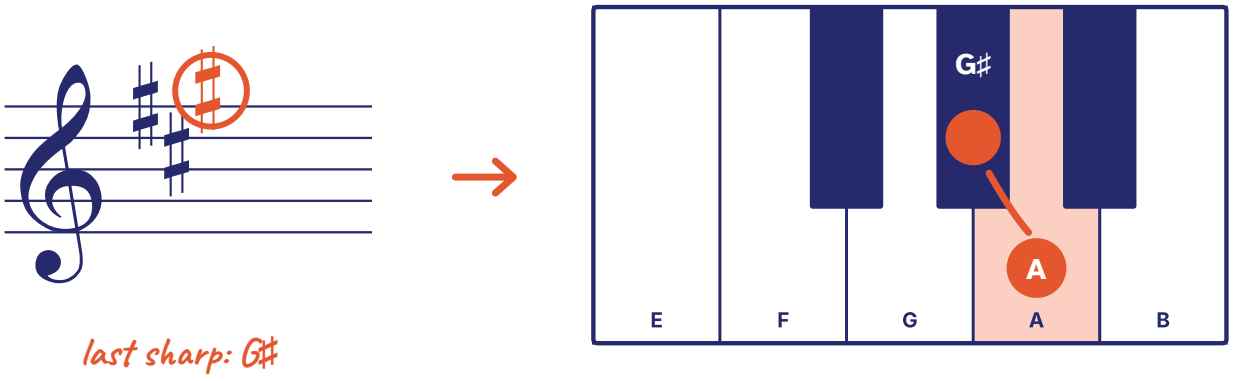
PART III · SCALES AND KEYS · 6

Finding the key

A key signature names two keys, a major and its relative minor. Two quick rules (and one to remember) find the major; one more finds the minor; then the music itself tells you which of the two it is.
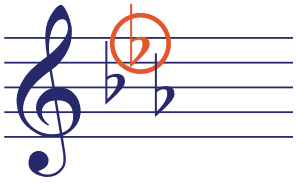
THE SECRET

Last sharp, one key up: the major key. Last flat but one: the major key.



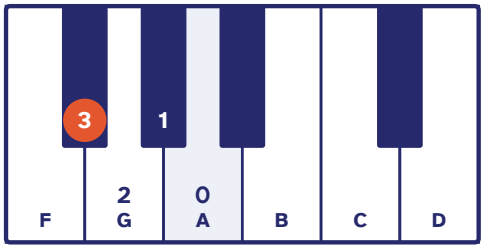
last sharp: G♯

Last sharp, one key up: A major



Flats: the second-last flat is the key.

B♭ **E♭** A♭ → **E♭ major**. One flat alone is F major.



Major to minor: three semitones down.

A, G♯, G, **F♯** → **F♯ minor**, the relative minor of A major (count every key, black and white).

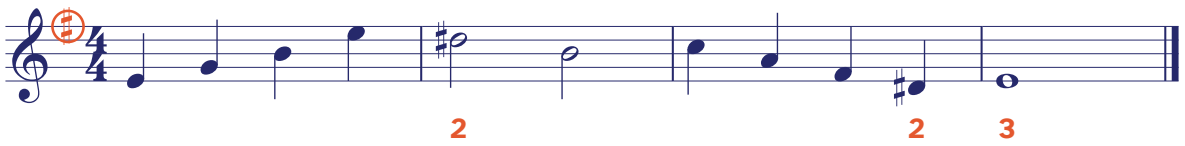
Minor keys and their clue notes

MINOR KEY	SIGNATURE	CLUE NOTE (RAISED 7TH)
A minor	none	**G♯**
E minor	1♯	**D♯**
D minor	1♭	**C♯**
B minor	2♯	**A♯**
G minor	2♭	**F♯**

PART III · SCALES AND KEYS · 6 · CONTINUED

EXAMPLE 1

What key is this tune in?

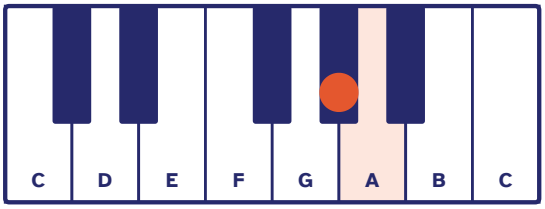


1. **The key signature:** one sharp, so G major or its relative, E minor.
2. **The clue notes:** D♯ keeps appearing. D♯ is the raised 7th of E minor; G major has no D♯.
3. **The last note:** E, the key note of E minor.

The key: **E minor**. Check it the same way with any piece you play.

EXAMPLE 2

Major or minor? No sharps or flats



1. **The signature:** none, so C major or A minor.
2. **The clue notes:** G♯ appears. G♯ is the raised 7th of A minor.
3. **The last note:** A.

No sharps or flats, G♯ clues, ending on A: **A minor**.

GAME
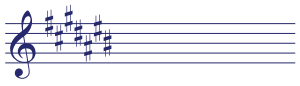
Name the major key

1	2	3	4
2 sharps	4 sharps	1 flat	3 flats

Sharps: the last sharp, one key up. Flats: the second-last flat is the key (one flat alone is F).

Answer: 1 D · 2 E · 3 F · 4 E♭.

TRY IT · 10 SECONDS



Three sharps, stop, step up.

Say “F C G”, stop: the last one is G♯. One key up is A. Three sharps is A major. Try it with two and four.

PART III · SCALES AND KEYS · 6 · CONTINUED

EXAMPLE 3

Read a flat key signature

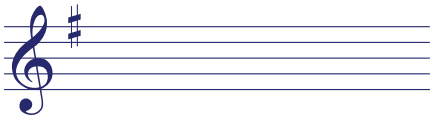


1. **Name the flats:** B♭, E♭, A♭.
2. **The second-last flat** is E♭.
3. **Its relative minor** is three semitones below: C.

Three flats: **E♭ major** or **C minor**.

EXAMPLE 4

Major or minor? Look for the raised 7th



1. **One sharp:** G major or E minor.
2. **E minor's 7th note is D;** raised, it becomes D♯.
3. **Look through the piece:** D♯ often, and an ending on E means E minor.

No D♯ and an ending on G: **G major**. D♯s and an ending on E: **E minor**.
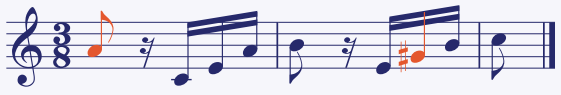
HOW TO

Find the key of a new piece

1. **The key signature:** name the major key and its relative minor.
2. **The clue notes:** look for the minor key's raised 7th.
3. **The last bass note:** pieces nearly always end on the key note.

YOU KNOW THIS

Für Elise (Beethoven)



No key signature, a G♯ as the clue, and the music keeps coming home to A: **A minor**.

Try it. Play A, then G♯ A: hear the G♯ lean into home.

THINK OF IT Detective work: the key signature narrows it to two suspects; the clue note and the last note name the culprit.

SEE ALSO
Keys and key signatures p. 75 · The circle of fifths p. 81

PRACTISE IT
Workbook Scales and keys 2 · Online: Library › Keys

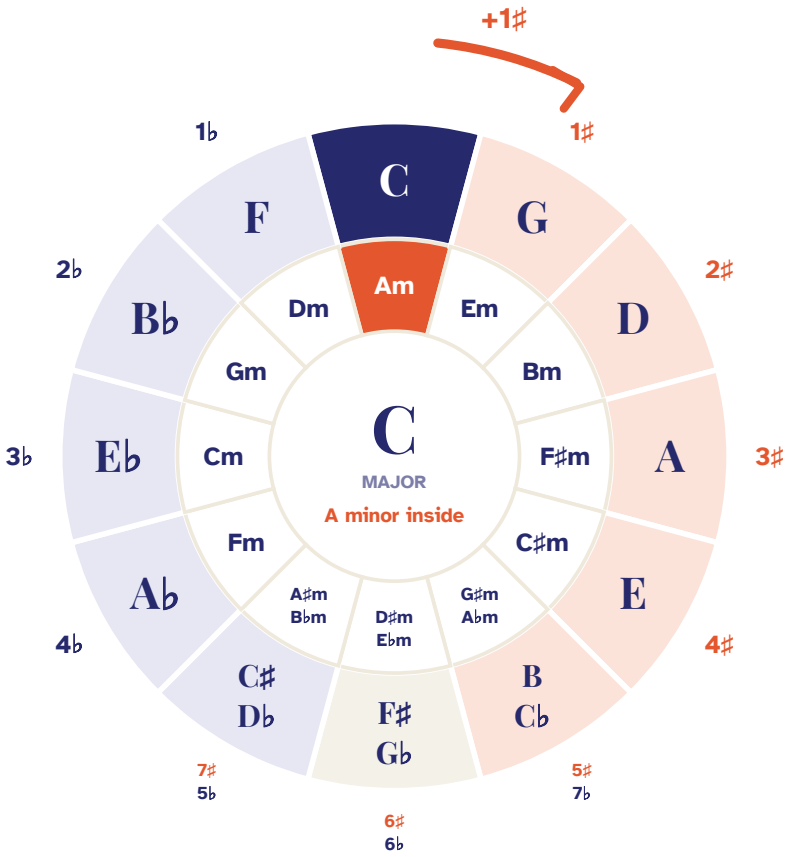
PART III · SCALES AND KEYS · 7

The circle of fifths

All the keys on one clock face. Going round clockwise, each key is a **5th higher** than the last and has **one more sharp** (or one fewer flat). Anticlockwise, each is a 4th higher, a 5th lower, with one more flat.

THE SECRET

Clockwise adds a sharp, anticlockwise adds a flat: every key on one clock.



Clockwise, one more sharp each step

How to read the clock

Outer ring: major keys. **Middle ring:** their relative minors (small letters). **Inner ring:** the key signature.

Neighbours are close

Keys next to each other share all but one note, so music moves between them easily: from C major to G major or F major, or to the relative minor. These are the **related keys**.

PART III · SCALES AND KEYS · 7 · CONTINUED

Six jobs for one circle

Key signatures
Count the steps from C: clockwise for sharps, anticlockwise for flats.

Relative minors
Look straight in from any major key.

The main chords
A key's two neighbours give its other main chords: in C, the chords of F and G.

Cadences
The strongest close, V to I, is one step anticlockwise: G to C.

Changing key
Pieces usually move one step round, or to the relative minor.

Scale practice
Go round the circle: one new sharp or flat each time.

The first steps round the circle

STEPS	CLOCKWISE (SHARPS)	ANTICLOCKWISE (FLATS)
1	G major: F♯	F major: B♭
2	D major: F♯ C♯	B♭ major: B♭ E♭
3	A major: F♯ C♯ G♯	E♭ major: B♭ E♭ A♭

C major's neighbours

KEY	WHERE ON THE CIRCLE	WHY IT MATTERS
G major	one step clockwise	the dominant: the strongest pull home
F major	one step anticlockwise	the subdominant: a gentle move away
A minor	straight in from C	the relative minor: the same notes
E minor, D minor	in from G and F	the relatives of the neighbours

EXAMPLE 1

Why do G major and D major sound so close?

1. **Find both:** D is one step clockwise from G.
2. **Compare the scales:** G major has C; D major has C♯.
3. **Count:** six notes the same, one different.

Neighbours share six of their seven notes, so music moves between them easily.

EXAMPLE 2

Find A major's key signature

1. **Start at C** at the top of the circle: no sharps.
2. **Go clockwise:** G, D, A is three steps.
3. **Three steps,** three sharps: F♯, C♯, G♯.

A major has **three sharps**.

PART III · SCALES AND KEYS · 7 · CONTINUED

GAME

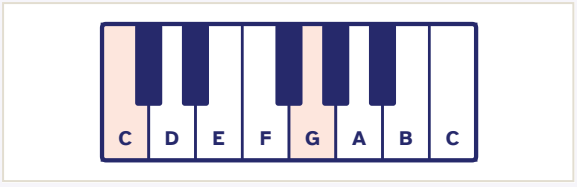
Neighbours on the clock

1	2	3	4
right of D	left of C	opposite C	inside G

Clockwise is right. Inside each major sits its relative minor.

Answer: 1 A · 2 F · 3 F♯ / G♭ · 4 E minor.

TRY IT · 10 SECONDS



Walk the circle: C G D A E.

Play each key note with one finger, a 5th up each time (drop an octave when you run out of hand). You just walked four steps clockwise.

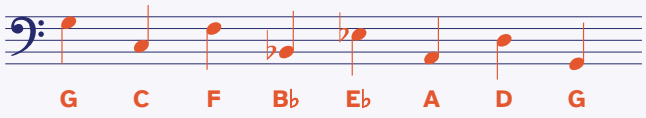
HOW TO

Learn the circle by heart

1. **Clockwise, out loud:** C G D A E B F♯. Each is a 5th above the last.
2. **Anticlockwise:** C F B♭ E♭ A♭ D♭ G♭. Each is a 5th below.
3. **Check yourself:** the number of steps from C is the number of sharps or flats.

YOU KNOW THIS

Passacaglia (Handel), the bass



Each bass note is a **5th below** the one before (or a 4th above): G C F B♭ E♭ A D G, one step round the circle each chord.

Try it. Play the bass line slowly and hear it pull towards the last G.

THINK OF IT Like a colour wheel: neighbours blend, opposites clash. C and G sit side by side; C and F♯ sit opposite.

TRAP

- × Going the wrong way round: clockwise adds sharps, anticlockwise adds flats.
- × Forgetting the relative minors on the middle ring.

SEE ALSO

Keys and key signatures p. 75 · Finding the key p. 78 · Chords in a key p. 92

PRACTISE IT

Workbook Scales and keys 2 · Online: Library › Related keys

PART III · SCALES AND KEYS · TIPS

When a note sounds wrong

You read the note right, yet it sounds sour. The slip is usually in the key, not the note. Name the slip first, then fix only that note.
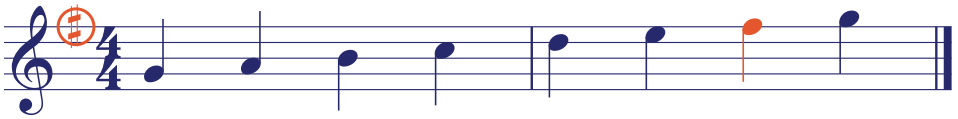
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
A note sounds sour, though you read it right	**the key signature**: is this note sharp or flat all the way through?	p. 75
Two notes sound too close or too far apart	**the interval**: count the letters, then the keys between	p. 63
A minor tune sounds wrong near the end	**the raised 7th**: harmonic and melodic minor sharpen it	p. 72
A scale sounds wrong in the middle	**the semitone steps**: 3–4 and 7–8 in every major scale	p. 69
A sharp is missing from the key signature	**the order of sharps**: F C G D A E B	p. 75
You are not sure which major key it is	**the last sharp**: the key is one note above it	p. 78

EXAMPLE

Find the slip



1. **Read the key signature:** one sharp, on the top line, F (ringed).
2. **The red note sits on that line**, so the sharp belongs to it too.
3. **No natural cancels it**, so this F is F♯, here and all through the piece.

Play **F♯**, the black key. Plain F is the most common slip in G major.

HOW TO

Three checks before you play

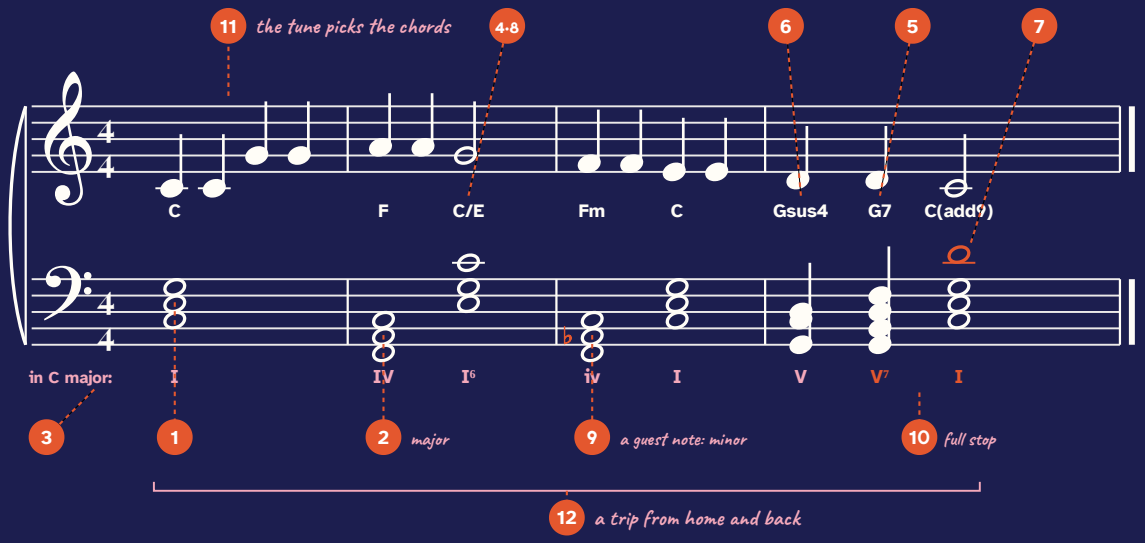
1. **The key signature:** say its sharps or flats aloud.
2. **The key:** name it from the last sharp or the last-but-one flat, then check the last note.
3. **The scale:** play it once, so your fingers know which black keys belong.

THINK OF IT Like proofreading: a spelling mistake jumps out once you know which words you usually get wrong. Keep a short list of your own slips, and check those first.

PART IV · WHICH NOTES SOUND TOGETHER

Harmony

A melody is the singer; harmony is the band behind it: the chords underneath. Every chapter of this part is in these four bars of Twinkle.



1 **Chords and triads** p. 86
three notes stacked like a snowman

2 **Four kinds of triad** p. 89
major 3 + 2, minor 2 + 3

3 **Chords in a key** p. 92
Roman numerals: I, IV and V do most of the work

4 **Inversions** p. 95
the same notes, a different one at the bottom

5 **Seventh chords** p. 98
one more floor: G7 pulls home

6 **Colour chords** p. 101
sus and add: a note moved or added

7 **The tower: 9, 11, 13** p. 104
keep skipping a letter

8 **Slash chords** p. 107
C/E: the chord, then the bass note

9 **Chords from outside the key** p. 110
a guest note: A♭ in C major

10 **Cadences** p. 113
V to I ends a phrase like a full stop

11 **Chords under a melody** p. 115
the tune's main notes pick the chord

12 **Progressions everywhere** p. 117
the same few trips, in every song

★ **Tips: when a chord sounds wrong** p. 119

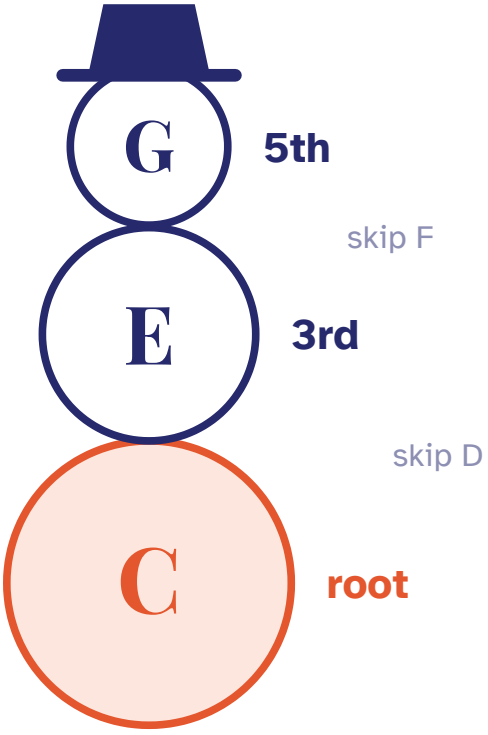
PART IV · HARMONY · 1

Chords and triads

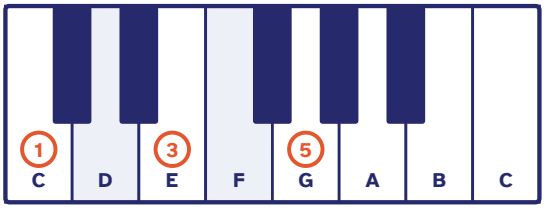
Press two or more notes at once and you have a **chord**. The favourite of all is the **triad**: three notes stacked like a snowman.

THE SECRET

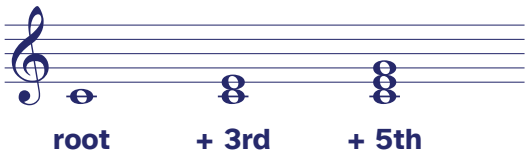
Play one, skip one, play one: the snowman rule.



1. Pick a note: the **root**, the big ball at the bottom. It holds the chord up and gives it its name.
2. Skip one letter, take the next: **E**, the **3rd**.
3. Skip again: **G**, the **5th**. C E G is the triad of C.



On the keys: play one, skip one (shaded), play one, skip one, play one.



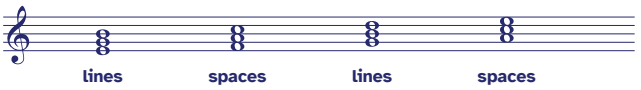
The snowman on the staff: three lines in a row (or three spaces).

On the staff, each ball sits on the **next line up** (or the next space up): the root, a **3rd** above it, a **5th** above the root.

First met: Level ½ (the tonic triad of C, G and F major).

All lines or all spaces

A stacked triad sits on **three lines in a row** or in **three spaces in a row**, in a neat stack. If one note breaks the pattern, it is not a triad.



Lines, spaces, lines, spaces: every one a triad.

PART IV · HARMONY · 1 · CONTINUED

Solid and broken

Played together, a chord is **solid** (or blocked). Played one note at a time, it is **broken**. Spread up and down the keyboard, a broken chord becomes an **arpeggio**. Left-hand accompaniments, the arpeggios in technical work and the chord names over a song are all triads.

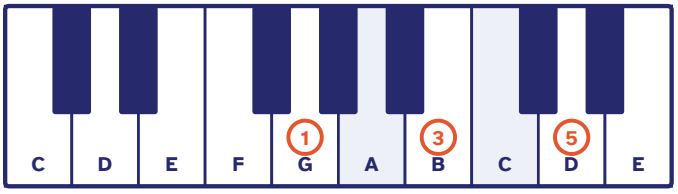
First met: Level 0; arpeggios from Level 1.



The same three notes, solid then broken.

EXAMPLE 1

Build a triad on G



1. **Start on G:** the root.
2. **Skip A, play B:** the 3rd.
3. **Skip C, play D:** the 5th.

G B D: the triad of **G**. Same shape as C E G, just moved up.

EXAMPLE 2

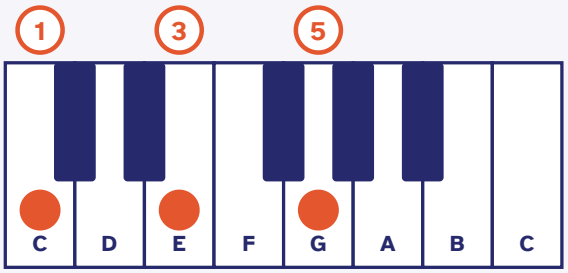
Is it a triad?



1. **Chord 1:** D F A, all in spaces, each a 3rd apart: **a triad**.
2. **Chord 2:** D E A: D to E is a 2nd. **Not a triad.**
3. **Chord 3:** D G C, stacked in 4ths. **Not a triad.**

Only chord 1 is a triad. Check that each note is a **3rd** above the one below.

TRY IT · 10 SECONDS



Thumb on C. Fingers 1, 3, 5 down together.

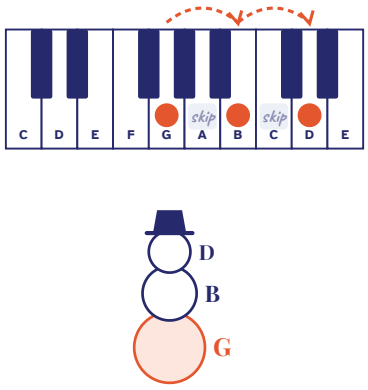
That is the C chord. Slide your hand to any white key and do it again: the snowman shape always works.

PART IV · HARMONY · 1 · CONTINUED

GAME

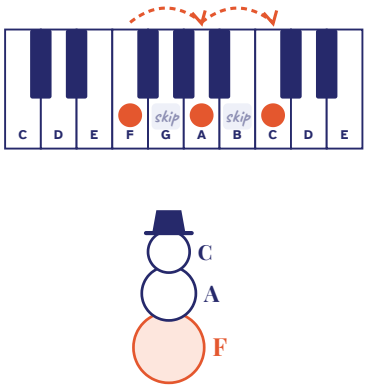
Build one yourself

G



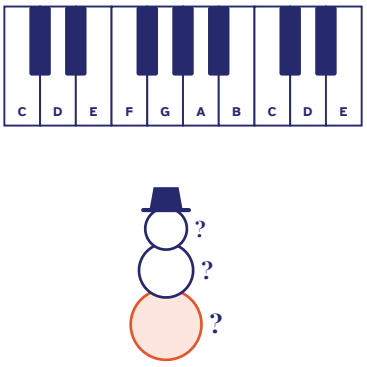
Start on G: **G B D**.

F



Start on F: **F A C**.
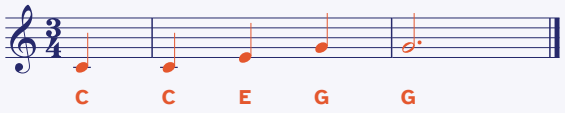
?



Your turn: start on **A**. Which three keys?

Build the snowman on A: play one, skip one, play one, skip one, play one.

Answer: A C E, the A minor chord.

YOU KNOW THIS

The Blue Danube (Johann Strauss II), here in C



The most famous waltz tune simply climbs the **C triad**: **C, C E G, G**. Root, 3rd, 5th, then it holds the top: that is why it sounds so open.

Try it. Play C E G one at a time, then hum the waltz over it.

THINK OF IT Solid: three friends shout “Hello!” together. Broken: each says it in turn.

TRAP

- × Stacking 2nds or 4ths: a triad skips exactly one letter each time.
- × Naming a chord from its top note: the root names it (at the bottom when the notes are stacked in 3rds).

SEE ALSO
Counting intervals p. 63 · Major scales p. 69

PRACTISE IT
Workbook Harmony 1 · Online: Library › Chords

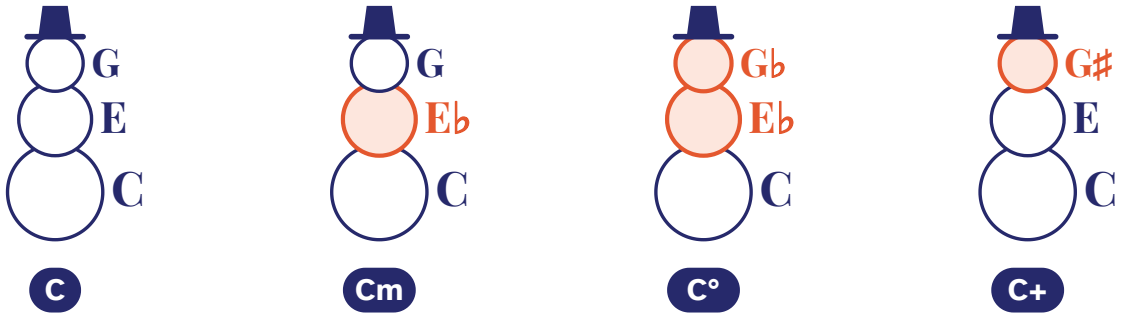
PART IV · HARMONY · 2

Four kinds of triad

Every triad is two 3rds stacked. To tell the kinds apart, count the keys **in between**, black and white, that you do not play: root to 3rd, then 3rd to 5th. Those two numbers name the kind, and its sound.

THE SECRET

Count the keys between: 3 + 2 is major, 2 + 3 is minor.

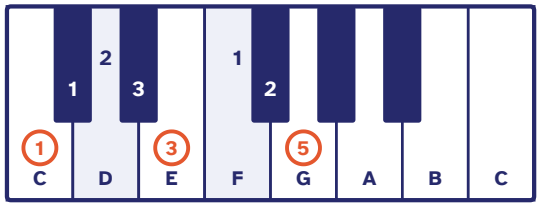


Count the keys between: 3 + 2 is major, 2 + 3 is minor

KIND	KEYS BETWEEN	ON C	CHORD NAME	FIRST MET
major	**3 + 2**	C E G	C	Level ½
minor	**2 + 3**	C E♭ G	Cm	Level 2
diminished	**2 + 2**	C E♭ G♭	C° or Cdim	Level 5
augmented	**3 + 3**	C E G♯	C+ or Caug	Level 6

Chord names: a letter alone means major (C); m means minor (Cm). You see them above the melody in songbooks and lead sheets: C | Am | F | G. In interval names, 3 keys between is a major 3rd and 2 is a minor 3rd (p. {S2}).

Count the keys in between



Play C and E and count the keys you skip, black and white: C♯, D, D♯ is **3**. Then E to G: F, F♯ is **2**. **3 + 2 is major.** Any root works the same way: from D, skip 3 to F♯, then 2 to A: D F♯ A, D major.

PART IV · HARMONY · 2 · CONTINUED

The seven white-key triads, sorted



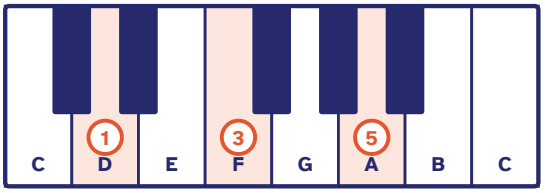
Always the same on the white keys: **three major** (C, F, G), **three minor** (D, E, A), **one diminished** (B), no augmented. In any major key, chords I, IV and V are the major ones (p. 92).

How they sound

KIND	SOUNDS	HEARD IN
major	bright, settled, open	most cheerful tunes, fanfares
minor	darker, softer, sadder	lullabies, laments, film tension
diminished	tense, unstable, wants to move	suspense, the chord before home
augmented	floating, dreamy, unsettled	dream scenes, a passing colour

EXAMPLE 1

Major or minor? D F A



1 **Root to 3rd:** D to F skips D♯ and E: **2**.

2 **3rd to 5th:** F to A skips F♯, G and G♯: **3**.

3 **2 + 3** is minor.

D F A is **D minor** (Dm). Three of the white-key triads are minor: D, E and A.

GAME

Which kind?



1 2 3

Count the keys between: 3 + 2 major, 2 + 3 minor, 2 + 2 diminished, 3 + 3 augmented.

Answer: 1 diminished · 2 major · 3 augmented.

PART IV · HARMONY · 2 · CONTINUED

TRY IT · 10 SECONDS

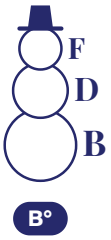


C E G, then C E♭ G, then C E♭ G♭.

Bright, dark, tense. Only the middle ball moves first, then the top one. Listen for the change each time.

EXAMPLE 2

Build a triad on B
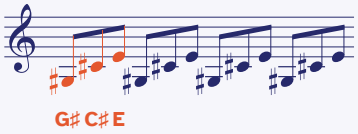


1. **Skip a letter:** B, D, F.
2. **Count the keys between:** B to D, 2 (C, C♯); D to F, 2 (D♯, E).
3. **2 + 2:** diminished.

B D F is **B diminished** (B°): the only diminished triad on the white keys.

YOU KNOW THIS

Beethoven, Moonlight Sonata (first movement)



The famous slow triplets are the notes of a **C♯ minor triad**: C♯, E and G♯, rocking gently. Minor is why it sounds so dark and sad.

THINK OF IT Two floors of a building: a tall floor (3 keys between) and a short one (2). Major: tall, then short, 3 + 2. Minor: short, then tall, 2 + 3. Diminished: two short, 2 + 2. Augmented: two tall, 3 + 3.

TRAP

- × Counting the keys you play: count only the keys in between, black and white.
- × Forgetting that in diminished and augmented triads the 5th changes too.
- × Spelling with the wrong letter: a triad skips one letter, so C minor is C E♭ G, never C D♯ G.

SEE ALSO
Chords and triads p. 86 · Major, minor and perfect p. 66

PRACTISE IT
Workbook Harmony 2 · Online: Library › Chords

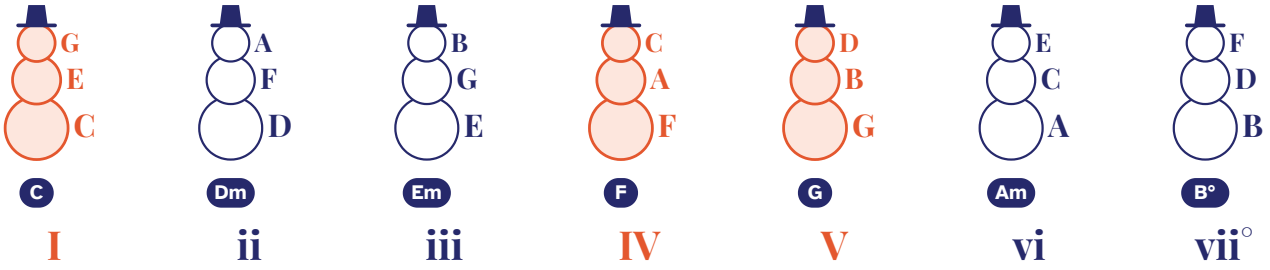
PART IV · HARMONY · 3

Chords in a key
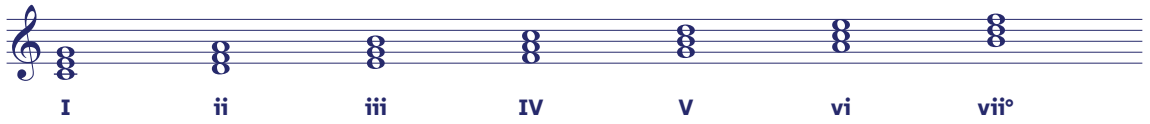
Build a triad on every note of a scale and you have the chords of that key. **Roman numerals** name each one by its place in the scale: I on the 1st note, IV on the 4th, V on the 5th. Capitals are major, small numerals are minor, and small with ° is diminished.

THE SECRET

Build a snowman on every note of the scale: I, IV and V are the strongest.



One snowman on every note: I, IV and V do most of the work



The seven chords of C major. Only white keys, so the kinds come out as on p. 90.

NUMERAL	NAME	KIND	KEYS BETWEEN	IN C	IN G
I	tonic	major	3 + 2	C	G
ii	supertonic	minor	2 + 3	Dm	Am
iii	mediant	minor	2 + 3	Em	Bm
IV	subdominant	major	3 + 2	F	C
V	dominant	major	3 + 2	G	D
vi	submediant	minor	2 + 3	Am	Em
vii°	leading note	diminished	2 + 2	B°	F♯°

The same in every major key: I, IV and V are major; ii, iii and vi are minor; vii° is diminished. First met: I, IV and V at Levels 1–2; the other chords from about Level 4.

The three primary chords
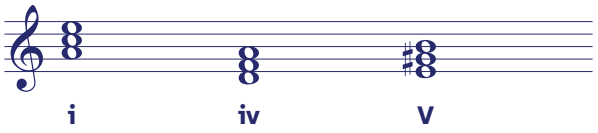
I, IV and V are the strongest chords of a key. Between them they hold **every note of the scale**: C E G, F A C, G B D. That is why so many tunes need only these three.

THINK OF IT I is home. IV is a visit next door. V is the walk back, already looking at the front door: it wants to go home to I. Play C, F, G, C and hear it.

PART IV · HARMONY · 3 · CONTINUED

In a minor key

In A minor the three main chords are **i** (A minor), **iv** (D minor) and **V**: E G♯ B, made major by the **raised 7th**, G♯, which pulls up to the home note A. Check: E G♯ B is 3 + 2, major.



A minor: i, iv and V. The G♯ makes V major.

Which chord holds the note?

NOTE IN C	C	D	E	F	G	A	B
Chord	I or IV	V	I	IV	I or V	IV	V

The same idea works in every key: find which of I, IV and V holds the note.

EXAMPLE 1

Find I, IV and V in G major



1. **Count up the scale:** G A B C D. The 1st note is G, the 4th C, the 5th D.
2. **Build a triad on each,** using the key signature: the F is F♯.
3. **Check with the keys between:** all three are 3 + 2, major.

In G major: **I = G**, **IV = C**, **V = D** (D F♯ A).

GAME

Which numeral?



All in C major. Count up the scale from C to find each chord's number.

Answer: 1 IV · 2 V · 3 vi · 4 ii.

TRY IT · 10 SECONDS



Left hand: I IV V I.

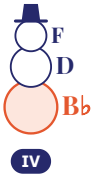
C, F, G, C, one chord per bar, four beats each. The three primary chords: nearly every nursery tune fits them.

PART IV · HARMONY · 3 · CONTINUED

EXAMPLE 2

Find I, IV and V in F major





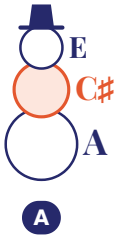


1. **Count up the scale:** F G A B♭ C. The 1st note is F, the 4th B♭, the 5th C.
2. **Build a triad on each,** with the key signature's B♭.
3. **Check:** all three are 3 + 2, major.

In F major: **I = F**, **IV = B♭**, **V = C**.

EXAMPLE 3

Find V in D minor



1. **The 5th note** of D minor is A.
2. **Build the triad:** A C E.
3. **Raise the 7th note of the scale,** C to C♯: A C♯ E, major.

V in D minor is **A major**: the C♯ pulls up to D.

YOU KNOW THIS

The twelve-bar blues

I C	I C	I C	I C
IV F	IV F	I C	I C
V G	IV F	I C	V G

Almost every blues, and a lot of rock 'n' roll, is just **I, IV and V** in twelve bars. Learn this chart and you can play along with hundreds of songs.

Try it. Play the chart in C with your left hand, one chord per bar, four beats each.

TRAP

- × Writing every numeral big: ii, iii and vi are minor, so they are written small.
- × Forgetting the raised 7th in a minor key: V in A minor is E G♯ B, not E G B.

SEE ALSO

Four kinds of triad p. 89 · Major scales p. 69 · Cadences p. 113

PRACTISE IT

Workbook Harmony 3 · Online: Library › Chords

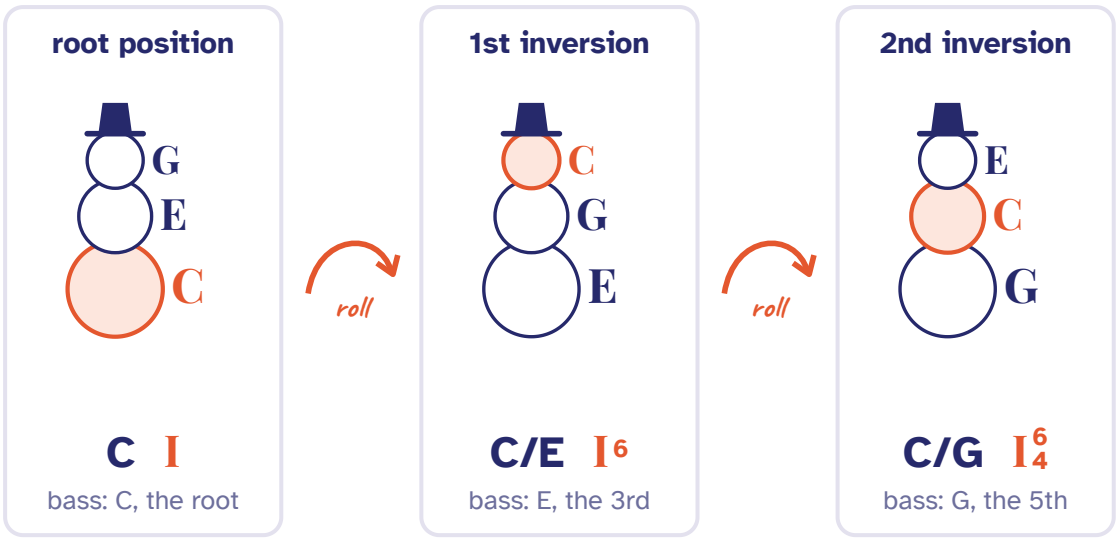
PART IV · HARMONY · 4

Inversions

Same three notes, a different one at the bottom. The chord keeps its name; only the bass changes. The red ball is the root, C: watch it travel.

THE SECRET

Move the bottom note to the top: same chord, new bass.

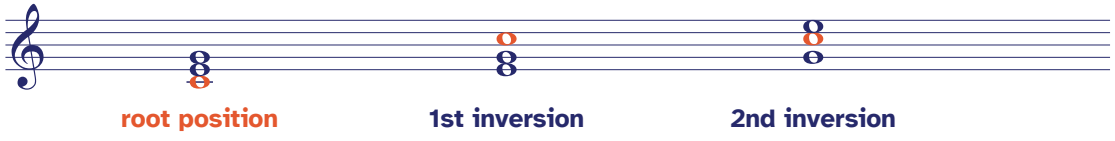


1. Start with the snowman **C E G**: root at the bottom.
2. Roll the bottom ball to the top: **E G C**. That is the 1st inversion.
3. Roll again: **G C E**, the 2nd inversion. Still **C major**.

Two ways to write it

	ROOT POSITION	1ST INVERSION	2ND INVERSION
Chord symbol pop, jazz	C	C/E	C/G
Harmony class numerals	I	I^6	I^6_4
Also written	Ia, 5/3	Ib, 6/3	Ic, 6/4
First met	Level ½	Level 3	Level 5

The little numbers count up from the bass. E G C: C is a **6th** above E, so I^6. G C E: E is a 6th and C a **4th** above G, so I^6_4. A slash chord just names the bass: C/E (more on p. 107).



The same three chords on the staff; the red note is the root, C. To find the root, restack the notes every other letter (like C E G): then the bottom ball is the root.

PART IV · HARMONY · 4 · CONTINUED

Why inversions?

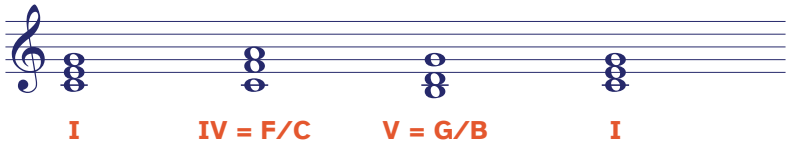
They let your hand **move as little as possible**: keep any note two chords share, and move the others to the nearest note of the next chord. Left-hand accompaniments and hymn parts are written this way.

Chords of C, every position

CHORD	ROOT POSITION	1ST INVERSION	2ND INVERSION
I (C)	C E G	E G C	G C E
IV (F)	F A C	A C F	C F A
V (G)	G B D	B D G	D G B
vi (Am)	A C E	C E A	E A C

EXAMPLE 1

Play I, IV, V, I smoothly



1. **I:** C E G. **IV:** keep C, move E up to F and G up to A: C F A.
2. **V:** no shared note. C steps down to B, F down to D, A down to G: B D G.
3. **I again:** B up to C, D up to E, G stays: C E G.

IV is F/C and V is G/B: **inversions** keep the hand in one place.

GAME

Which position?

1	2	3
C E G	E G C	G C E

Same three notes. Look at the bottom one: root, 3rd or 5th?

Answer: 1 root position · 2 first inversion (C/E) · 3 second inversion (C/G).

TRY IT · 10 SECONDS



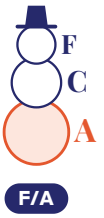
Climb the chord.

Play C E G, then E G C, then G C E, then C E G an octave up. Same snowman, turning over as it climbs.

PART IV · HARMONY · 4 · CONTINUED

EXAMPLE 2

Write F major in first inversion



1. **F major:** F A C.
2. **First inversion:** the 3rd, A, goes to the bottom.
3. **Stack it:** A C F.

First inversion of F: **A C F**, written **F/A** or **IV⁶** in C.

EXAMPLE 3

Find the second inversion of G
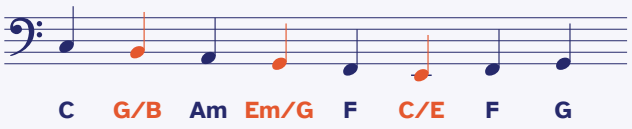


1. **G major:** G B D.
2. **Second inversion:** the 5th, D, goes to the bottom.
3. **Stack it:** D G B.

Second inversion of G: **D G B**, written **G/D** or **V⁶₄** in C.

YOU KNOW THIS

Pachelbel's Canon, the bass (here in C)



The famous bass walks **straight down the scale**: C B A G F E. It can, because some chords are **inversions**: G/B, Em/G and C/E put a different note in the bass.

Try it. Play the bass slowly and hum the famous violin tune over it.

THINK OF IT The same three friends in a photo, swapping places: still the same group (the same chord), but whoever stands at the front (the bass) changes the picture.

TRAP

- × Naming a chord from its bass note: in an inversion the bass is not the root. Restack it in 3rds first.
- × Thinking an inversion is a new chord: E G C is still C major.

SEE ALSO

Chords and triads p. 86 · Chords in a key p. 92 · Cadences p. 113

PRACTISE IT

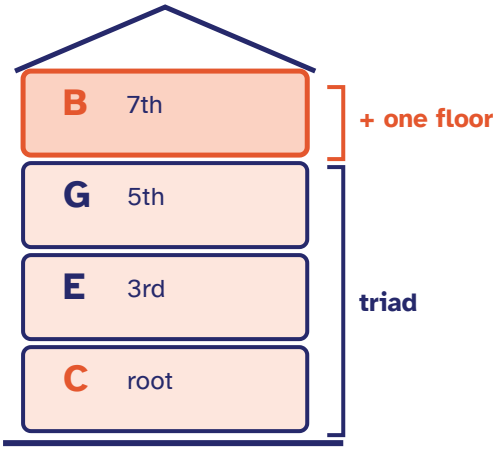
Workbook Harmony 4 · Online: Library › Chords

PART IV · HARMONY · 5

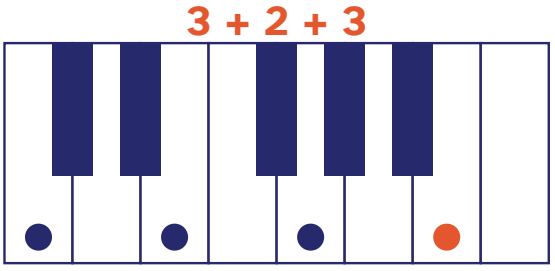
Seventh chords

Add one more floor to a triad and the chord changes colour: dreamy, bluesy, spooky. Same rule as before: skip one note, stack the next. The new note is the **7th**.

THE SECRET
One more floor on the snowman and the chord changes colour.

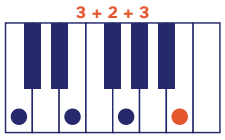


1. Build the triad: **C E G**.
2. Skip one, add the next: **B**.
3. Count the keys between: **3 + 2 + 3**. That is **Cmaj7**.



Five flavours, all on C

Cmaj7

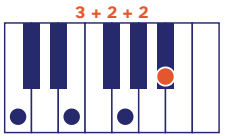


C E G B

dreamy

bossa nova, film

C7

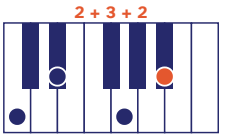


C E G B♭

bluesy

the blues

Cm7

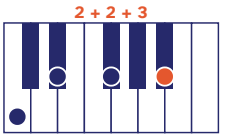


C E♭ G B♭

mellow

soul, jazz ballads

Cø7

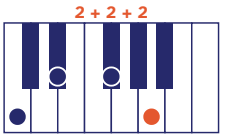


C E♭ G♭ B♭

searching

Bø7 is ii in A minor

C°7
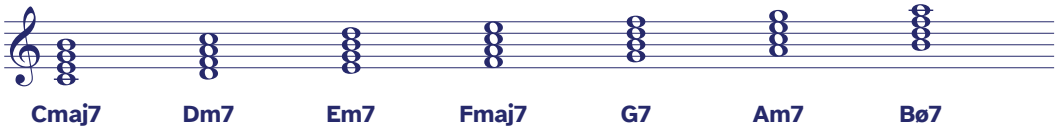


C E♭ G♭ B𝄫 (= A)

spooky

old film villains

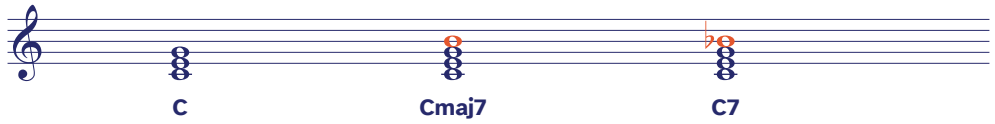
How the names work: major triad + major 7th = **maj7** · major triad + minor 7th = **7** · minor triad + minor 7th = **m7** · diminished triad + minor 7th = **ø7** · diminished triad + diminished 7th = **°7**. Also written: CM7 or CΔ = Cmaj7, C–7 = Cm7, Cm7♭5 = Cø7, Cdim7 = C°7.

A seventh on every floor of C major

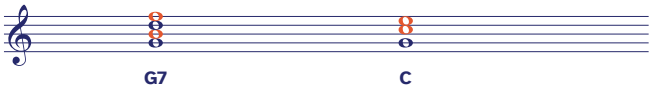


Only white keys. G7 is the one with the strongest pull home.

PART IV · HARMONY · 5 · CONTINUED



One note decides: B makes Cmaj7, B♭ makes C7. Everything below stays the same.



Why G7 pulls home: its B wants to rise to **C** and its F wants to fall to **E**. Two small steps, and you are home in C.

Spot the difference

EXAMPLE 1

Which seventh is it? G B D F



1. **Count the keys between:** G to B is 3, B to D is 2, D to F is 2.
2. **3 + 2 + 2** is the bluesy one.
3. **The root is G:** G7.

G B D F is **G7**, the chord that pulls hardest back to C.

GAME

Which seventh?

1	2	3
G B D F	C E G B	D F A C

A triad with one more floor. Name each chord.

Answer: 1 G7 (dominant 7th) · 2 Cmaj7 · 3 Dm7.

TRY IT · 10 SECONDS



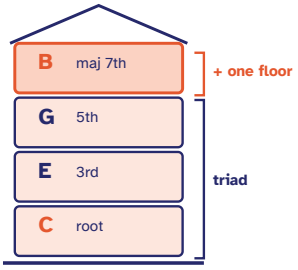
Play G7, then C.

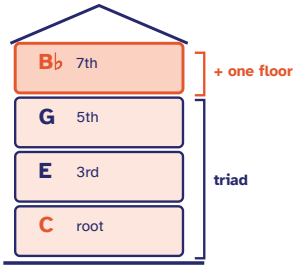
G B D F, hold it, then C E G. The F in G7 leans on E; the B leans up to C. That lean is the pull home.

PART IV · HARMONY · 5 · CONTINUED

EXAMPLE 2

Build Cmaj7, then C7





1. **Start with C major:** C E G.
2. **Add B**, one key below the top C: Cmaj7, dreamy.
3. **Lower it to B♭:** C7, the bluesy one that wants to move.

One key decides it: **B** is Cmaj7, **B♭** is C7.

HOW TO

Hear the four sevenths

1. **Play C7, Cmaj7, Cm7 and Cm7♭5** slowly, one after another.
2. **Name the feeling** of each: bluesy, dreamy, soft, uneasy.
3. **Then play one** without looking and guess which it is.

YOU KNOW THIS

The blues, with sevenths

I C7	I C7	I C7	I C7
IV F7	IV F7	I C7	I C7
V G7	IV F7	I C7	V G7

Swap the triads for **7th chords** and the twelve-bar chart (p. 93) turns into the real blues: the 7th is the blue note.

Try it. Play the chart with plain C, F and G first (p. 93), then add the red 7ths: B♭, E♭ and F.

THINK OF IT A triad is plain toast; the 7th is the topping: same bread, new taste.

TRAP

- × Writing C7 when you mean Cmaj7: "7" alone means the bluesy one, with B♭.
- × Stacking the 7th from the wrong note: count a 7th up from the root, not from the 5th.

SEE ALSO
Four kinds of triad p. 89 · Chords in a key p. 92

PRACTISE IT
Workbook Harmony 5 · Online: Library › Chords

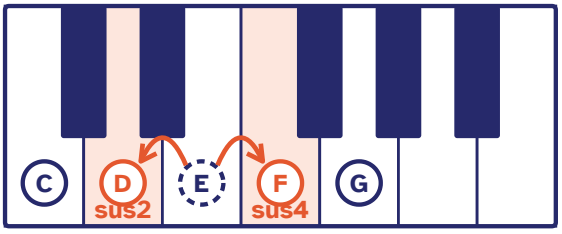
PART IV · HARMONY · 6

Colour chords

The 3rd is the chord's mood: major or minor, happy or sad. Slide it aside, take it away, or add one note, and a plain chord takes on a new colour. Five colours, all from C E G.

THE SECRET

Move the 3rd, take it away, or add one note: five new colours.

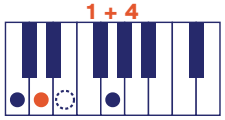


1. Start from **C E G**. The **E** (the 3rd) makes it major.
2. Slide the E up to the next letter, **F**: **Csus4**.
3. Or down to the letter below, **D**: **Csus2**. With no 3rd, the chord is neither major nor minor: it **hangs in the air**.

Five colours, all on C

Csus2

1 + 4



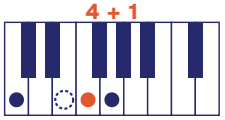
C D G

open

the 3rd steps down

Csus4

4 + 1



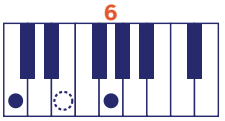
C F G

floating

the 3rd steps up

C5

6



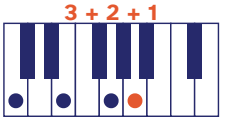
C G

strong

no 3rd at all

C6

3 + 2 + 1



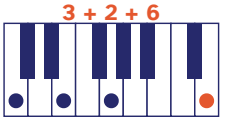
C E G A

warm

add the 6th

Cadd9

3 + 2 + 6

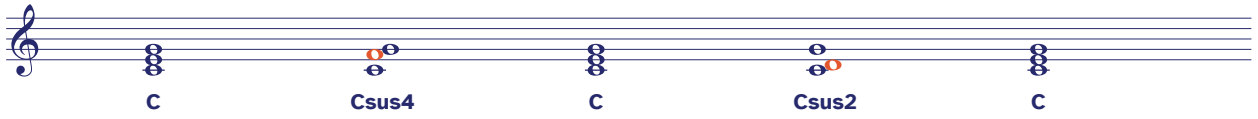


C E G D

sparkly

add the 9th on top

Sus wants to come home



The red note leans, then slides back to E: F steps down, D steps up. Play it slowly: C, Csus4, C, Csus2, C. Guitarists love this shimmer.

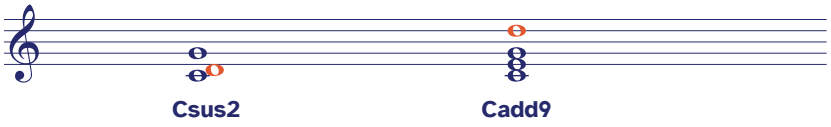
On any root, the counts stay the same

COLOUR	KEYS BETWEEN	ON C	ON G	ON F	ON D
sus2	1 + 4	C D G	G A D	F G C	D E A
sus4	4 + 1	C F G	G C D	F B♭ C	D G A
5	6	C G	G D	F C	D A
6	3 + 2 + 1	C E G **A**	G B D **E**	F A C **D**	D F♯ A **B**
add9	3 + 2 + 6	C E G **D**	G B D **A**	F A C **G**	D F♯ A **E**

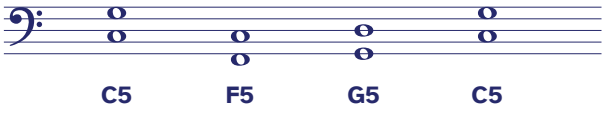
PART IV · HARMONY · 6 · CONTINUED

sus: the 3rd held up or down · **5**: no 3rd at all · **6** and **add9**: one note added on top. Csus means Csus4.

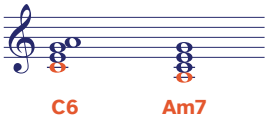
Spot the difference



The same red D, two jobs. Csus2 throws the E out; Cadd9 keeps the E and puts the D on top.



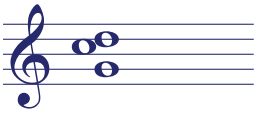
The power chord: just the root and the 5th, played low in the left hand. With no 3rd it is neither happy nor sad, only **strong**: the sound of rock guitar. Play C5, F5, G5, C5 and drum the rhythm.



C6 or Am7? The same four notes, C E G A. The red bottom note decides: C at the bottom is **C6**; A at the bottom is **Am7**. More on the bass note in Slash chords, p. 107.

EXAMPLE 1

Which colour is it? G C D



1. **Count the keys between:** G to C is 4, C to D is 1.
2. **4 + 1** is the sus4 card: no B (no 3rd), and C has taken its place.
3. **The root is G:** Gsus4.

G C D is **Gsus4**. Let the C fall to **B** and you have G major.

GAME

sus2, sus4 or add9?

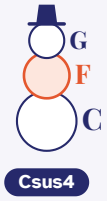
1	2	3
C D G	C F G	C E G D

In a sus chord the 3rd has moved; in an add chord nothing moved and a note was added.

Answer: 1 Csus2 · 2 Csus4 · 3 Cadd9.

PART IV · HARMONY · 6 · CONTINUED

TRY IT · 10 SECONDS

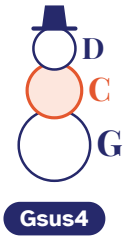


Csus4, then C.

Play C F G, then let the F slide down to E. The sus chord waits; the plain chord arrives.

EXAMPLE 2

Build Gsus4

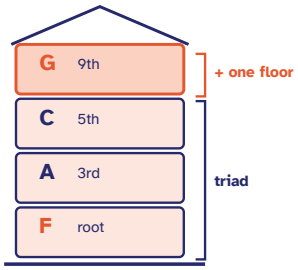


1. **Start with G major:** G B D.
2. **Lift the 3rd, B, up a step** to C.
3. **G C D** waits; let the C fall back to B to finish.

Gsus4: **G C D**, then home to G B D.

EXAMPLE 3

Build Fadd9



1. **Start with F major:** F A C.
2. **Keep every note.**
3. **Add the 9th:** G, the note just above F, placed on top.

Fadd9: **F A C G**. Nothing moved; one note was added.

THINK OF IT Dressing the snowman: a scarf (the 6th) or a hat (the 9th) and it is still the same snowman. Sus moves its middle ball; a power chord takes the middle ball away.

TRAP

- × Writing C2: some books mean Csus2, others Cadd9. Check whether the E is there.
- × Calling Cadd9 a ninth chord: C9 also has the 7th (B♭). Add9 is just the triad plus the 9th (D on C), nothing else (The tower, p. 104).

SEE ALSO
Four kinds of triad p. 89 · Seventh chords p. 98

PRACTISE IT
Workbook Harmony 6 · Online: Library › Chords

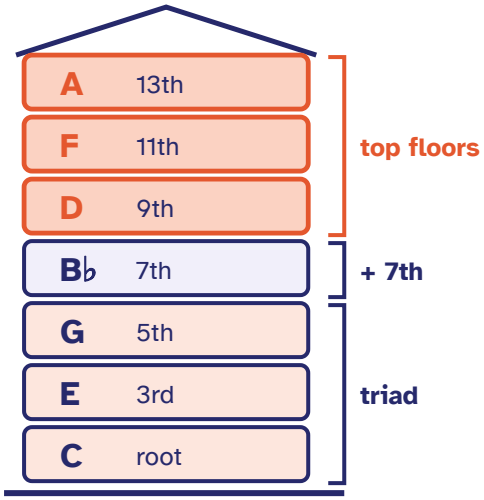
PART IV · HARMONY · 7

The tower: 9, 11, 13

Why stop at the 7th? Keep stacking 3rds and the building grows three more floors: the 9th, the 11th and the 13th. At the top you have used **all seven letters, A to G**.

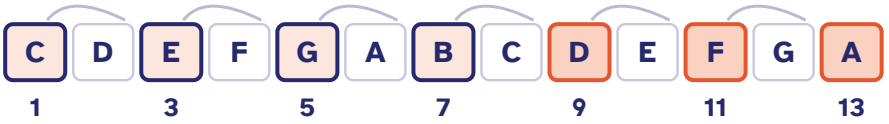
THE SECRET

Keep skipping letters above the 7th: 9, 11, 13 are 2, 4, 6 an octave up.



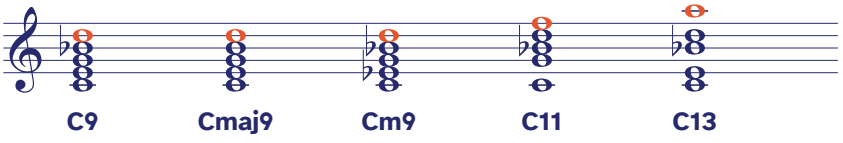
1. Build the seventh chord: **C E G B♭** (C7).
2. Keep skipping one letter: **D**, **F**, **A**. Those floors are the **9th, 11th and 13th**.
3. Take 7 away to find the letter: 9 → 2nd (D), 11 → 4th (F), 13 → 6th (A). Same letters, an octave up.

Skip a letter, all the way up



Start on C and take every other letter: C E G B D F A (letters only: in C7 the B is B♭). After the 13th comes C again: the tower is full.

Five tall chords on C

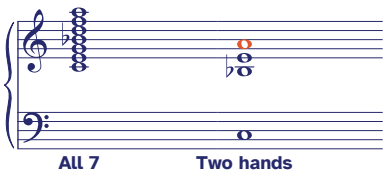


The red note is the top floor that gives each chord its name. C11 and C13 leave some notes out: see p. 105.

CHORD	BUILT AS	NOTES	SOUNDS	HEARD IN
C9	C7 + 9th	C E G B♭ **D**	soulful	funk, soul
Cmaj9	Cmaj7 + 9th	C E G B **D**	dreamy	film, ballads
Cm9	Cm7 + 9th	C E♭ G B♭ **D**	smooth	neo-soul, R&B
C11	C9 + 11th, no 3rd	C G B♭ D **F**	airy	gospel, pop
C13	C7 + 13th	C E B♭ (D) **A**	rich	jazz, blues endings

PART IV · HARMONY · 7 · CONTINUED

Too many notes? Leave some out



All seven notes sound muddy. Two hands play C13 with four: C in the left hand, B♭ E A in the right.

Ten fingers, seven floors: you choose. **Keep** the 3rd and the 7th (they say what kind of chord it is) and the red top note (its colour). **Leave out** the 5th first. The right hand can drop the root too, when the left hand or a bass player plays it.

With a major 3rd, the 11th (F) rubs against E. So C13 leaves out the F. C11 does the opposite: it keeps the F and leaves out the E.

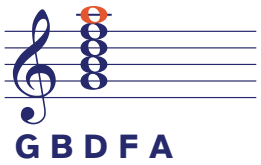
6 or 13? add9 or 9?

SAME TOP NOTE	WITHOUT THE 7TH	WITH THE 7TH (B♭)
A on C	**C6**: C E G A	**C13**: C E B♭ A
D on C	**Cadd9**: C E G D	**C9**: C E G B♭ D

Look for the 7th. No 7th: a colour chord (p. {H6}). A 7th: a tower chord.

EXAMPLE 1

Name this tall chord: G B D F A



1. **Find the root:** each note skips one letter, so the bottom note, G, is the root.
2. **The first four floors**, G B D F, count 3 + 2 + 2: G7.
3. **One more floor:** A is the 2nd above G, an octave up: the 9th.

G B D F A is **G9**: G7 with the 9th on top.

GAME

Which floor?

1 — the 9th of C

2 — the 11th of C

3 — the 13th of C

Keep skipping one letter above the 7th: C E G B♭ D F A.

Answer: 1 D · 2 F · 3 A.

PART IV · HARMONY · 7 · CONTINUED

TRY IT · 10 SECONDS

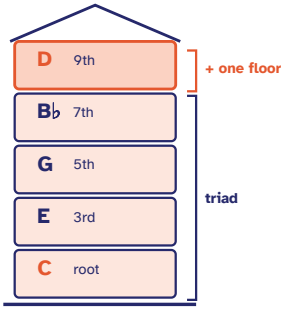


Play C E G B♭ D.

Both hands if you need them: left hand C E G, right hand B♭ D. A C9: the tower with one floor above the 7th.

EXAMPLE 2

Build C9



1. **Build C7:** C E G B♭.
2. **Skip one more letter** above B♭: D.
3. **Five notes:** share them between the hands if you need to.

C9: **C E G B♭ D**, the tower with one floor above the 7th.

HOW TO

Build any tall chord

1. **Start on the root** and skip one letter each time: 1 3 5 7 9 11 13.
2. **Stop** at the number in the name.
3. **Leave out** the 5th, then the 11th, when it is too many notes for your hands.

THINK OF IT A tower block has a lift: you don't stop at every floor. Get out at the 3rd, the 7th and the top floor with the view.

TRAP

- × Hunting far up for the 9th: it is just the 2nd, an octave higher (D on C).
- × Calling C13 a C6: the 13th chord has the 7th (B♭) as well.
- × Playing F and E together in C13 or C11: they clash. C13 drops the F; C11 drops the E.

SEE ALSO
Seventh chords p. 98 · Colour chords p. 101

PRACTISE IT
Workbook Harmony 7 · Online: Library › Chords

PART IV · HARMONY · 8

Slash chords

C/E means one chord with a chosen bass note: play a C chord, with E as the lowest note. It is not a fraction, and not two chords.

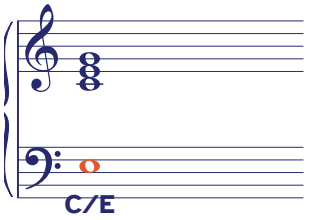
THE SECRET
Chord before the slash, bass note after it.

C/E

the chord
right hand

the bass
left hand, lowest note

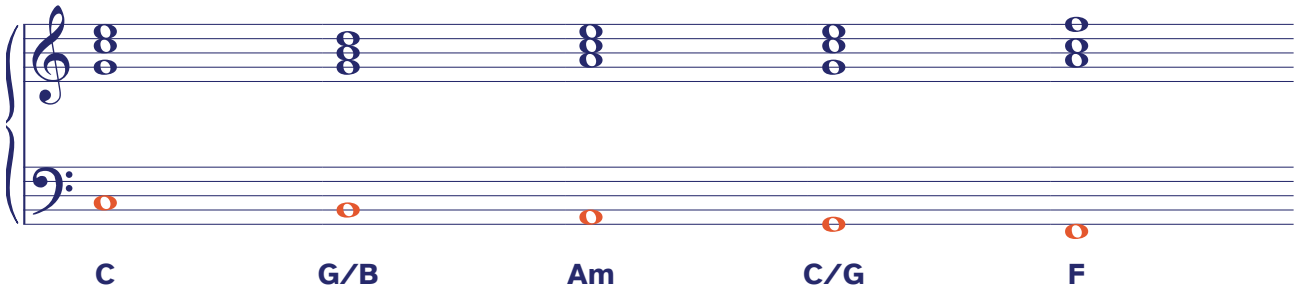
1. Before the slash: the **chord**. C means C E G, in the right hand.
2. After the slash: the **bass**. E goes in the left hand, below everything.
3. Play them together: **C/E**, “C over E”.



Four jobs for a slash chord

JOB	EXAMPLE	THE BASS NOTE IS	WHAT YOU HEAR
Inversion	C/E · C/G	a note of the chord	smoother changes (p. 95)
Walk-down	C · G/B · Am	one step down each time	a bass line that walks down
Pedal	F/C · G/C	the same note, held	chords move over a floor
Colour	F/G · D/C	not in the chord	open, modern: pop, gospel

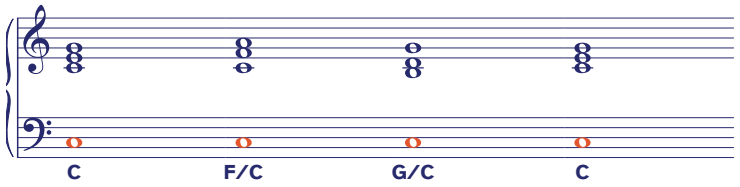
Job 2: a bass that walks down



The red bass walks straight down, C B A G F, while the right hand moves a step at most. G/B and C/G make the steps possible.

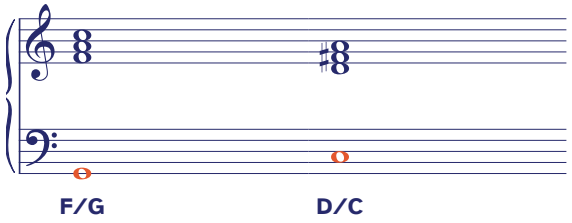
PART IV · HARMONY · 8 · CONTINUED

Job 3: a pedal, one bass under everything



The bass holds C while the chords change above it: C, F/C, G/C, C. The chords pull away from the C, then come home. (It is called a pedal after the organ's foot pedals, which hold long low notes.)

Job 4: a bass from outside the chord

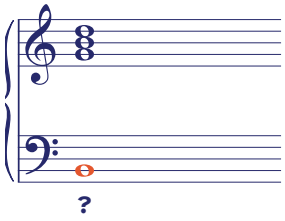


F/G: F A C over a G. G is not in the F chord, so it floats, open and unfinished, ready to land on C. Pop and gospel players love it.

D/C: D F♯ A over a C. The C is not in the D chord, so it rubs, then usually steps down to B (D/C to G/B).

EXAMPLE 1

Name it: G B D over a low B



1. **The right hand:** G B D, the G chord.
2. **The lowest note:** B, in the left hand.
3. **Chord, slash, bass:** G/B. B is in the G chord: an inversion.

It is **G/B**, "G over B": G major, 1st inversion.

GAME

Read the slash

1	2	3	4
G/B	C/G	F/A	Am/C

Before the slash: the chord. After it: the bass note. Which note goes in the left hand?

Answer: 1 B · 2 G · 3 A · 4 C.

PART IV · HARMONY · 8 · CONTINUED

TRY IT · 10 SECONDS



Right hand C, left hand E.

Play C E G in the right hand and a low E in the left: C/E. Now move the left hand to G: C/G. Same chord, new floor.

EXAMPLE 2

Play C/G



1. **Before the slash:** the chord, C E G, in the right hand.
2. **After the slash:** the bass note, a low G, in the left hand.
3. **Play them together.**

C/G: the C chord standing on a **G**.

EXAMPLE 3

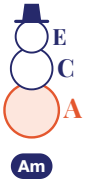
Walk the bass down: C, C/B, Am



1. **Right hand** holds C E G for two chords.



2. **Left hand** walks down: C, then B.



3. **Then A**, under A C E: Am.

The bass walks **C B A**, one step at a time, while the chords hardly move.

THINK OF IT A house and the ground it stands on: the slash says what to build (the chord) and where to build it (the bass). Same house, new ground, new view.

TRAP

- × Writing it back to front: the chord comes first, the bass second. E/C would be an E chord over C.
- × Letting the right hand dip below the bass: the letter after the slash must be the lowest note you play.
- × Forgetting the chord itself: in C/E the right hand still plays the whole C chord, C E G, not just C.

SEE ALSO
Inversions p. 95 · Colour chords p. 101

PRACTISE IT
Workbook Harmony 8 · Online: Library › Chords

PART IV · HARMONY · 9

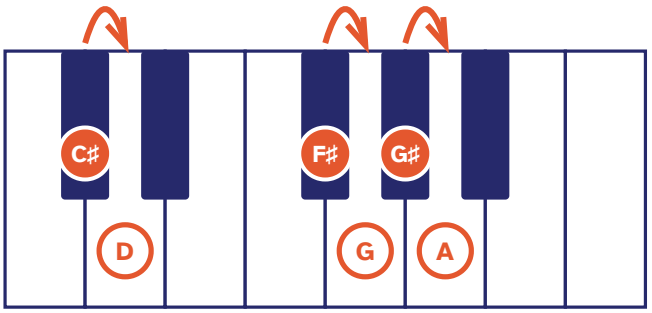
Chords from outside the key

C major lives on the white keys. A black key in a chord is a **guest**. Guests never stay: each one moves **one key**, and shows you where the music goes next.

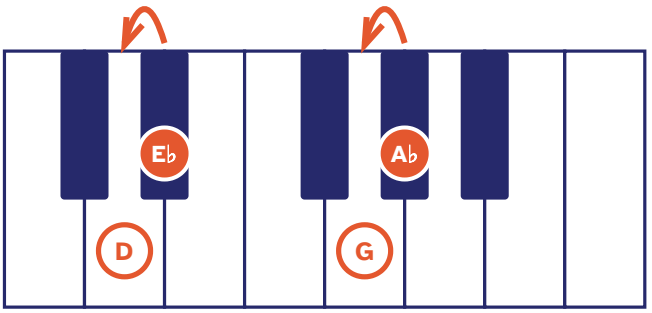
THE SECRET

Sharp guests push up, flat guests sigh down: follow them one key and you find the next chord.

Sharp guests push up

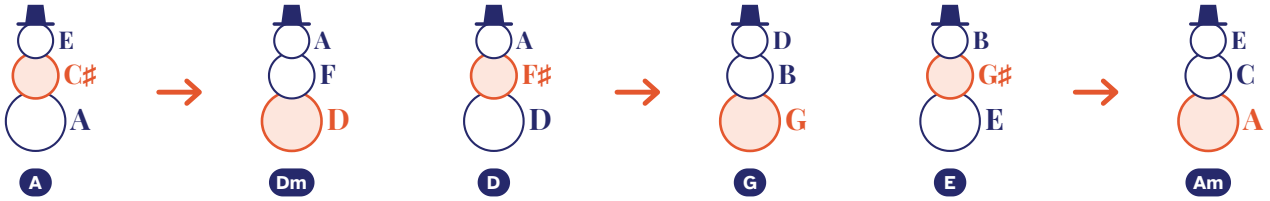


Flat guests sigh down

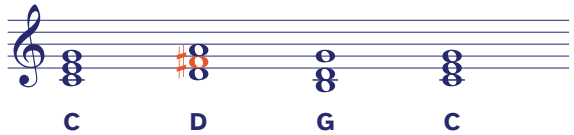


1. Find the **black key** in the chord: that is the guest.
2. A **sharp**? Go **up** one key. A **flat**? Go **down** one key.
3. You have found a note of the **next chord**. Play it and hear it land.

Sharp guests: up one key names the next chord

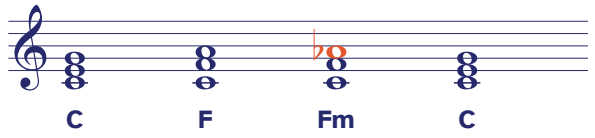


C♯ goes up to **D**, so the next chord is D minor. F♯ goes up to **G**: G. G♯ goes up to **A**: A minor.



Try it in a song: C, D, G, C. The **F♯** pushes up into G, then G leads home to C.

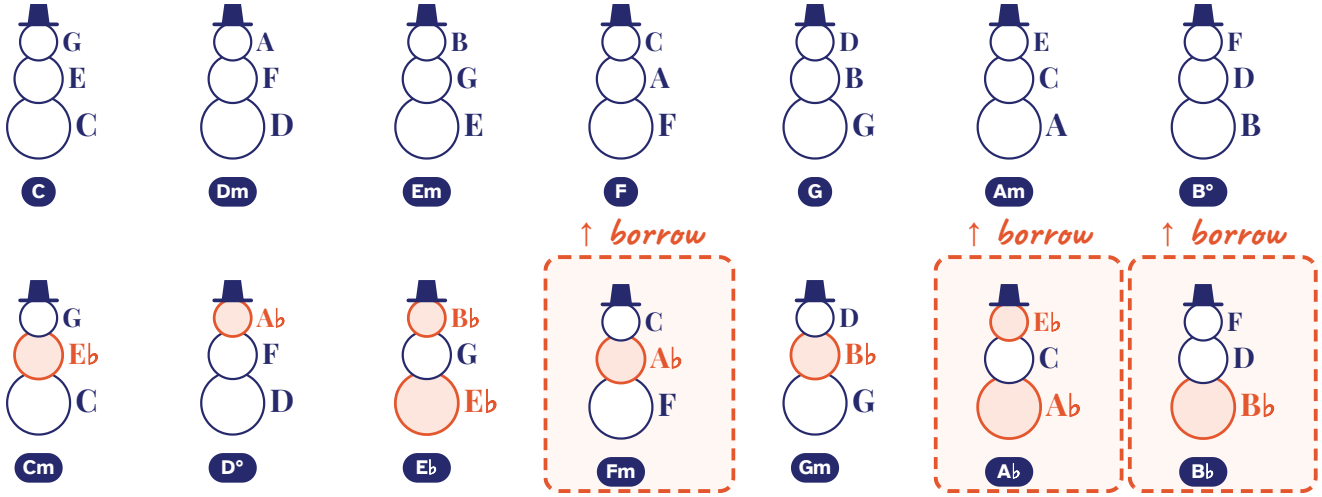
Flat guests: borrowed from the twin, they sigh down



C major and C minor are **twins**: the same home note, C, in different clothes. A flat guest is one piece of the twin's outfit. In C, F, Fm, C the **A♭** sighs down to G.

PART IV · HARMONY · 9 · CONTINUED

C major *at home*

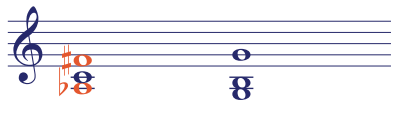


C minor *the twin: same home note, darker clothes*

Their names, for later

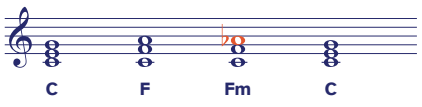
GUEST	NOTES	MOVES	NEXT CHORD	ITS NAME
sharp guest	D F♯ A	F♯ up to G	G	V of V
sharp guest	A C♯ E	C♯ up to D	Dm	V of ii
sharp guest	E G♯ B	G♯ up to A	Am	V of vi
flat guest	F A♭ C	A♭ down to G	C	borrowed iv
flat guest	A♭ C E♭	A♭ down to G	G or B♭	borrowed ♭VI
flat guest	B♭ D F	B♭ on to C	C	borrowed ♭VII

B♭ is the one flat guest that walks **up** instead, on to C: the rock and film ending B♭ → C. The bigger guests (the Neapolitan, Level 7) follow the same idea.



An advanced guest (Extra): the augmented 6th has two guests at once. The flat **A♭** falls and the sharp **F♯** rises, and both land on G. The rule still works.

TRY IT · 10 SECONDS



Play C, F, Fm, C slowly.

Hear the A♭ sigh down to G? Now play C, D, G, C: the F♯ pushes up instead. One sighs, one pushes.

PART IV · HARMONY · 9 · CONTINUED

EXAMPLE

Find the guest: A C♯ E in C major



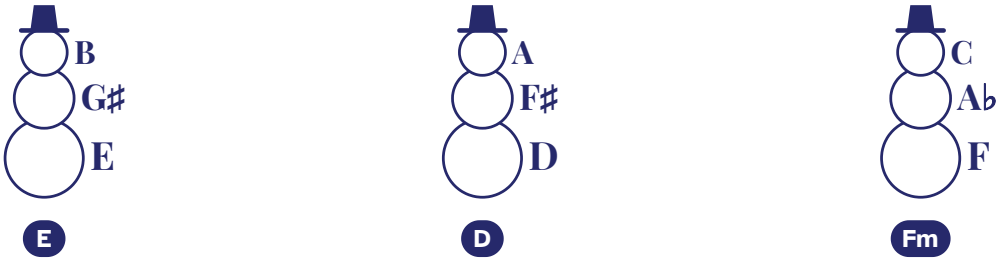
A C♯ E

1. **The black key:** C♯. That is the guest.
2. **It is a sharp,** so it goes up one key: to D.
3. **D names the next chord:** D minor (D F A).

A C♯ E leads to **D minor**. Play them one after the other and hear C♯ slide into D.

GAME

Where does the guest go?



In C major: find each guest, follow it one key, and write the chord that comes next.

Answer: 1 G♯ up to A: A minor · 2 F♯ up to G: G · 3 A♭ down to G: C.

THINK OF IT A surprise guest at dinner: one new face changes the whole conversation, and then everyone goes home.

TRAP

- × Following a sharp downwards: sharps push **up**, flats sigh **down**.
- × Learning the names first: play it, hear where the guest goes, then name it.
- × Mixing up the relatives: C minor is C major's twin (same home note). A minor is its neighbour (same key signature). We borrow from the twin.

SEE ALSO

Chords in a key p. 92 · Slash chords p. 107 · Cadences p. 113

PRACTISE IT

Workbook Harmony 9 · Online: Library › Chromatic chords

PART IV · HARMONY · 10

Cadences

A phrase is a musical sentence, and it needs punctuation. The last two chords of a phrase are its **cadence**. They tell you: finished, taking a breath, or surprise!

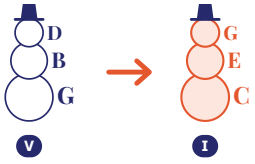
THE SECRET

Music has full stops and commas too: listen to the last two chords of a phrase.

.

Perfect

V → I

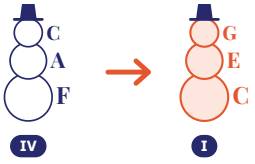


Home. The end.

Amen

Plagal

IV → I

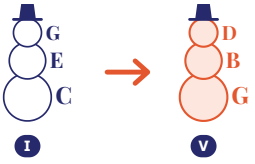


A soft, gentle end.

,

Imperfect

… → V

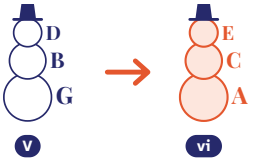


To be continued…

?!

Interrupted

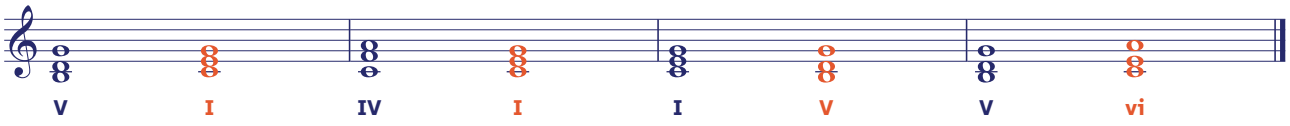
V → vi



Surprise! Not home.

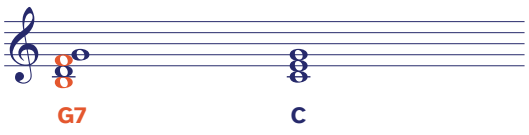
1. Find the end of a phrase: often a long note, or a breath.
2. Name its **last two chords** with numerals: V I, IV I, anything then V, or V vi.
3. Match the punctuation: full stop (.), Amen, comma (,) or ?!

The four cadences in C



Perfect, plagal, imperfect, interrupted. The red chord is where each one lands.

CADENCE	CHORDS	FEELS LIKE	ALSO CALLED
Perfect	V → I	a full stop: finished	authentic, full close
Plagal	IV → I	the “A-men” sung at the end of a song in church	Amen cadence
Imperfect	any → V	a comma: waiting	half close, half cadence
Interrupted	V → vi	a surprise: not home yet	deceptive

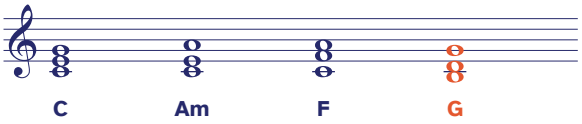


Make the full stop stronger: add the 7th to V. In G7 the **F** falls to E and the **B** rises to C: two small steps home (Seventh chords, p. 98).

PART IV · HARMONY · 10 · CONTINUED

EXAMPLE

How does this phrase end? C Am F G



1. **The last two chords:** F, then G.
2. **As numerals:** IV, then V.
3. **It lands on V:** the comma.

An **imperfect cadence**: the music takes a breath, and the next phrase answers it.

GAME

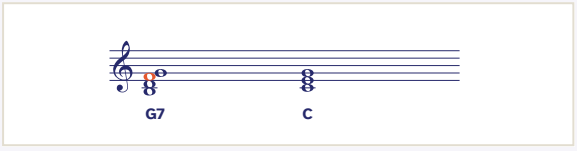
Punctuate the phrases



Each pair ends a phrase in C major. Turn the letters into numerals first (C is I, F is IV, G is V, Am is vi), then write full stop, comma, Amen or ?! beside each.

Answer: 1 Amen (plagal), 2 ?! (interrupted), 3 comma (imperfect), 4 full stop (perfect).

TRY IT · 10 SECONDS



Play G7, then C. The end!

Now play G7, then A minor: “huh?!” That is the interrupted cadence. Then play F, then C, and sing “A-men”.

THINK OF IT Punctuation in a sentence: a full stop ends it, a comma makes you wait, and a surprise makes you gasp. Music ends its phrases the same way.

TRAP

- × Calling every ending on I (C) perfect: a perfect cadence comes **from V**. F to C is plagal.
- × Mixing up imperfect and perfect: imperfect **ends on** V; perfect **starts on** V and lands on I.

SEE ALSO

Chords in a key p. 92 · Seventh chords p. 98 · Chords under a melody p. 115

PRACTISE IT

Workbook Harmony 10 · Online: Library › Cadences

PART IV · HARMONY · 11

Chords under a melody

The melody is the singer; the chords are the band. To choose a chord for each bar, look at the melody's main notes and pick the snowman holding them.

THE SECRET

Look at the melody's main notes, then pick the snowman that holds them.



I



IV



V
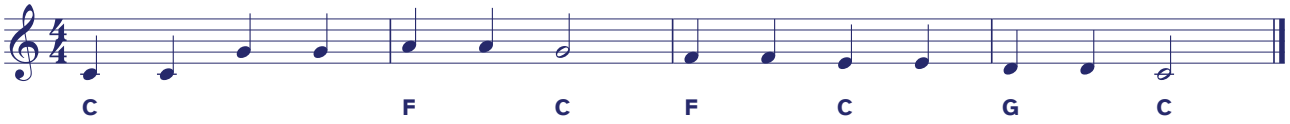
C	D	E	F	G	A	B
I, IV	V	I	IV	I, V	IV	V

Which snowman holds the tune's main notes?

Every note of C major lives in I, IV or V. C and G live in two: let your ear choose.
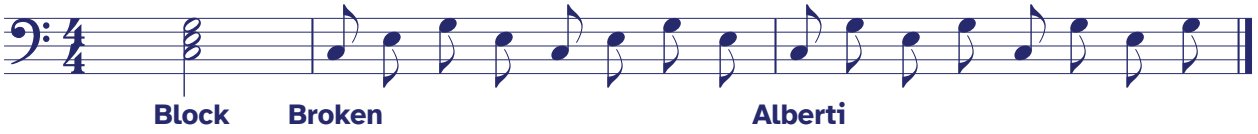
1. Look at the **main notes** of the bar: the long ones, and the ones on beats 1 and 3.
2. Find the snowman that holds them: **I, IV or V**.
3. Play that chord in the left hand while the right hand plays the tune.

Twinkle, Twinkle, with its chords



C C G G sit in I (C); A A in IV (F), then G in I; F F in IV, E E in I; D D in V (G), and the last C in I. The chord changes halfway through bars 2, 3 and 4.

Three ways to play the chord in the left hand



Block: all three at once. Broken: one at a time, up and down. Alberti: low, high, middle, high, named after Domenico Alberti, who loved it.

PART IV · HARMONY · 11 · CONTINUED

EXAMPLE

Choose the chord: E D C D | E E E

E D C D | E E E

red = beats 1 and 3

1. **Bar 1:** the main notes are E and C (beats 1 and 3).
2. **E and C** both live in I.
3. **Bar 2:** E E E lives in I too.

Both bars take **I** (C). That is the start of Mary Had a Little Lamb.

GAME

Pick the snowman
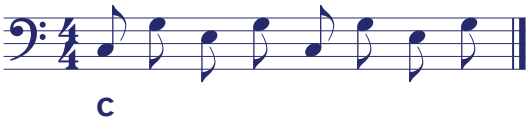
F A F A D B G B G E C E

1 2 3

Each bar is in C major. Which snowman holds its notes: I, IV or V?

Answer: 1 IV (F A C) · 2 V (G B D) · 3 I (C E G).

TRY IT · 10 SECONDS



Left hand: C G E G, over and over.

That is the Alberti bass on the C chord. Keep it going while you hum Twinkle, then try it on F (F C A C) and G (G D B D).

YOU KNOW THIS

Mozart, Sonata in C, K. 545: the left hand



The opening left hand is an **Alberti bass**: C G E G is just the C chord, broken into low, high, middle, high; D G F G is G7 (G B D F, without the B). Mozart wrote it an octave higher, starting on middle C.

Try it. Play this left hand alone, then add the famous tune: C, E G, B, C D C.

THINK OF IT Like picking a coat for the weather: when the tune changes, change the chord.

TRAP

- × Choosing a chord for a passing note: use the main notes (strong beats, long notes).
- × Playing the left hand louder than the tune: the melody is the star, the chords are the stage.

SEE ALSO

Chords in a key p. 92 · Cadences p. 113

PRACTISE IT

Workbook Harmony 11 · Online: Library › Chords

PART IV · HARMONY · 12

Progressions everywhere

A **progression** is a trip from chord to chord. A handful of trips turn up in song after song. Learn them by their numbers and you can play them in any key.

THE SECRET
Most songs take the same few trips: home, away, a pull, and home again.

I home · vi iii home's cousins · IV ii away · V the pull back home

The pop four
countless pop songs and ballads

I C → V G → vi Am → IV F

The fifties
old rock 'n' roll and doo-wop

I C → vi Am → IV F → V G

The jazz three-step
jazz standards (as ii7, V7, Imaj7)

ii Dm → V G → I C

The canon
Pachelbel's Canon and its many children

I C → V G → vi Am → iii Em → IV F → I C → IV F → V G

1. Learn the trip by **numbers**: I V vi IV.
2. Pick a key and turn the numbers into chords: in C, **C G Am F**.
3. Play it round and round. Change key: the numbers stay the same.

Same trip, any key

The pop four	I	V	vi	IV
in C	C	G	Am	F
in G	G	D	Em	C
in F	F	C	Dm	B♭
in D	D	A	Bm	G

EXAMPLE

Play the pop four in G major

I G → V D → vi Em → IV C

1. **I** is the key note: G.
2. **V** is the 5th note of the G scale: D. **vi** is the 6th: E, minor, so Em.
3. **IV** is the 4th note: C.

G, D, Em, C: **the pop four in G**. Same trip, new key.

PART IV · HARMONY · 12 · CONTINUED

GAME

Name the trip

1 **C · Am · F · G**

2 **Dm · G · C**

3 **C · G · Am · F**

All in C major. Turn each into numbers, then name the trip: the pop four, the fifties or the jazz three-step.

Answer: 1 the fifties (I vi IV V) · 2 the jazz three-step (ii V I) · 3 the pop four (I V vi IV).

TRY IT · 10 SECONDS
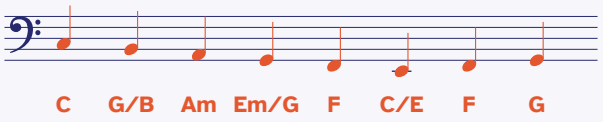


Play C, G, Am, F, four beats each.

Go round four times without stopping, left hand plays the chord's name note (C for C, A for Am), right hand the whole snowman. Hum any tune you like over it.

YOU KNOW THIS

Pachelbel's Canon: the trip under the tune



The famous bass walks down, and above it the chords take the **canon trip**: C G Am Em F C F G. Countless songs borrow it.

Try it. Play the bass slowly with your left hand and the chords in your right.

THINK OF IT A walk round the block: leave home, wander, feel the pull, and turn the corner back home.

TRAP

- × Learning songs one by one: learn the trip by numbers, then change key.
- × Mixing up the pop four and the fifties: same four chords, different order. Listen to which chord comes second.

SEE ALSO

Chords in a key p. 92 · Cadences p. 113 · Chords under a melody p. 115

PRACTISE IT

Workbook Harmony 12 · Online: Library › Chords

PART IV · HARMONY · TIPS

When a chord sounds wrong

A chord that sounds wrong usually has one small slip. Name the slip, fix that one note, and the chord comes right.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
A chord sounds sour	**the kind**: keys between, 3 + 2 or 2 + 3	p. 89
Your hand jumps all over	**an inversion**: keep the shared note still	p. 95
A big chord sounds muddy	**leave notes out**: drop the 5th first	p. 104
A sharp or flat surprises you	**a guest**: sharps go up, flats go down	p. 110
The end does not feel finished	**the cadence**: end with V → I	p. 113
The left hand fights the tune	**the main notes**: pick the snowman holding them	p. 115

Four ways to learn chords fast

1. **Build, then play**
   Say the snowman (C E G) before your fingers go down.

2. **Hands alone first**
   Left hand chords alone, then the tune alone, then together.

3. **One trip a week**
   Learn one progression, then play it in three keys.

4. **Listen for the kind**
   Play major and minor side by side: bright, then dark.

EXAMPLE

Fix a sour chord



1. **It sounds dark**, but the music wants a bright C chord.
2. **Count the keys between:** 2, then 3. That is minor.
3. **Move one note:** E♭ up to E makes 3 + 2. Bright again.

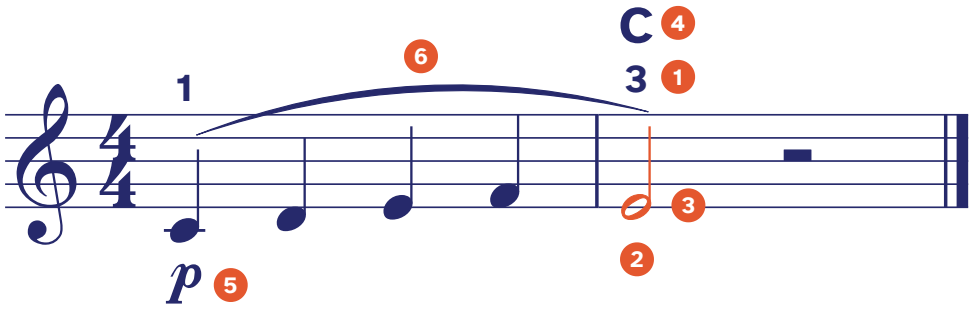
One note was off: **fix the middle of the snowman**.

THINK OF IT Like a mechanic: one odd sound points to one part. Fix that part and the engine runs smoothly.

PUTTING IT TOGETHER · 1

Everything one note tells you

Look at the red note. It carries six pieces of information at once, and each one can be read straight off the page. Read all six and the note is right; miss one and it is only nearly right.



1 The finger number

3: the middle finger plays it. Fingering is planned, not guessed.

PLAYING AND LISTENING (LATER)

2 The beat

Beat 1 of bar 2, and it is held for **two beats**.

PART II · TIME · P. 39

3 The note name

E: the white key just right of the two black keys.

PART I · PITCH · P. 22

4 The chord

The **C** above the staff: this E belongs to the C major chord.

PART IV · HARMONY · P. 86

5 The marks

p and the slur: play it softly and smoothly.

EXPRESSION (LATER)

6 The phrase

The slur ends here, so the note lands gently, like the end of a sentence.

SHAPE AND STYLE (LATER)

THE SAME SIX, IN ANY MUSIC

A CLASSICAL SCORE
Most of it is written in: finger numbers, dynamics, slurs. Your job is to read every one.

A POP LEAD SHEET
A melody and chord names only. You choose the fingers, the marks and the shape yourself.

A JAZZ CHART
Chords and a style word such as Swing: the beat and the shape are yours to feel.

Try it on any piece: point to one note and find all six. Finger numbers count as much as the note name.

THINK OF IT A note is like a word said aloud: its spelling, its timing, its tone of voice and its place in the sentence all change what it means.

PUTTING IT TOGETHER · 2

When a passage won't come right

Name the problem first, then fix that one thing. Most hard bars are hard for one reason, and the six things a note tells you point straight to it.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
You stumble in the same place every time	**the finger number**	Playing and listening
It rushes, drags or the counting slips	**the beat**	Part II · Time
A note sounds wrong	**the note name** and the key	Part I · Pitch, Part III · Scales and ke
A chord sounds odd or thin	**the chord**	Part IV · Harmony
It is correct but flat and dull	**the marks**	Expression
It does not seem to go anywhere	**the phrase**	Shape and style

Six ways to fix a hard bar

1. **Slow it right down**
   Slow enough to be right three times in a row, then a little faster.

2. **Hands separately**
   Each hand on its own until it is easy, then together, slowly.

3. **Start from the end**
   Play the last beat, then the last two, then three: you always finish on what you know best.

4. **Practise the join**
   The last note of one bar and the first of the next: most slips live there.

5. **Climb the metronome**
   Up four clicks each time it is clean; down eight if it falls apart.

6. **Put it back**
   Play two bars before and two after, so the fixed bar lives inside the music again.

Before you play a new piece: a thirty-second look

Key signature
Which sharps or flats, all the way through?

Time signature
How many beats in a bar, and which note is one beat?

Tempo and mood
The words at the top: how fast, what character?

Patterns
Bars that repeat or nearly repeat: learn them once.

THINK OF IT Like a doctor: name the symptom first, then treat that one thing. Practising everything harder rarely fixes the one bar that is wrong.